

61. Capture Engine Unit (CEU)

61.1 Overview

The Capture Engine Unit (CEU) is a capture module that fetches image data externally input and transfers it to the memory. The CEU is connected to the system bus via bus bridge modules.

Lists the features of CEU as follows.

(1) Image data fetch

- Captures an image output from an external module and writes YCbCr data to the memory with it separated into Y data and CbCr data.
- Fetches image data other than YCbCr data, for example, JPEG data, RGB565, from an externally connected module, such as a camera, and sequentially writes the image data to the memory.
- Fetches an interlace source image in both-field units or one-field units and writes it to the memory. In both-field capture, an image can be stored in the memory as a frame image.

(2) Filter processing

- Performs scale-down and removal of high-frequency components (only in the horizontal direction) for an image using internal filters. Note that the scaled-down image must not exceed VGA. The filter processing can be applied to only YCbCr input data.

(3) Format conversion

- Converts image data input in the YCbCr422 format into the YCbCr420 format and writes it to the memory. Note that the conversion algorithm is simple thinning in which the chrominance component (CbCr) of the evennumbered lines is thinned.

The functional overview of the CEU is shown in [Table 61.1](#) and the main functions and their details are shown in [Table 61.2](#).

Table 61.1 Functional overview of CEU (1 of 2)

Classification	Item	Function	Description	Note
Connectable camera	Size ^{*1}	5 megapixels	2,560 pixels × 1,920 lines	Image capture mode or data synchronous fetch mode Horizontal : 8-pixel units Vertical : 4-line units Data enable fetch mode 32-byte units The range of the image size that can be input is as follows. Horizontal: 2,560 pixels to 128 pixels Vertical: 1,920 pixels lines to 96 lines
		3 megapixels	2,048 pixels × 1,536 lines	
		2 megapixels	1,632 pixels × 1,224 lines	
		UXGA	1,600 pixels × 1,200 lines	
		SXGA (1)	1,280 pixels × 1,024 lines	
		SXGA (2)	1,280 pixels × 960 lines	
		WXGA	1,280 pixels × 768 lines	
		XGA	1,024 pixels × 768 lines	
		SVGA	800 pixels × 600 lines	
		WVGA	800 pixels × 480 lines	
		VGA	640 pixels × 480 lines	
		CIF	352 pixels × 288 lines	
		WQVGA	480 pixels × 240 lines	
		QVGA	320 pixels × 240 lines, 240 pixels × 320 lines	
		QCIF	176 pixels × 144 lines	
		QQVGA	160 pixels × 120 lines	
		Sub-QCIF	128 pixels × 96 lines	
	Input format	YCbCr422 8 bits	Cb ₀ , Y ₀ , Cr ₀ , Y ₁ ...	Supports clock ratio of 1:1
			Cr ₀ , Y ₀ , Cb ₀ , Y ₁ ...	
			Y ₀ , Cb ₀ , Y ₁ , Cr ₀ ...	
			Y ₀ , Cr ₀ , Y ₁ , Cb ₀ ...	
		YCbCr422 16 bits	{Y ₀ , Cb ₀ }, {Y ₁ , Cr ₀ }, ...	
			{Y ₀ , Cr ₀ }, {Y ₁ , Cb ₀ }, ...	
		Binary data	Specified amount to be fetched on edges of the sync signal	Written sequentially
			Data is fetched with the horizontal sync signal as an enable signal	
	Horizontal and vertical sync signal polarities	Arbitrary	High-active and low-active	
	Capture start location	Arbitrary	Can be specified in camera input clock units	Horizontal: 1-cycle units Vertical: 1-HD (horizontal sync signal) units
	Number of captured pixels	Arbitrary	Can be specified in 8-pixel units horizontally and in 4-line units vertically	
	Interlace	Both-field capture	Stored as a field image	Capture: 2-VD (vertical sync signal) units
			Stored as a frame image	
		One-field capture	Top field or bottom field can be specified	Capture: 1-VD units
Memory write	Output format	YCbCr422 YCbCr420	YCbCr420 is realized by simple thinning	

Table 61.1 Functional overview of CEU (2 of 2)

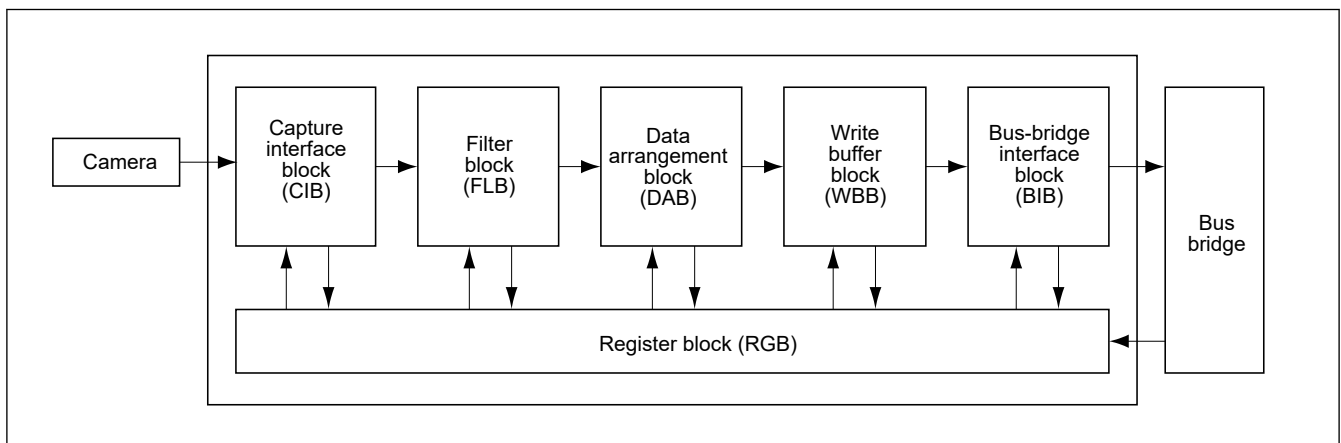
Classification	Item	Function	Description	Note
Filter function	No scaling or scale-down	Scale-down of captured display	Desired scaling factor from 1/16 to 1 (scaled-down display must not exceed VGA)	
	Low-pass filter		Removal of high-frequency components	Only in the horizontal direction
Module-stop function	Module-stop state can be set to reduce power consumption.			
TrustZone Filter	Security and Privilege attribution can be set.			

Note 1. The range of the image size that can be input depends on the AC characteristics and the frame rate of the connected device, and the transfer speed to the destination RAM.

Table 61.2 Main functions of CEU and their details

Main function	Detailed description
Image data fetch	- Captures an image output from an external module and writes YCbCr data to the memory with it separated into Y data and CbCr data. - Fetches image data other than YCbCr data, e.g. JPEG data, from an externally connected module, such as a camera, and sequentially writes the image data to the memory. - Fetches an interlace source image in both-field units or one-field units and writes it to the memory. In both-field capture, an image can be stored in the memory as a frame image.
Filter processing	Performs scale-down and removal of high-frequency components (only in the horizontal direction) for an image using internal filters. Note that the scaled-down image must not exceed VGA. The filter processing can be applied to only YCbCr input data.
Format conversion	Converts image data input in the YCbCr422 format into the YCbCr420 format and writes it to the memory. Note that the conversion algorithm is simple thinning in which the chrominance component (CbCr) of the even-numbered lines is thinned.

Figure 61.1 shows a block diagram of the CEU.

**Figure 61.1 Block diagram of CEU**

The pin configuration of the CEU is shown in Table 61.3.

Table 61.3 Pin configuration of CEU (1 of 2)

Pin name	Function	I/O	Description
VIO_D15 to VIO_D0 ^{*1}	CEU data bus	Input	Camera image data input to the CEU
VIO_CLK	CEU clock	Input	Camera clock input to the CEU
VIO_VD	CEU vertical sync	Input	Camera vertical sync signal input to the CEU

Table 61.3 Pin configuration of CEU (2 of 2)

Pin name	Function	I/O	Description
VIO_HD	CEU horizontal sync	Input	Camera horizontal sync signal input to the CEU
VIO_FLD*2	Field signal	Input	Field identification signal to the CEU

Note 1. When the distinction according to the bus width for the data bus is not needed, VIO_D is used in this manual. Otherwise, VIO_D15 to VIO_D0 are used.

Note 2. The VIO_FLD pin is valid only in interlace input mode (CAIFR.IFS=1).

61.2 Register Descriptions

The register configuration of the CEU is shown in [Table 61.4](#).

Most CEU registers have a 2-plane configuration (plane A and plane B). The CEU switches the planes when using these 2-plane registers. A mirror address, which is an address that can always access the register on the unused plane, is provided for each 2-plane register. [Figure 61.2](#) shows the timing to switch the register planes. The CEU switches the register planes at the same time a VD interrupt is asserted.

In the following register descriptions, "during operation" indicates the period that begins when the CEU is activated by the CE bit in the capture start register (CAPSR) and ends when a capture end interrupt (CPE) of the capture event flag clear register (CETCR) occurs. In the read-only bits in each register, the write value should always be 0. If a value other than 0 is written to any of these bits, correct operation cannot be guaranteed.

Table 61.4 Register configuration of CEU (1 of 2)

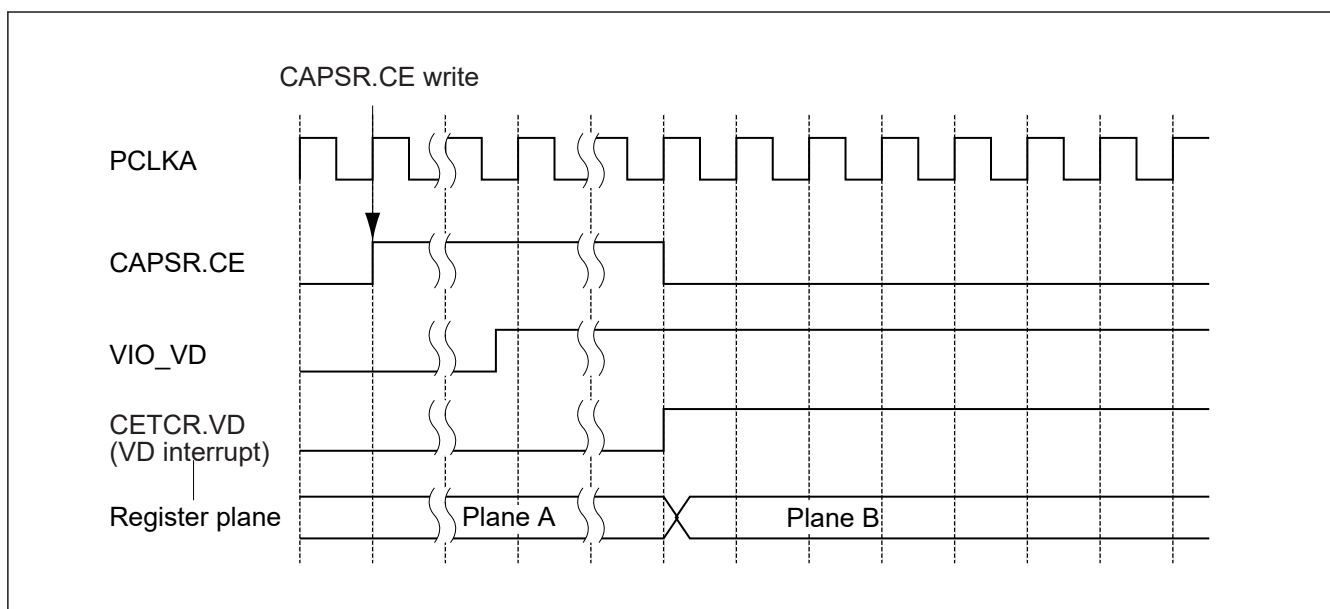
Register name	Abbr.	R/W	Addresses			Access size
			Address (Plane A)	Address (Plane B)	Mirror Address	
Capture Start Register	CAPSR	R/W	0x4034_8000	—	—	32
Capture Control Register	CAPCR	R/W	0x4034_8004	—	—	32
Capture Interface Control Register*1	CAMCR	R/W	0x4034_8008	—	—	32
Capture Interface Cycle Register*1	CMCYR	R/W	0x4034_800C	—	—	32
Capture Interface Offset Register	CAMOR	R/W	0x4034_8010	0x4034_9010	0x4034_A010	32
Capture Interface Width Register	CAPWR	R/W	0x4034_8014	0x4034_9014	0x4034_A014	32
Capture Interface Input Format Register	CAIFR	R/W	0x4034_8018	—	—	32
CEU Register Control Register	CRCNTR	R/W	0x4034_8028	—	—	32
CEU Register Forcible Control Register	CRCMPR	R/W	0x4034_802C	—	—	32
Capture Filter Control Register	CFLCR	R/W	0x4034_8030	0x4034_9030	0x4034_A030	32
Capture Filter Size Clip Register	CFSZR	R/W	0x4034_8034	0x4034_9034	0x4034_A034	32
Capture Destination Width Register	CDWDR	R/W	0x4034_8038	0x4034_9038	0x4034_A038	32
Capture Data Address Y Register	CDAYR	R/W	0x4034_803C	0x4034_903C	0x4034_A03C	32
Capture Data Address C Register	CDACR	R/W	0x4034_8040	0x4034_9040	0x4034_A040	32
Capture Data Bottom-Field Address Y Register	CDBYR	R/W	0x4034_8044	0x4034_9044	0x4034_A044	32
Capture Data Bottom-Field Address C Register	CDBCR	R/W	0x4034_8048	0x4034_9048	0x4034_A048	32
Capture Bundle Destination Size Register	CBDSR	R/W	0x4034_804C	0x4034_904C	0x4034_A04C	32
Firewall Operation Control Register	CFWCR	R/W	0x4034_805C	—	—	32
Capture Low-Pass Filter Control Register	CLFCR	R/W	0x4034_8060	0x4034_9060	0x4034_A060	32
Capture Data Output Control Register	CDOCR	R/W	0x4034_8064	0x4034_9064	0x4034_A064	32
Capture Event Interrupt Enable Register	CEIER	R/W	0x4034_8070	—	—	32
Capture Event Flag Clear Register	CETCR	R/W	0x4034_8074	—	—	32

Table 61.4 Register configuration of CEU (2 of 2)

Register name	Abbr.	R/W	Addresses			Access size
			Address (Plane A)	Address (Plane B)	Mirror Address	
Capture Status Register	CSTSR	R	0x4034_807C	—	—	32
Capture Data Size Register	CDSSR	R/W	0x4034_8084	—	—	32
Capture Data Address Y Register 2	CDAYR2	R/W	0x4034_8090	0x4034_9090	0x4034_A090	32
Capture Data Address C Register 2	CDACR2	R/W	0x4034_8094	0x4034_9094	0x4034_A094	32
Capture Data Bottom-Field Address Y Register 2	CDBYR2	R/W	0x4034_8098	0x4034_9098	0x4034_A098	32
Capture Data Bottom-Field Address C Register 2	CDBCR2	R/W	0x4034_809C	0x4034_909C	0x4034_A09C	32
CEU Bufferable Write Enable Register	CBWER	R/W	0x4034_80A0	—	—	32

Note: The address shown in this table are Secure region. The address corresponding to the Non-secure region is when b28 of address is set to 1.

Note 1. After changing the setting of a register (CAMCR or CMCYR) that is determined by the external module characteristics, do not start capture until at least 10 external input clock cycles have elapsed.

**Figure 61.2 Register plane switching timing (VD polarity is high-active in data enable fetch mode)**

61.2.1 CAPSR : Capture Start Register

Base address: CEU = 0x4034_8000
CEU_NS = 0x5034_8000

Offset address: 0x0000

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	CPKIL
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	CE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	CE	Capture enable - Single-capture This bit reserves capturing of the next frame. When 1 is written to this bit, the capture of one frame starts from the next VD input, and stops when the one-frame capture end interrupt (CPE bit in CETCR) is asserted (Figure 61.7). To perform capture again, write 1 to this bit. After the VD or HD polarity is changed, do not write 1 to this bit until the next VD interrupt is asserted. - Continuous capture When this bit is set to 1 while the CTNCP bit in CAPCR is set to 1, continuous capture starts from the next frame (Figure 61.8). Note that this bit is not cleared to 0 but remains as 1. To stop capturing, clear this bit to 0; capturing stops after the current frame is completed. Continuous capture operations are possible in only image capture mode. The start address of the memory to which the captured data is written to must be set for each frame. 0: Stops capturing 1: Starts capturing	R/W
15:1	—	These bits are read as 0. The write value should be 0.	R/W
16	CPKIL	Software reset of capturing Write 1 to this bit to perform a software reset of capturing. At a software reset, capturing ends immediately without completing capture operation until the end of a frame. Clear the CE bit to 0 when writing 1 to this bit. Processing of the capture software reset is indicated by this bit being set to 1. When CPKIL = 1, do not start capturing since reset processing is in progress. When restarting capture operations, after referring to the CPTON bit in CSTSR to ensure that the CEU is halted (in the idle state), wait until this bit is cleared to 0. The timing of restarting capture operations is shown in Figure 61.6. When a software reset is generated by this bit, a capture end interrupt (CPE bit in CETCR) may be output immediately after the software reset. However, such kind of interrupt should be ignored. Also, even if the capture end interrupt is not output, the interrupt source (CPE bit) must be cleared before capturing of the next frame. 0: Normal state 1: Software reset of capturing	R/W
31:17	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

CAPSR captures data input to the CEU from an external module.

When both the VD (vertical sync signal) and HD (horizontal sync signal) polarities are high-active, one frame is defined as a period from a VD rising edge to the next VD rising edge, and one line as a period from an HD rising edge to the next HD rising edge. Figure 61.3 shows the timing of one frame (when both the VD and HD polarities are high-active).

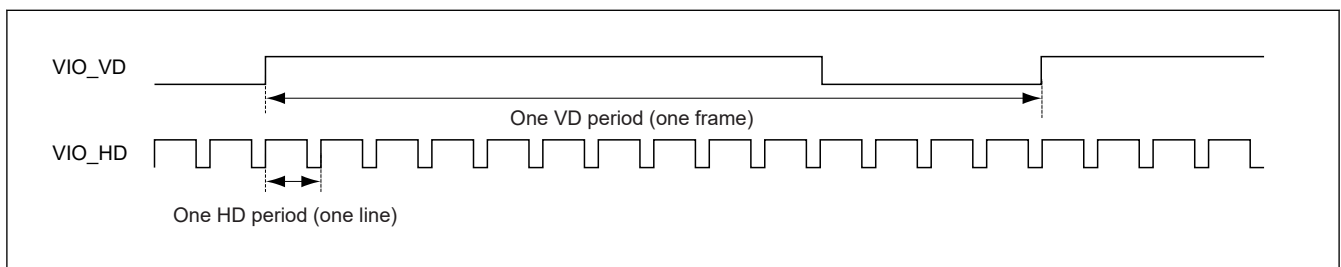


Figure 61.3 Frame timing

When both the VD and HD polarities are high-active, similar to one frame, one field is defined as follows:

- Period from a VD rising edge to the next VD rising edge
- One line is a period from an HD rising edge to the next HD rising edge

The field identification signal FLD should be fixed for at least 1-HD period from a VD input. [Figure 61.4](#) shows the timing of one field (when both the VD and HD polarities are high-active).

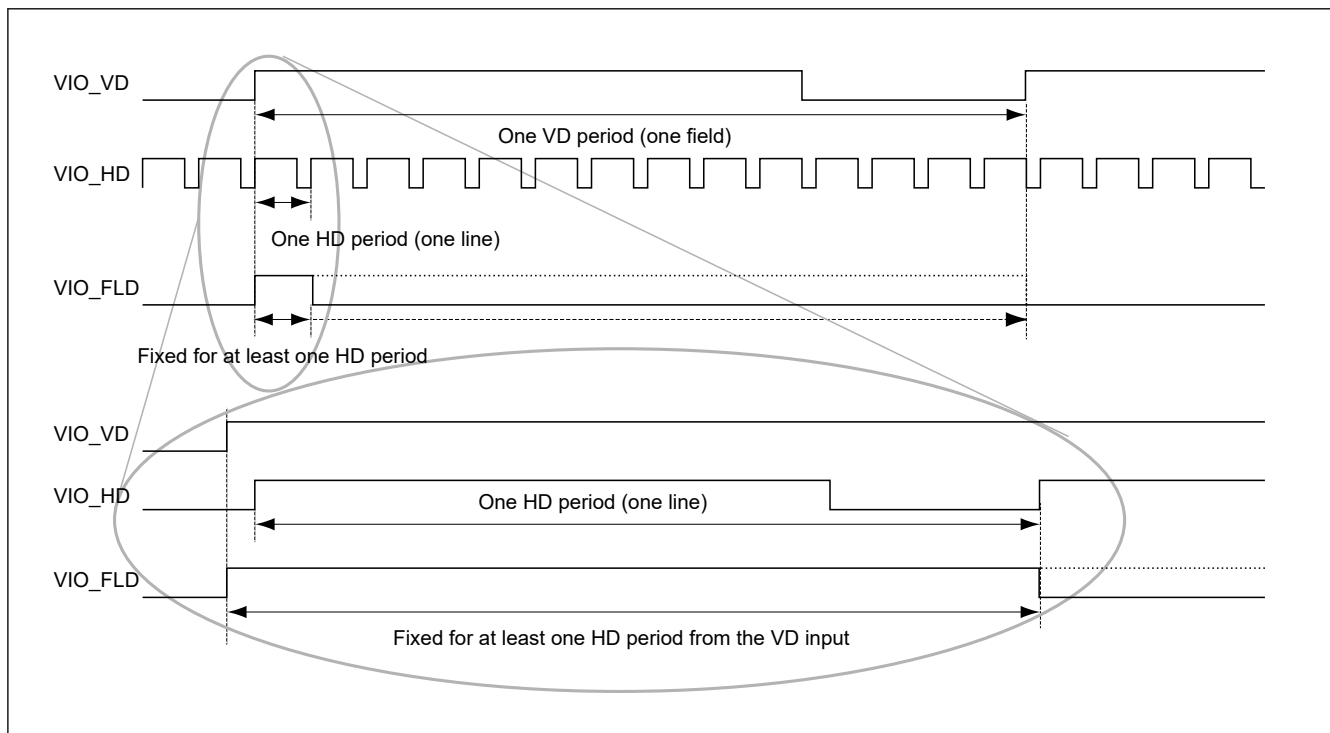


Figure 61.4 One field timing

In data enable fetch, one frame is defined as a period from a VD rising edge to the VD falling edge. With the HD as an enable signal (positive polarity), data of a cycle in which the HD is asserted is fetched while the VD is high. [Figure 61.5](#) shows the timing of one frame for data enable fetch.

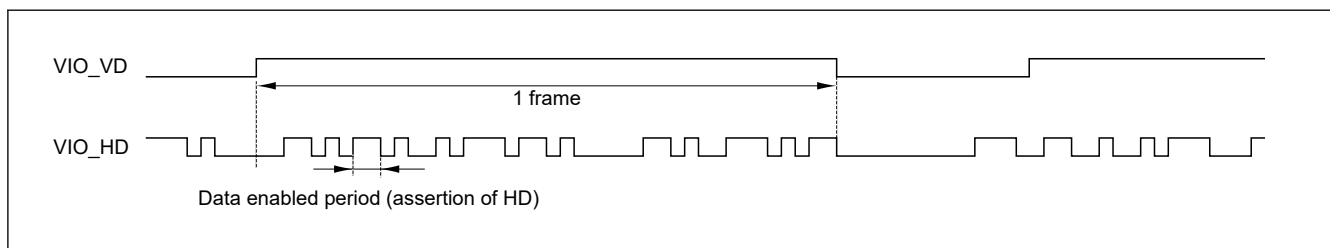


Figure 61.5 Frame timing (data enable fetch)

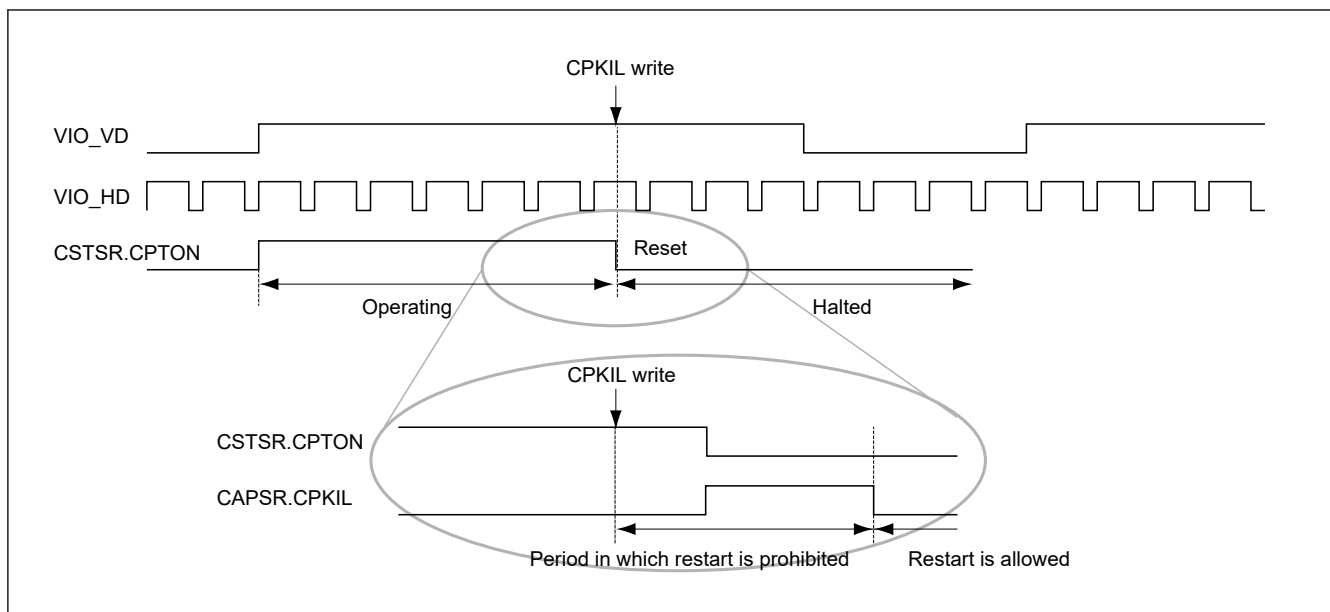


Figure 61.6 Timing of software reset and restart of capturing

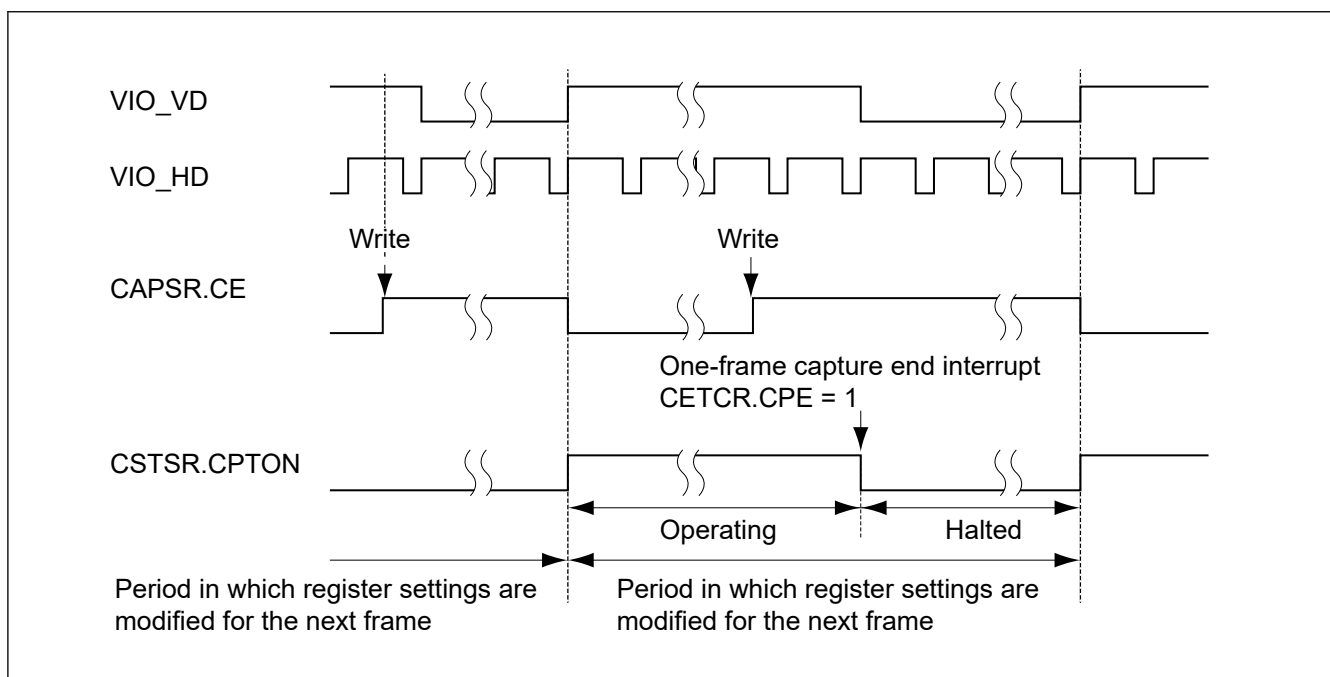


Figure 61.7 Timing of modifying CE bit and register setting in one frame capture

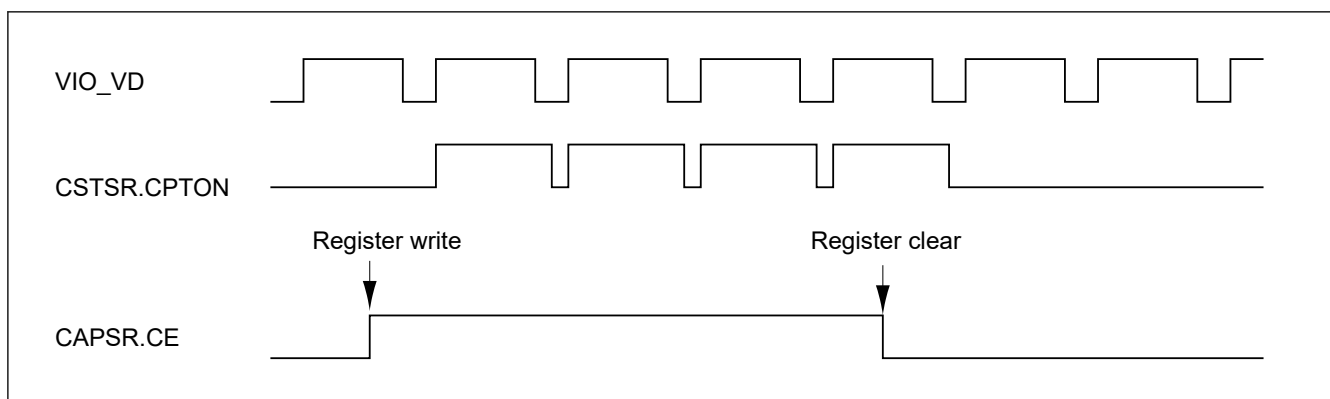


Figure 61.8 Continuous-frame capture

61.2.2 CAPCR : Capture Control Register

Base address: CEU = 0x4034_8000
CEU_NS = 0x5034_8000

Offset address: 0x0004

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	FDRP[7:0]								—	—	MTCM[1:0]	—	—	—	—	CTNCP
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	—	These bits are read as 0. The write value should be 0.	R/W
16	CTNCP	Continuous capture When capturing is started with this bit set to 1, capturing continues until the CE bit in CAPSR is cleared to 0 or a software reset is initiated by the CPKIL bit in CAPSR (see Figure 61.8). Continuous capture must be set before capturing is started. This bit is modified only after 0 is written to the CE bit to stop capturing. If this bit is modified during capturing, correct operation cannot be guaranteed. In data fetch mode, clear this bit to 0. 0: One-frame capture when CAPSR.CE = 1 1: Continuous capture until CAPSR.CE = 0	R/W
19:17	—	These bits are read as 0. The write value should be 0.	R/W
21:20	MTCM[1:0]	Specify the unit for transferring data to a bus bridge module. The efficiency of writing image data can be improved by continuously accessing the addresses. To improve the write efficiency, set these bits to 11. The setting of these bits appear to be unchanged from the outside. 0 0: Image capture: Y data and C data are transferred in 32-byte units Data fetch : Data is transferred in 32-byte units 0 1: Image capture: Y data and C data are transferred in 64-byte units Data fetch : Data is transferred in 64-byte 1 0: Image capture: Y data and C data are transferred in 128-byte units Data fetch : Data is transferred in 128-byte 1 1: Image capture: Y data and C data are transferred in 256-byte units Data fetch : Data is transferred in 256-byte	R/W
23:22	—	These bits are read as 0. The write value should be 0.	R/W
31:24	FDRP[7:0]	Set the frame drop interval in continuous-frame capture. When these bits are cleared to 0, frame drop is not performed, and all frames are captured. Figure 61.9 shows the value set in these bits and the timing of captured frames. The frame drop interval unit differs according to the capture setting. Table 61.5 shows the relationship between the capture setting and frame drop interval unit. The image of the frame drop timing for each capture setting when these bits are set to 2 is shown in Figure 61.10. In both-field capture, capturing is performed continuously for 2-VD periods, regardless of whether the second field is the top field or bottom field. In addition, in both-field capture, the frame drop counter is incremented when the first field has been identified as the top field or bottom field, regardless of whether the second field is the top field or bottom field. When 0 is written to the CE bit in CAPSR, capturing terminates after the current frame has been captured for a capture frame. However, for a drop frame, capturing is forcibly terminated in the CEU so no capture end interrupt (CPE bit in CETCR) is output. While CAPSR.CE = 1, do not change the setting of these bits. Note: Do not change the setting of these bits during continuous capture operations. To change the setting of these bits, stop continuous capture (CAPSR.CE = 0), clear the CTNCP bit (continuous capture) in CAPCR to 0, and then restart continuous capture. Continuous capture is performed during the period of CAPSR.CE = 1 shown in Figure 61.9.	R/W

Note: S-TYPE-3, P-TYPE-3

CAPCR sets continuous-frame capture and the frame drop intervals.

Do not modify this register during operation. If this register is modified during operation, correct operation cannot be guaranteed. In addition, the IGRW bit (interrupt source) in CETCR is set to 1.

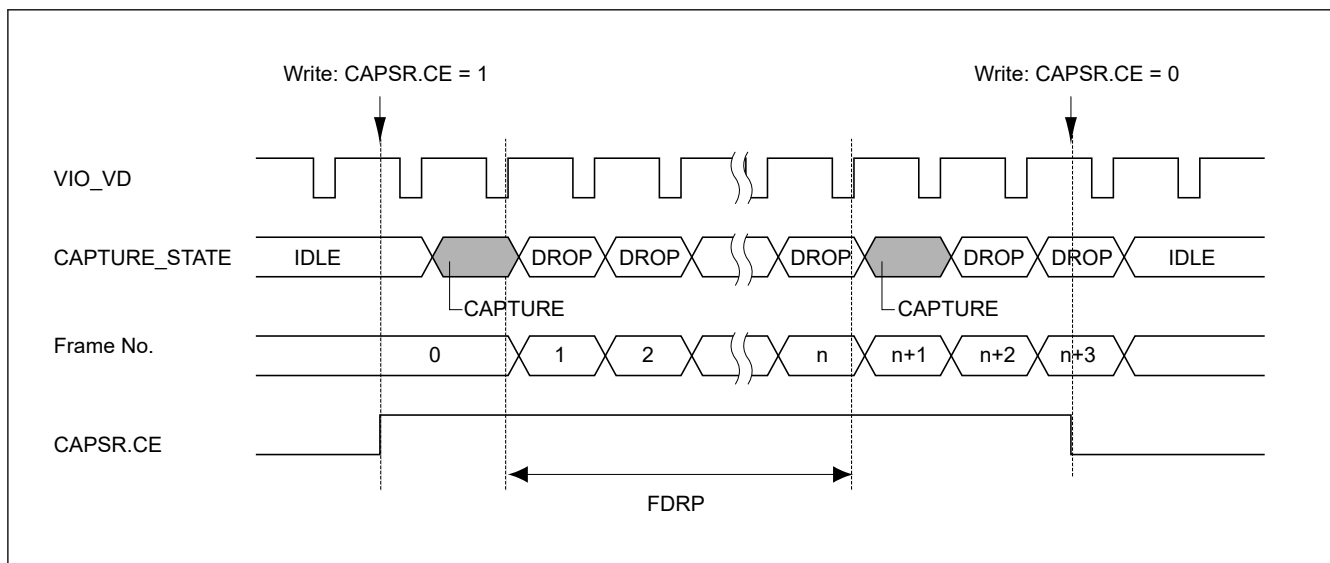


Figure 61.9 Setting of FDRP bits and frame drop timing

Table 61.5 Relationship between capture setting and frame drop interval unit

Input Mode	Captured Image	First Captured Image	Frame Drop Interval Unit	Capture Setting
Progressive	Frame	Frame immediately after capture start	Frame	A
Interlace	Both-field (2-VD capture)	Field immediately after capture start	2 fields (first capture field count)	B
				C
		Top field	2 fields (top-field count)	D
		Bottom field	2 fields (bottom-field count)	E
	One-field (1-VD capture)	Field immediately after capture start	First capture field	F
				G
		Top field	Top field	H
		Bottom field	Bottom field	I

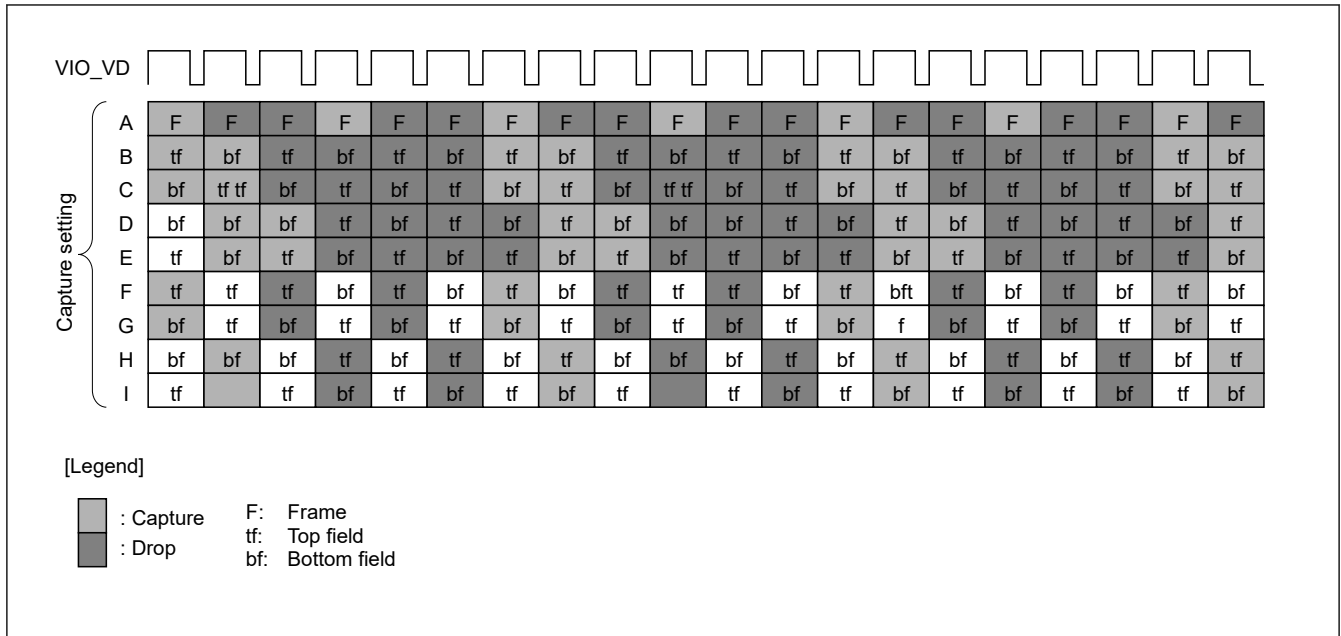


Figure 61.10 Image of frame drop timing for capture settings (FDRP[7:0] = 0x2)

61.2.3 CAMCR : Capture Interface Control Register

Base address: CEU = 0x4034_8000
CEU_NS = 0x5034_8000

Offset address: 0x0008

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	VDSE L	HDSE L	FLDS EL	DSEL	—	—	—	—	—	—	—	FLDP OL
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	DTIF	—	—	DTARY[1:0]	—	—	—	JPG[1:0]	—	—	—	VDPO L	HDPO L
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	HDPOL	Sets the polarity for detection of the horizontal sync signal (HD) input from an external module. Figure 61.16 shows the relationship between the HD and HD interrupt when high-active is selected. Since an HD interrupt may occur when this bit value is modified, the HD bit in CETCR must always be cleared to 0 when this bit value is changed. In data enable fetch mode, set this bit to 0. 0: Horizontal sync signal (HD) from an external module is detected as high active 1: Horizontal sync signal (HD) from an external module is detected as low active	R/W
1	VDPOL	Sets the polarity for detection of the vertical sync signal (VD) input from an external module. Figure 61.14 and Figure 61.15 show the relationship between the VD, HD and VD interrupt when high-active is selected. Since a VD interrupt may occur when this bit value is modified, the VD bit in CETCR must always be cleared to 0 when this bit value is changed. In data enable fetch mode, this bit is not used and the sense for detection is always active high. 0: Vertical sync signal (VD) from an external module is detected as high active 1: Vertical sync signal (VD) from an external module is detected as low active	R/W
3:2	—	These bits are read as 0. The write value should be 0.	R/W

Bit	Symbol	Function	R/W
5:4	JPG[1:0]	These bits select the fetched data type. 0 0: Image capture mode (input data are separated into the luminance component data (Y) and the chrominance component data (CbCr) for output to the memory) 0 1: Data synchronous fetch mode (specified size of input data are output to the specified memory addresses in order of input and in synchronization with the sync signal) 1 0: Data enable fetch mode (input data are fetched with HD as an enable signal and output to the specified addresses in memory in order of input) 1 1: Setting prohibited	R/W
7:6	—	These bits are read as 0. The write value should be 0.	R/W
9:8	DTARY[1:0]	Set the input order of the luminance component and chrominance component. The order in which Y and CbCr are input from an external module differs among modules. The CEU supports the input orders shown in Figure 61.12 . Set the corresponding value in these bits. In data fetch mode, set these bits to 00. 0 0: 8-bit interface: Image input data is fetched in the order of Cb ₀ , Y ₀ , Cr ₀ , and Y ₁ 16-bit interface: Image input data is fetched in the order of {Cb ₀ , Y ₀ } and {Cr ₀ , Y ₁ } 0 1: 8-bit interface: Image input data is fetched in the order of Cr ₀ , Y ₀ , Cb ₀ , and Y ₁ 16-bit interface: Image input data is fetched in the order of {Cr ₀ , Y ₀ } and {Cb ₀ , Y ₁ } 1 0: 8-bit interface: Image input data is fetched in the order of Y ₀ , Cb ₀ , Y ₁ , and Cr ₀ 16-bit interface: Image input data is fetched in the order of {Y ₀ , Cb ₀ } and {Y ₁ , Cr ₀ } 1 1: 8-bit interface: Image input data is fetched in the order of Y ₀ , Cr ₀ , Y ₁ , and Cb ₀ 16-bit interface: Image input data is fetched in the order of {Y ₀ , Cr ₀ } and {Y ₁ , Cb ₀ }	R/W
11:10	—	These bits are read as 0. The write value should be 0.	R/W
12	DTIF	Sets the digital image input pins from which data is to be captured. 0: Data input to 8-bit digital image input pins is captured 1: Data input to 16-bit digital image input pins is captured	R/W
15:13	—	These bits are read as 0. The write value should be 0.	R/W
16	FLDPOL	Sets the polarity of the field identification signal (FLD) from an external module. 0: When the FLD signal is high-active, the field is detected as the top field and when low-active, the field is detected as the bottom field. 1: When the FLD signal is low-active, the field is detected as the top field and when high-active, the field is detected as the bottom field.	R/W
23:17	—	These bits are read as 0. The write value should be 0.	R/W
24	DSEL	Sets the edge for fetching the image data (D15 to D0) from an external module. 0: D15 to D0 are fetched at the rising edge of the camera clock. 1: D15 to D0 are fetched at the falling edge of the camera clock.	R/W
25	FLDSEL	Sets the edge for capturing FLD from an external module. 0: FLD is captured at the rising edge of the camera clock. 1: FLD is captured at the falling edge of the camera clock.	R/W
26	HDSEL	Sets the edge for capturing HD from an external module. 0: HD is captured at the rising edge of the camera clock. 1: HD is captured at the falling edge of the camera clock.	R/W
27	VDSEL	Sets the edge for capturing VD from an external module. 0: VD is captured at the rising edge of the camera clock. 1: VD is captured at the falling edge of the camera clock.	R/W
31:28	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

CAMCR sets the capture interface.

The following items are set by CAMCR.

- Selection between the rising edge or falling edge of the camera clock for signal capture or data fetch operation
- Selection between image capture operation or data fetch operation
- Polarities of the vertical and horizontal sync signals
- Input order of image data components (Y, Cb, and Cr) (only for image capture mode)
- Selection of digital image input pins (8 bits or 16 bits)

- Polarity of the field identification signal

CAMCR must be set according to the module connected. In data fetch mode, set the DTARY bits to 0. Do not modify this register during operation. If this register is modified during operation, correct operation cannot be guaranteed. In addition, the IGRW bit (interrupt source) in CETCR is set to 1.

Note: After changing the setting of this register, do not start capture until at least 10 external input clock cycles have elapsed.

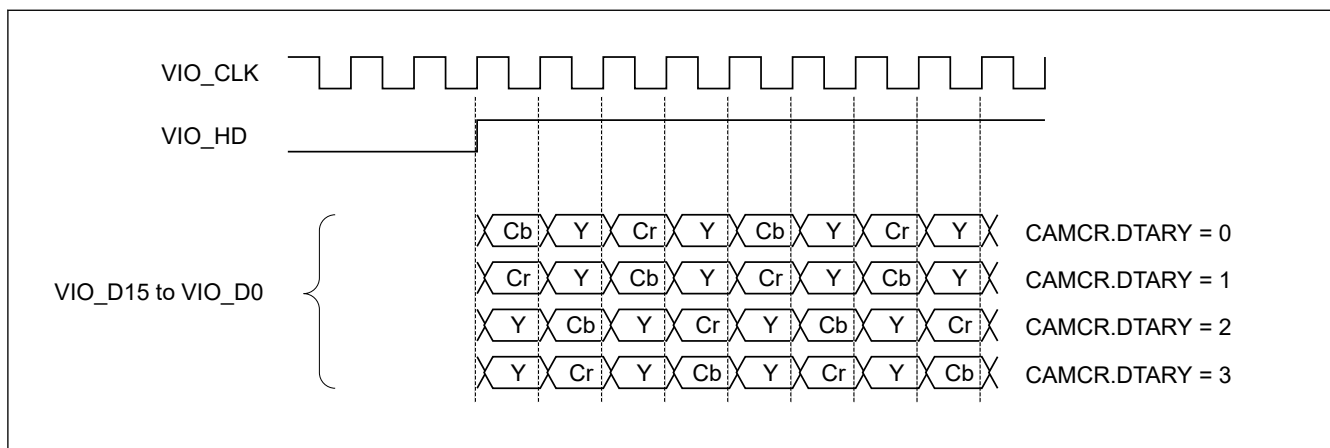


Figure 61.11 Input order of image data

The JPG bit in CAMCR selects whether digital image data is fetched or data such as JPEG is fetched. In addition, when data such as JPEG is fetched, select whether the specified amount of data is continuously fetched in synchronization with the sync signal or data is fetched while the horizontal sync signal is enabled.

In data enable fetch mode, one frame is defined as a period from the rising edge to the falling edge of the vertical sync signal (VD) for data fetching. The horizontal sync signal (HD) is enabled only when the VD is high and treated as an enable signal. Data input in the cycle in which the HD is asserted (high) is fetched and output to the memory continuously.

This module starts fetching data at the rising edge of the VD and stops fetching data at the falling edge of the VD in data enable fetch mode. Thus, if the VD remains high and does not go low, end processing does not start. In addition, if the VD remains high and the HD also remains asserted, data continues to be fetched.

Figure 61.12 and Figure 61.13 show the interface timing in data enable fetch mode.

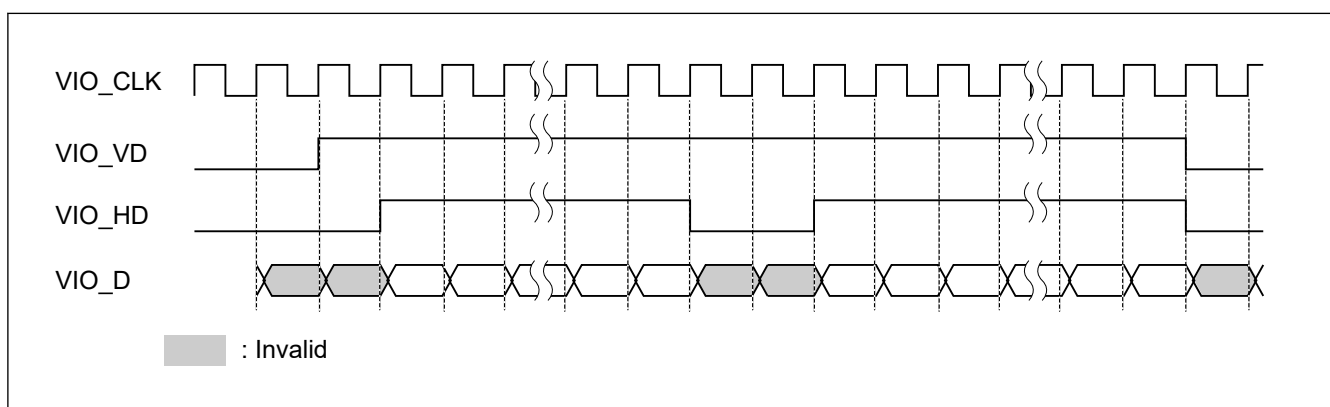


Figure 61.12 Data enable fetch timing (HD asserted (high) while VD is high)

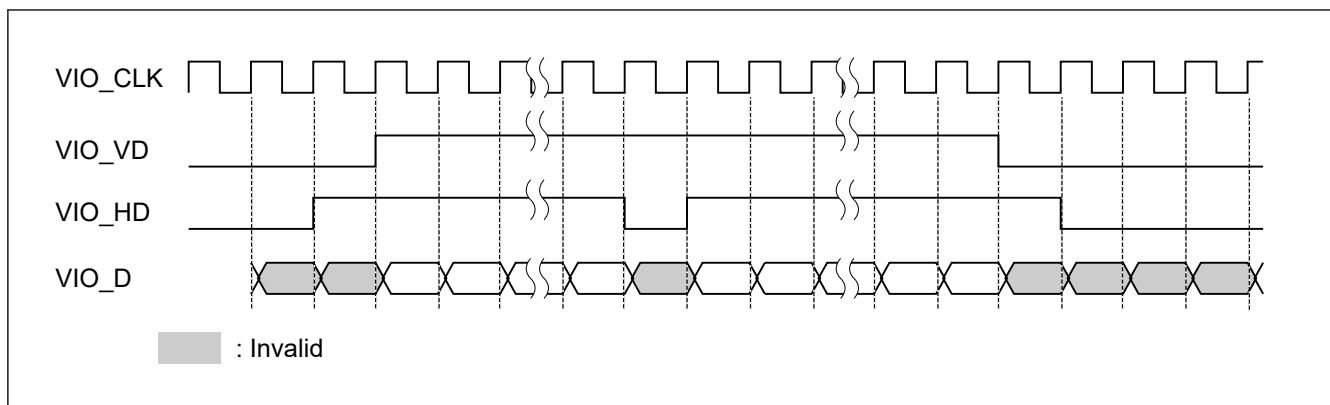


Figure 61.13 Data enable fetch timing (HD asserted (high) when VD is not high)

In data enable fetch mode, this module generates a VD interrupt in response to detection of the active level of VD. In image capture mode and data synchronous fetch mode, this module generates a VD interrupt in response to the first detection of the active level of HD following detection of the active level of VD. Note that, when VD and HD are asserted and detected at the same time, this module generates a VD interrupt at that time.

Figure 61.14 to Figure 61.16 show the relationships between the VIO_VD signal and the VD interrupt, the VIO_VD and VIO_HD signals and the VD interrupt, and the VIO_HD signal and the HD interrupt.

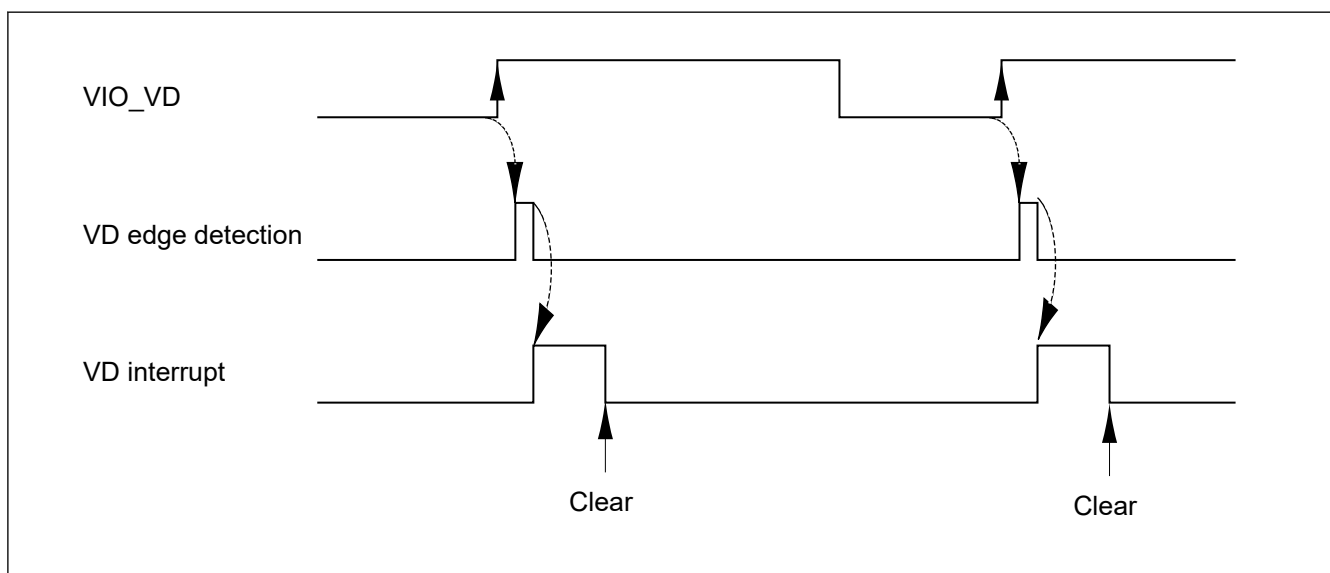


Figure 61.14 Relationship between VIO_VD and VD interrupt when VD is high-active (in data enable fetch mode)

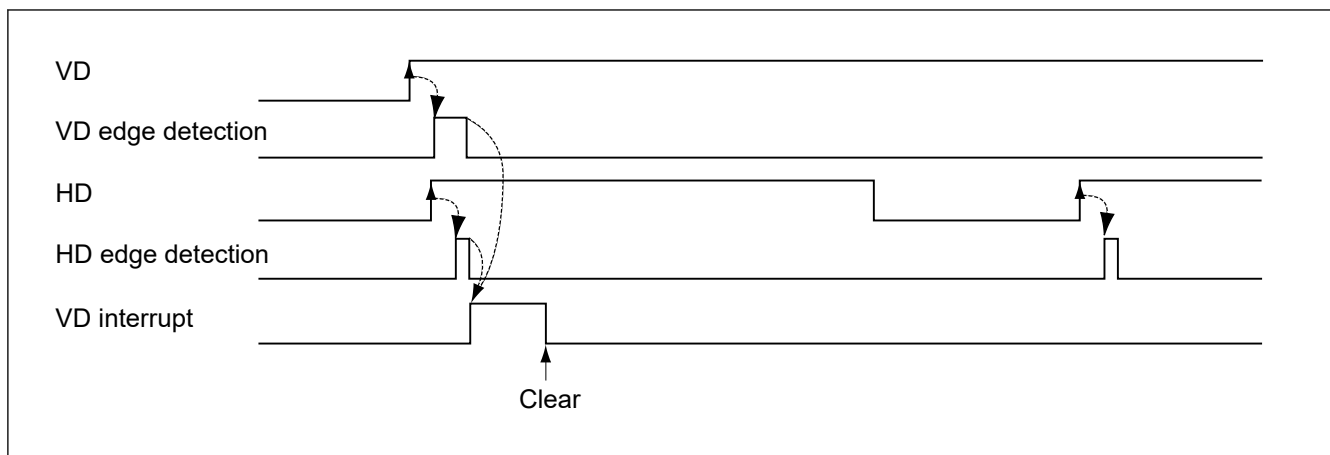


Figure 61.15 Relationship between the VIO_VD and VIO_HD signals and the VD Interrupt when VD and HD are high-active (in Image capture mode or data synchronous fetch mode)

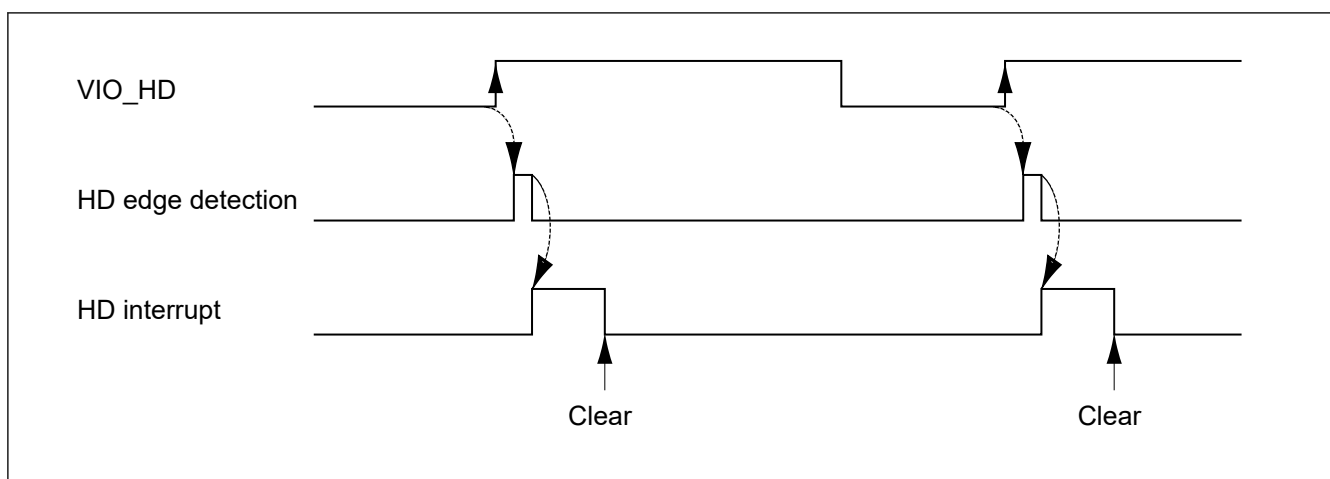


Figure 61.16 Relationship between VIO_HD and HD interrupt when HD is high-active

61.2.4 CMCYR : Capture Interface Cycle Register

Base address: CEU = 0x4034_8000
CEU_NS = 0x5034_8000

Offset address: 0x000C

Bit position:	31	29		16	13		0
Bit field:	—	—	VCYL[13:0]	—	—	HCYL[13:0]	
Value after reset:	0	0	0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0	0	0	0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0	0

Bit	Symbol	Function	R/W
13:0	HCYL[13:0]	Horizontal Cycle Count of External Module These bits set the number of HD cycles of an external module. The interrupt source bit IGHS in CETCR is set to 1 when the actual number of HD cycles input from the external module differs from this setting. Set these bits for detecting an illegal HD. When these bits are all cleared to 0, the interrupt source bit IGHS in CETCR is not set to 1. Though the interrupt source bit IGHS in CETCR may be set to 1 after the HDPOL bit (HD polarity) in CAMCR is changed, this interrupt should be ignored.	R/W
15:14	—	These bits are read as 0. The write value should be 0.	R/W

Bit	Symbol	Function	R/W
29:16	VCYL[13:0]	Vertical HD Count of External Module These bits set the number of VD cycles of an external module with the number of HD inputs. The interrupt source bit IGVS in CETCR is set to 1 when the actual number of VD cycles input from the external module differs from this setting. Set these bits for detecting an illegal VD. When these bits are all cleared to 0, the interrupt source bit IGVS in CETCR is not set to 1. Though the interrupt source bit IGVS in CETCR may be set to 1 after the VDPOL bit (VD polarity) in CAMCR is changed, this interrupt should be ignored.	R/W
31:30	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

CMCYR is used to detect an illegal VD and an illegal HD. For HD, the number of cycles from a rising edge of HD to the next rising edge is set (falling edges when low-active is selected for HD). For VD, the number of HD inputs from a rising edge of VD to the next rising edge is set (falling edges when low-active is selected for VD).

Do not modify this register during operation. If this register is modified during operation, correct operation cannot be guaranteed. In addition, the IGRW bit (interrupt source) in CETCR is set to 1.

Set 0 in all bits of this register, during data enable fetch mode.

Note: After changing the setting of this register, do not start capture until at least 10 external input clock cycles have elapsed.

61.2.5 CAMOR, CAMOR x : Capture Interface Offset Register (x = B, M)

Base address: CEU = 0x4034_8000
CEU NS = 0x5034_8000

Offset address: 0x0010 (CAMOR)
0x1010 (CAMOR_B)
0x2010 (CAMOR_M)

Bit position: 31 27 16 12 0

Bit field:	—	—	—	—	VOFST[11:0]	—	—	—	HOFST[12:0]
------------	---	---	---	---	-------------	---	---	---	-------------

Value after reset: 0

Bit	Symbol	Function	R/W
12:0	HOFST[12:0]	Specify the capture start location in terms of the number of clock cycles from a horizontal sync signal (1-cycle units). The blanking period from a horizontal sync signal differs among external modules. Therefore, the horizontal capture start location must be specified by these bits in terms of external input clock cycles from a horizontal sync signal so that an image can be captured from the valid image area. This is similar in data synchronous fetch mode (see Figure 61.19). Some external modules output a horizontal sync signal as a data enable signal. In this case, there is no blanking period so these bits must be cleared to 0 (see Figure 61.20). Note: The first HD (horizontal sync signal) being input simultaneously or after the first VD (vertical sync signal) is the operating condition of the CEU. These inputs are affected by the polarities (set by the VDPOL and HDPOL bits in CAMCR).	R/W
15:13	—	These bits are read as 0. The write value should be 0.	R/W
27:16	VOFST[11:0]	Specify the capture start location in terms of the HD count from a vertical sync signal (1-HD units). The blanking period from a vertical sync signal differs among external modules. Therefore, the vertical capture start location must be specified by these bits in terms of the HD count from a vertical sync signal so that an image can be captured from the valid image area (see Figure 61.17). Some external modules output a vertical sync signal as a data enable signal. In this case, there is no blanking period so these bits must be cleared to 0 (see Figure 61.18).	R/W
31:28	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

CAMOR sets the location to start capturing when capturing images.

Since the number of HD (horizontal sync signal) inputs from a VD (vertical sync signal) input to the start of a valid image period, and the number of clock cycles from an HD input to the start of a valid image period differ among external modules, these must be set in CAMOR. By setting a value greater than the valid image area, part of the image can be clipped for capture. In data synchronous fetch mode, the setting of this register becomes the number of cycles (HD count) up to the start of a valid data period.

This register is not used, during data enable fetch mode.

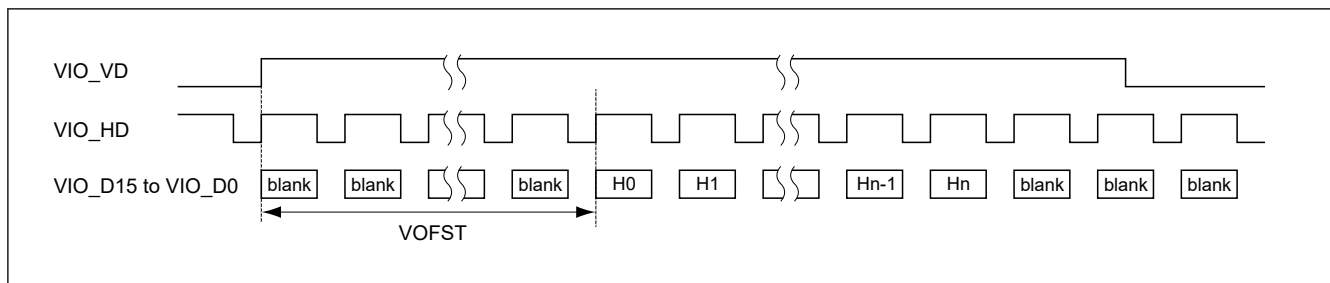


Figure 61.17 Vertical offset

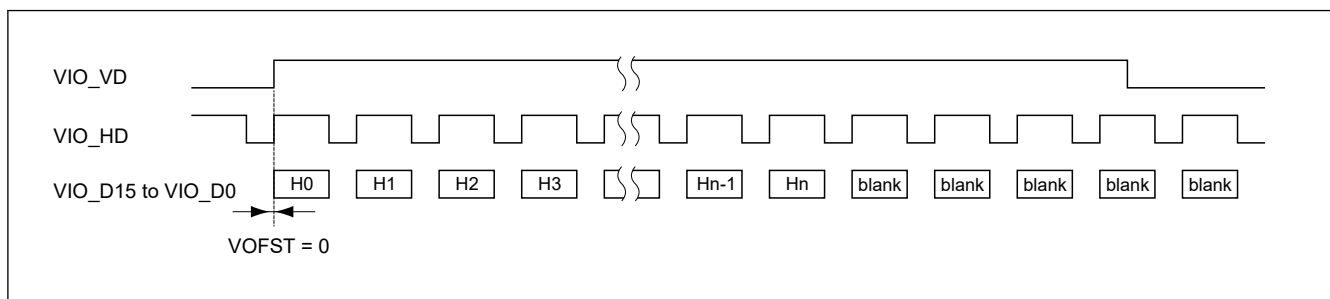


Figure 61.18 Timing when VD is data enable signal

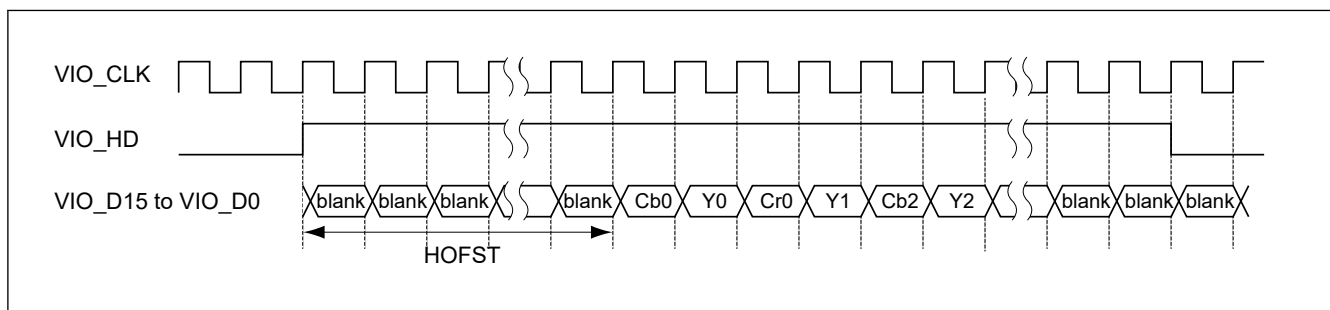


Figure 61.19 Horizontal offset

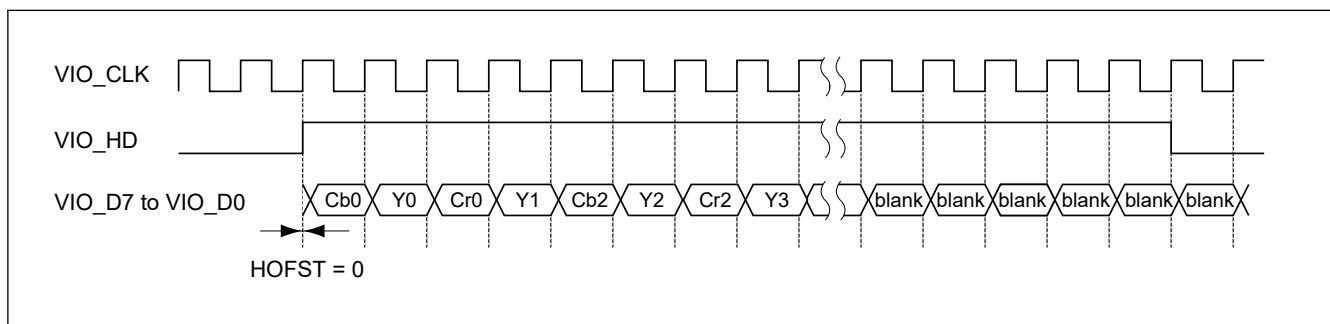
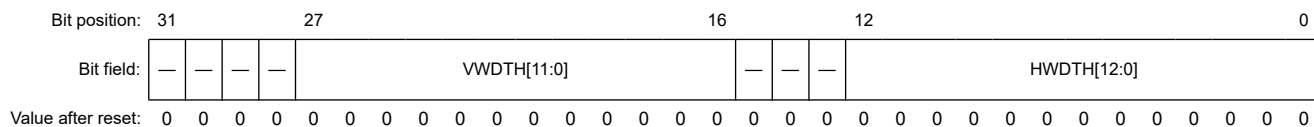


Figure 61.20 Timing when HD is data enable signal (8-bit interface)

61.2.6 CAPWR, CAPWR_x : Capture Interface Width Register (x = B, M)

Base address: CEU = 0x4034_8000
CEU_NS = 0x5034_8000

Offset address: 0x0014 (CAPWR)
0x1014 (CAPWR_B)
0x2014 (CAPWR_M)



Bit	Symbol	Function	R/W
12:0	HWDTH[12:0]	Specify the horizontal capture period. Specify the number of cycles to be captured from the location specified by the HOFST bits. Figure 61.22 shows the timing when the horizontal blanking period is 0. The CEU captures for only the number of cycles specified by these bits in the horizontal direction. Make a similar setting for data synchronous fetch. The maximum value to be set is as follows: 8-bit interface Image capture (16-cycle units) : 5,120 cycles (2,560 pixels) Data synchronous fetch (8-cycle units) : 2,560 cycles (2,560 bytes) 16-bit interface Image capture (8-cycle units) : 2,560 cycles (2,560 pixels) Data synchronous fetch (4-cycle units) : 1,280 cycles (2,560 bytes) The lower 2 bits should be written 0. Note: In data synchronous fetch mode, set CFSZR and CDWDR according to the values set in this register. For details, see the descriptions on CFSZR and CDWDR.	R/W
15:13	—	These bits are read as 0. The write value should be 0.	R/W
27:16	VWDTH[11:0]	Specify the vertical capture period (4-HD units). Specify the number of lines (HD count) to be captured from the location specified by the VOFST bits in CAMOR. Figure 61.21 shows the timing when the vertical blanking period is 0. The CEU captures only the number of lines (HD count) specified by these bits in the vertical direction. Make a setting in the same way to obtain data synchronization. The maximum value to be set is 1,920 HD (5 megapixels). The value of writing are ignored on the lower 2 bits, so write 0 to the lower 2 bits.	R/W
31:28	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

CAPWR sets the fetch (capture) cycle width when capturing images.

The cycle width unit differs according to the interface and the data type to be captured. For each setting unit, see [Table 61.6](#).

This register is not used, during data enable fetch mode.

Table 61.6 Unit for setting fetch (capture) cycle width

Interface	Vertical Direction		Horizontal Direction	
	Image Capture	Data Synchronous Fetch	Image Capture	Data Synchronous Fetch
8-bit interface	4 HD	4 HD	16 cycles	8 cycles
16-bit interface	4 HD	4 HD	8 cycles	4 cycles

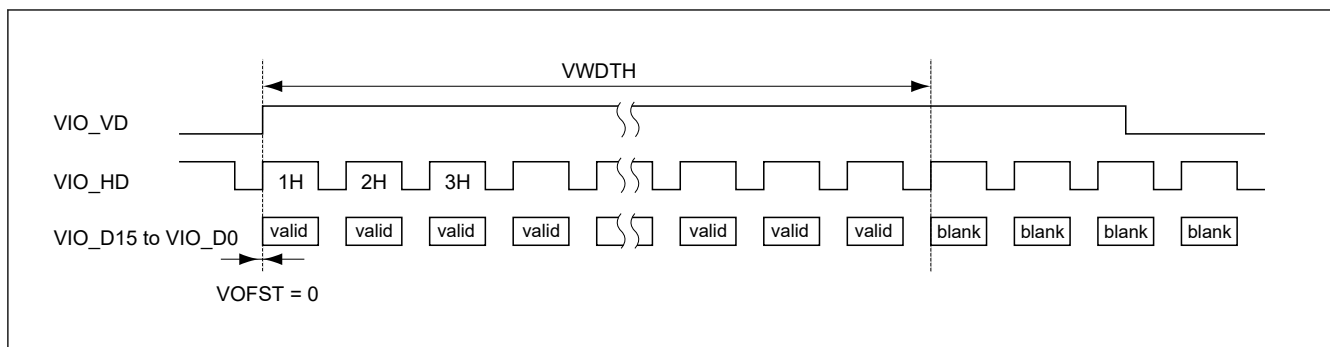


Figure 61.21 Vertical capture timing

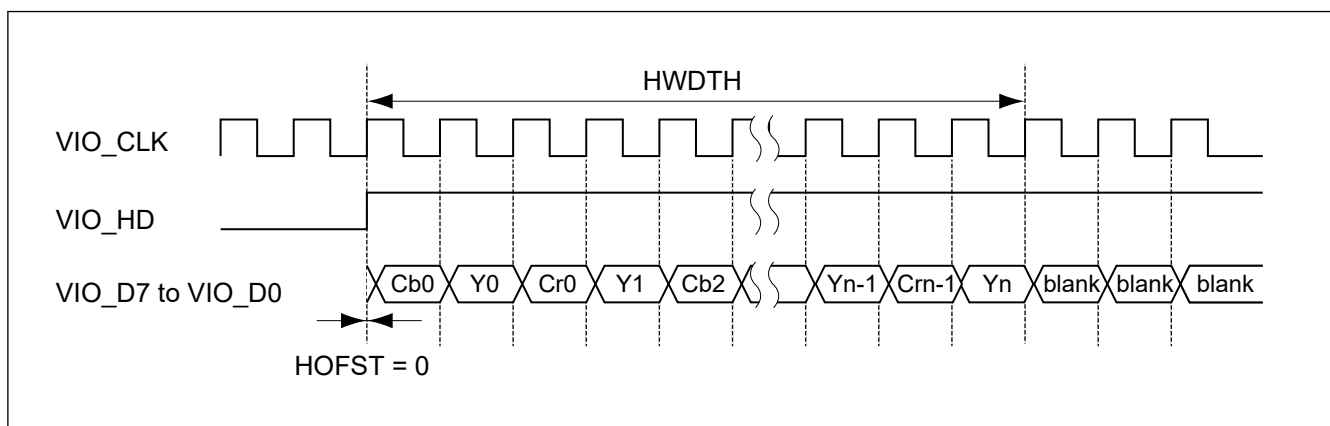


Figure 61.22 Horizontal capture timing (image capture with 8-bit interface)

61.2.7 CAIFR : Capture Interface Input Format Register

Base address: CEU = 0x4034_8000
CEU_NS = 0x5034_8000

Offset address: 0x0018

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	IFS	—	—	—	CIM	—	—	—	FCI[1:0]
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
1:0	FCI[1:0]	Set the timing to start capturing. The timing to start capturing is set by specifying the image to be captured first. Set these bits to 00 when the input mode is progressive. 0 0: Capture starts from the VD input immediately after the CEU activation regardless of it being a top or bottom field 0 1: After the CEU activation, input of a top-field image is waited, and then capture starts from the top field 1 0: After the CEU activation, input of a bottom-field image is waited, and then capture starts from the bottom field 1 1: Setting prohibited	R/W
3:2	—	These bits are read as 0. The write value should be 0.	R/W

Bit	Symbol	Function	R/W
4	CIM	Sets the images to be captured. Clear this bit to 0 when the input mode for image capture is progressive (frame image) or when the input mode for image capture is interlace for continuous capture of both the top and bottom fields. Set this bit to 1 when the input mode for image capture is interlace for capture of only a one-field image. 0: Capture of frame image (1 VD) or both-field image (2 VD) 1: Capture of one-field image (1 VD)	R/W
7:5	—	These bits are read as 0. The write value should be 0.	R/W
8	IFS	Sets the input mode for capturing images. 0: Progressive 1: Interlace	R/W
31:9	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

CAIFR sets the input mode (progressive or interlace) for capturing images, the images to be captured (frame, both-field, or one-field), the image from which capturing starts (top field or bottom field), etc. CAIFR is not used in data fetch mode.

Do not modify this register during operation. If this register is modified during operation, correct operation cannot be guaranteed. In addition, the IGRW bit (interrupt source) in CETCR is set to 1.

The items set by CAIFR are listed in [Table 61.7](#).

Table 61.7 CAIFR Setting Items

Input Mode	IFS Bit	Captured Image	CIM Bit	Image to Start Capture	FCI Bits
Progressive	0	Frame	0	Frame immediately after activation	00
Interlace	1	Both-field (2-VD capture)	0	Field immediately after activation	00
				Top field	01
				Bottom field	10
				Setting prohibited	11
		One-field (1-VD capture)	1	Field immediately after activation	00
				Top field	01
				Bottom field	10
				Setting prohibited	11

In frame image capture and one-field image capture, a one-frame capture end interrupt occurs when capture for 1 VD finishes. In both-field image capture, a one-field capture end interrupt occurs when capture for 1 VD finishes and an oneframe capture end interrupt occurs when capture for 2 VD finishes. At this time, a one-field capture end interrupt occurs simultaneously with a one-frame capture end interrupt. [Figure 61.23](#) shows the timing of a one-frame capture end interrupt and one-field capture end interrupt in both-field image capture.

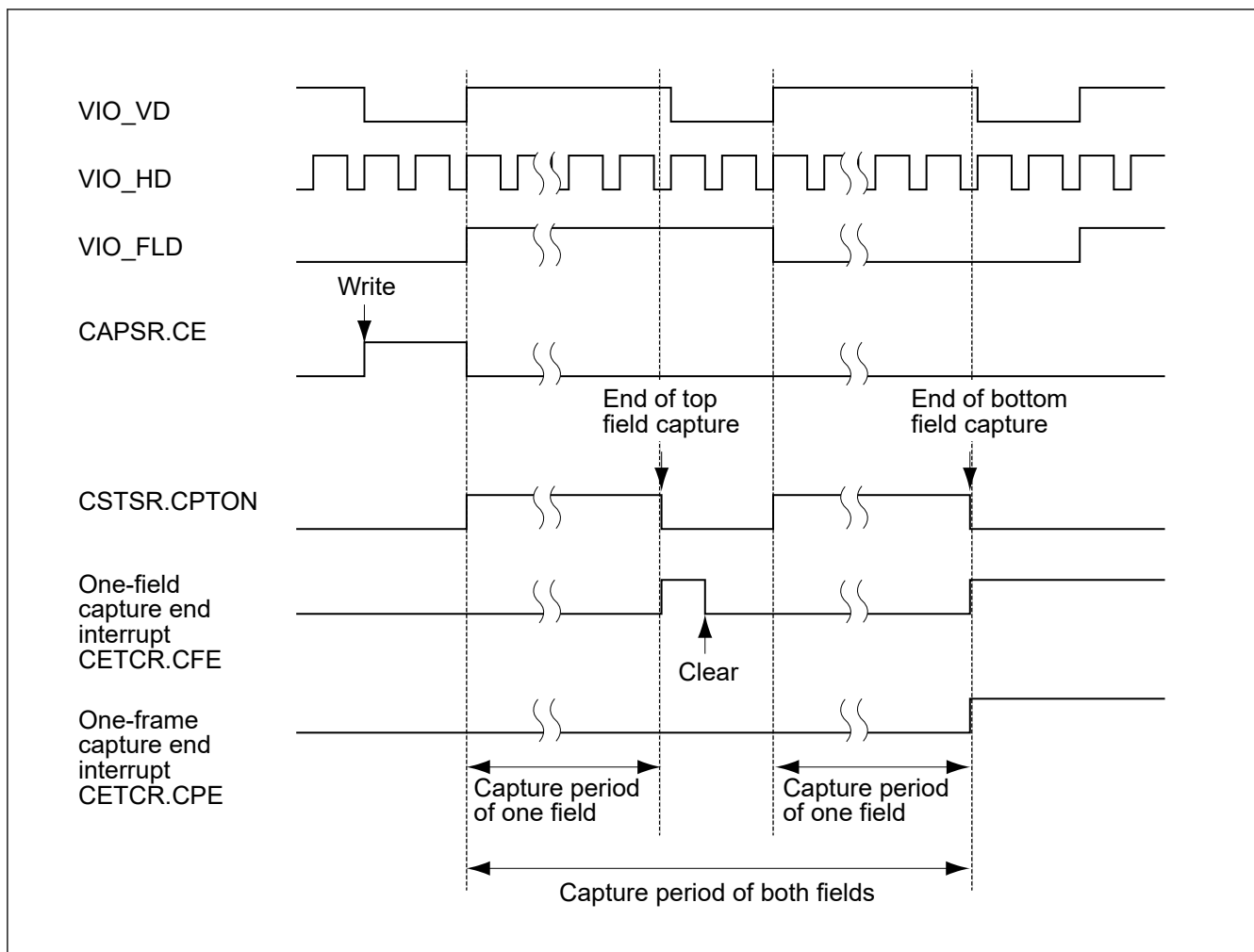


Figure 61.23 One-frame capture end interrupt and one-field capture end interrupt in both-field image capture

A captured frame image or captured one-field image is stored in the memory from the addresses set in CDAYR and CDACR (Figure 61.24). Captured both-field images are stored in different memory areas depending on whether it is a top-field or bottom-field image. A top-field image is stored in the memory from the addresses set in CDAYR and CDACR whereas a bottom-field image is stored in the memory from the addresses set in CDBYR and CDBCR (Figure 61.25).

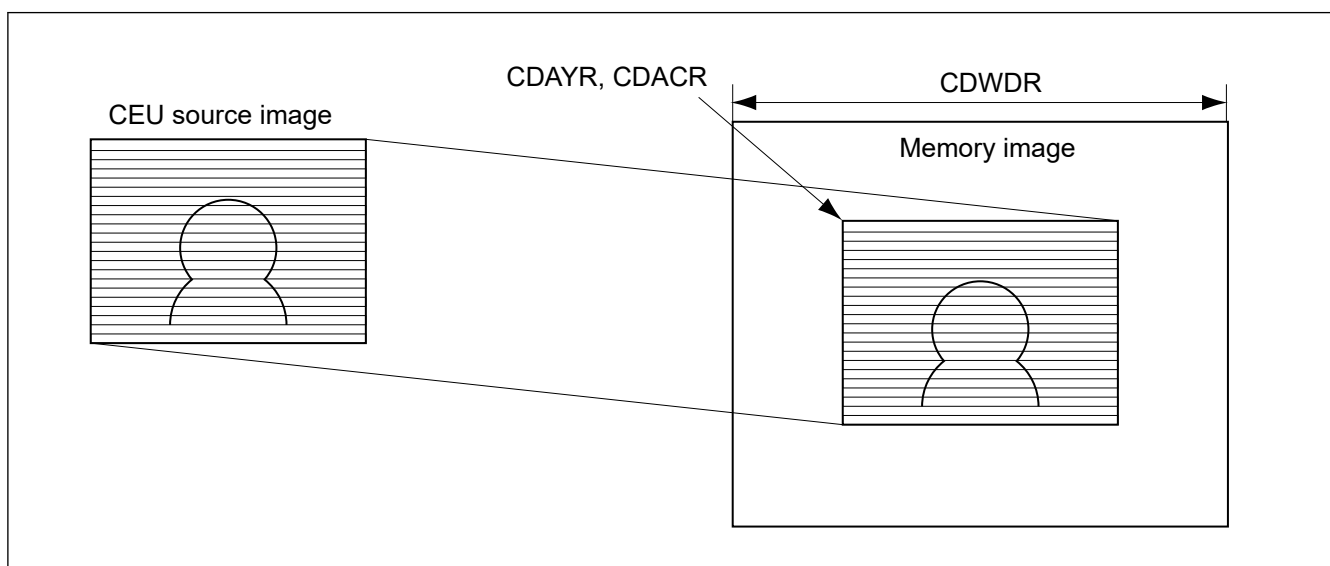


Figure 61.24 Image of storing captured frame image or captured one-field image in memory

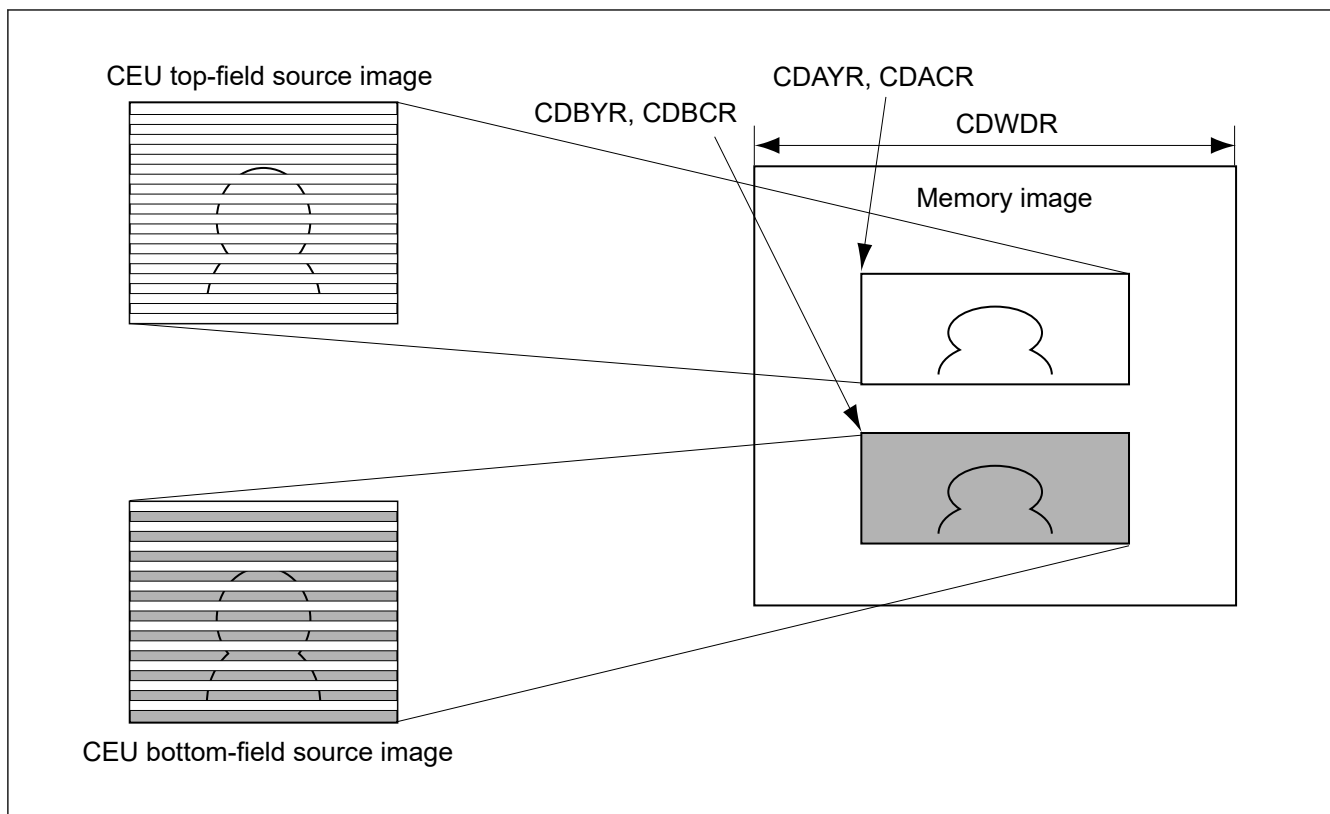


Figure 61.25 Image of storing captured both-field images in memory

If the FCI bits are set to 00b for continuous capture in interlace input mode, images are continuously captured for 2 VD with the first captured field as the reference in both-field image capture (Figure 61.26). In one-field image capture, only the first captured field is continuously captured for 1 VD (Figure 61.27).

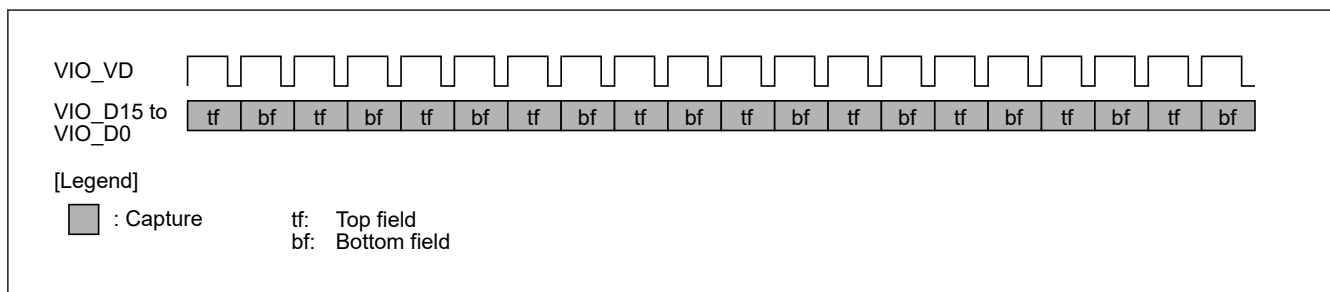


Figure 61.26 Continuous both-field capture in interlace mode (Image Immediately after Activation is Top Field (FCI[1:0] = 00b))

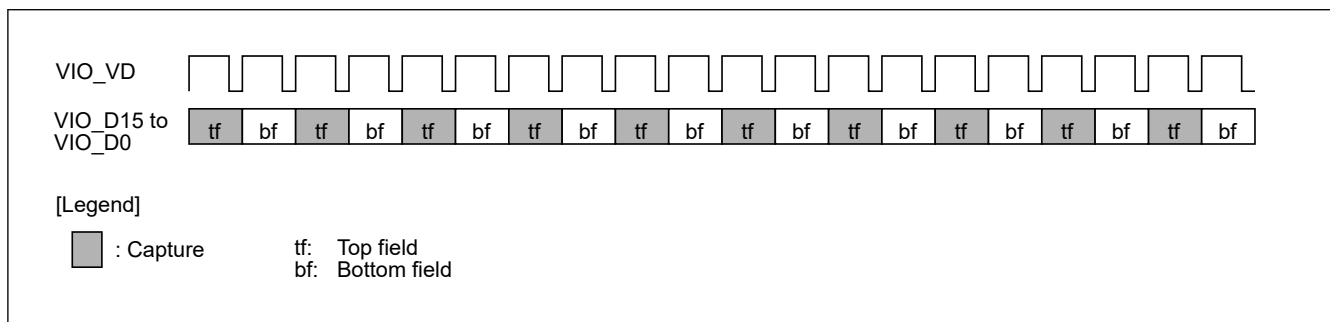


Figure 61.27 Continuous one-field capture in interlace mode (Image Immediately after Activation is Top Field (FCI[1:0] = 00b))

(1) Storage of Interlace Input as Frame Image

The CEU can store an interlace source image in the memory as a frame image. To store an interlace source image as a frame image, make the following register settings:

Input mode: Interlace (IFS = 1)

Capture image: Both-field (CIM = 0)

Image to start capture: Any setting other than the prohibited setting (FCI[1:0] = as desired)

Figure 61.28 shows a memory image of capturing both fields of an interlace input and storing it as a frame image in the memory. Set the start addresses of the memory destination for the captured top-field image in CDAYR and CDACR, and the start addresses of the memory destination for the captured bottom-field image in CDBYR and CDBCR. When storing an interlace image as a frame image in the memory, set the horizontal image size of the memory area in CDWDR with the top-field image and bottom-field image placed next to each other as shown in Figure 61.28. In addition, set the number of captured lines of the field image in the VWDTH bits in CAPWR.

A memory image of folding the horizontal image size of the memory area in Figure 61.28 at CDWDR/2 is shown in Figure 61.29. Setting the registers to form the image in Figure 61.28 enables an interlace image to be stored as a frame image in the memory as shown in Figure 61.29.

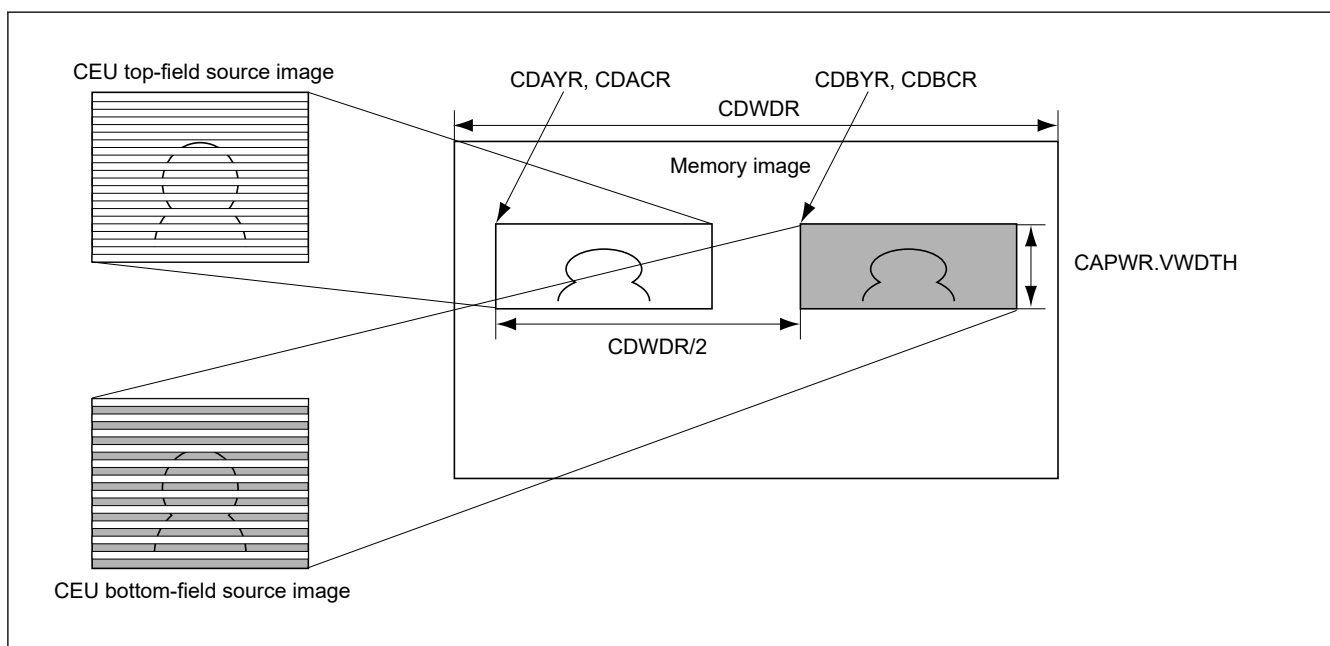


Figure 61.28 Image of storing captured both-fields of interlace input in memory

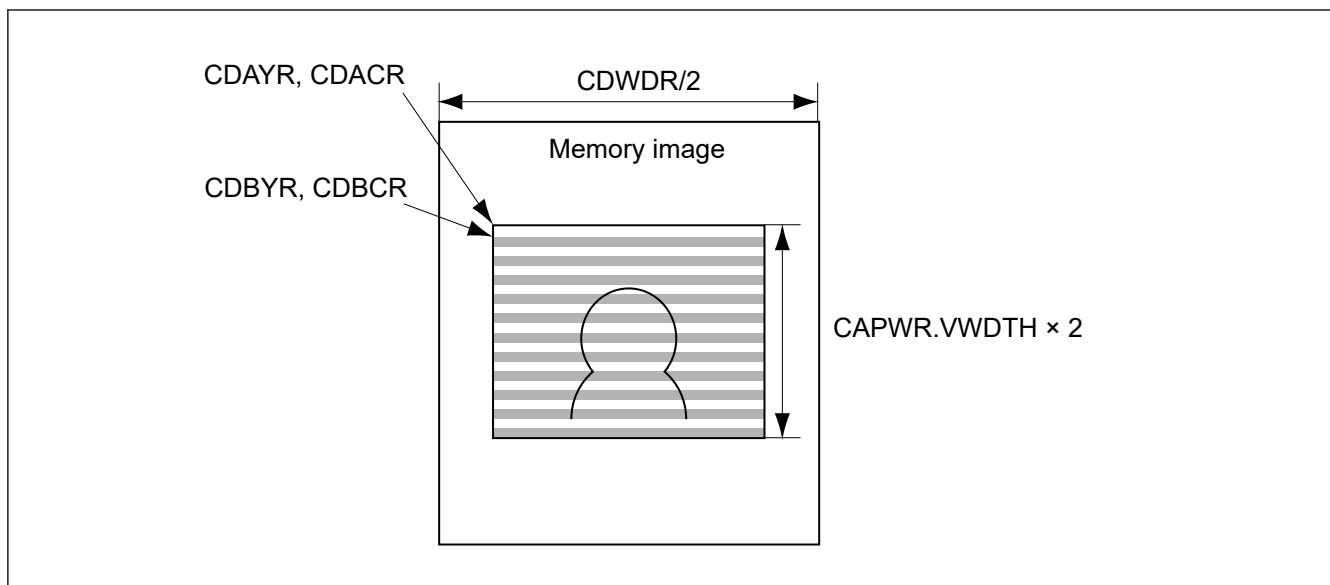


Figure 61.29 Image of storing Interlace input as frame image in memory

61.2.8 CRCNTR : CEU Register Control Register

Base address: CEU = 0x4034_8000
CEU_NS = 0x5034_8000

Offset address: 0x0028

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	RVS	—	—	RS	RC
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	RC	Specifies switching of the register plane used by the CEU in synchronization with VD. If the register plane is not switched, the register plane specified by the RS bit is used. 0: Uses the specified register plane in synchronization with VD 1: Switches the register plane in synchronization with VD	R/W
1	RS	Specifies which register plane is used by the CEU in synchronization with VD. The setting of this bit is valid only when RC = 0. 0: Uses plane A of the register 1: Uses plane B of the register	R/W
3:2	—	These bits are read as 0. The write value should be 0.	R/W
4	RVS	Sets the timing to switch the register plane in both-field capture. The setting of this bit is valid only when RC = 1 in both-field capture. 0: Switches the register plane every 2 VD 1: Switches the register plane every 1 VD	R/W
31:5	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

CRCNTR controls switching of the planes of registers with a 2-plane configuration.

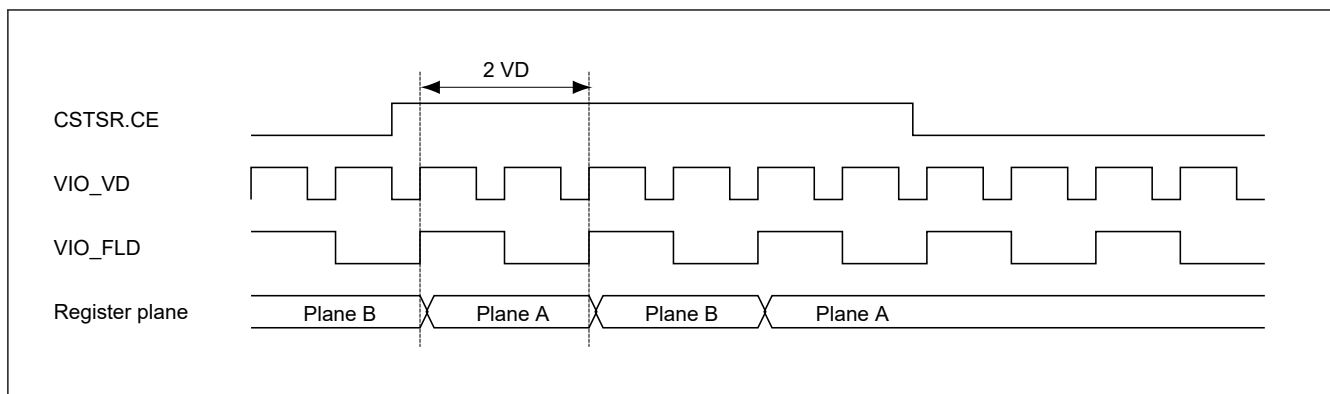


Figure 61.30 Timing for register plane switching when RVS = 0

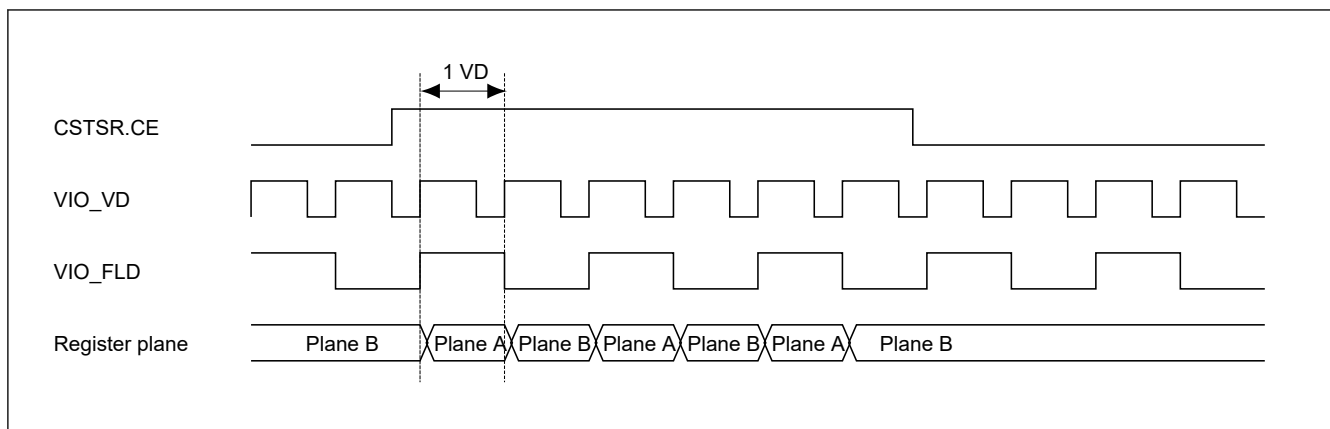


Figure 61.31 Timing for register plane switching when RVS = 1

61.2.9 CRCMPR : CEU Register Forcible Control Register

Base address: CEU = 0x4034_8000
CEU_NS = 0x5034_8000

Offset address: 0x002C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	RA
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	RA	Indicates the register plane currently specified. This register value automatically changes in synchronization with VD for starting capture. To start capture with plane A of the register when a setting to switch the register plane in synchronization with VD has been made (CRCNTR.RC = 1), specify plane B of the register using this bit. 0: Specifies plane A of the register 1: Specifies plane B of the register	R/W
31:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

CRCMPR forcibly controls switching of the planes of registers with a 2-plane configuration. Setting this register enables direct control of register plane switching.

Do not modify this register during operation. If this register is modified during operation, correct operation cannot be guaranteed. In addition, the IGRW bit (interrupt source) in CETCR is set to 1.

61.2.10 CFLCR, CFLCR x : Capture Filter Control Register (x = B, M)

Base address: CEU = 0x4034_8000
CEU NS = 0x5034 8000

Offset address: 0x0030 (CFLCR)
0x1030 (CFLCR_B)
0x2030 (CFLCR_M)

Bit position: 31 28 27 16 15 12 11 0

Bit field:	VMANT[3:0]	VFRAC[11:0]	HMANT[3:0]	HFRAC[11:0]
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Value after reset: 0

Bit	Symbol	Function	R/W
11:0	HFRAC[11:0]	Fraction Part of Horizontal Scale-Down Factor The specifiable range is 0x000 to 0xFF8. The fraction of the scale-down factor that cannot be set with only the HMANT bits must be specified with these bits. The value of writing are ignored on the lower 3 bits, so write 0 to the lower 3 bits.	R/W
15:12	HMANT[3:0]	Mantissa Part of Horizontal Scale-Down Factor The specifiable range is 0x0 to 0xF. When 0x0 is set for the HMANT bits and 0x000 is set for the HFRAC bits, the scale-down filter is not used.	R/W
27:16	VFRAC[11:0]	Fraction Part of Vertical Scale-Down Factor The specifiable range is 0x000 to 0xFF8. The fraction of the scale-down factor that cannot be set with only the VMANT bits must be set with these bits. The value of writing are ignored on the lower 3 bits, so write 0 to the lower 3 bits.	R/W
31:28	VMANT[3:0]	Mantissa Part of Vertical Scale-Down Factor The specifiable range is 0x0 to 0xF. When 0x0 is set for the VMANT bits and 0x000 is set for the VFRAC bits, the scale-down filter is not used.	R/W

Note: S-TYPE-3, P-TYPE-3

CFLCR sets the scale-down factor for the filter to scale images down.

The CEU has an image scale-down filter which can be used to scale down the captured images before storing them in the memory. Set CFLCR to 0 when not performing scale-down (same size output). If a value other than 0 is set in CFLCR, scale-down is performed. In data fetch mode, set CFLCR to 0.

When handling an interlace source image as a frame image, set CFLCR to 0 not to use the filter.

An image scale-down filter is installed in the CEU, and the captured images can be scaled down and stored in the memory.

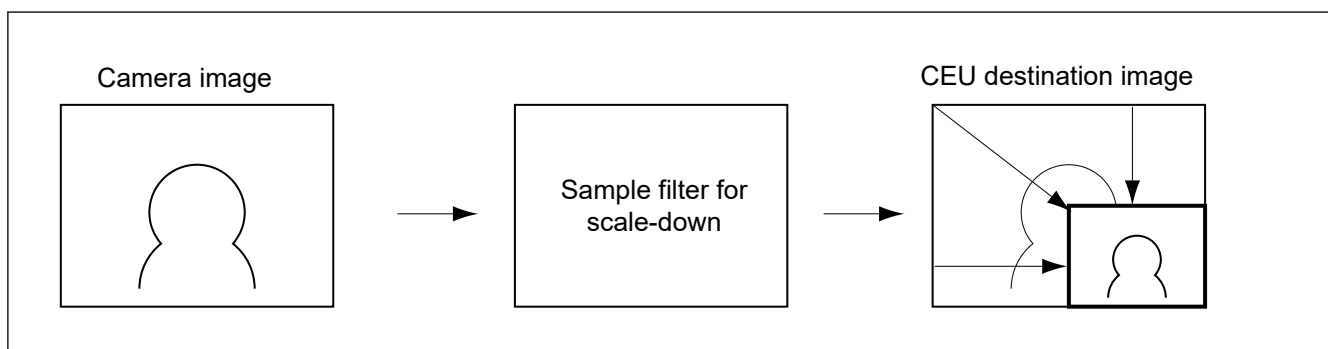


Figure 61.32 **Scale-down of captured image**

The formulas for obtaining the MANT (VMANT or HMANT) and FRAC (VFRAC or HFRAC) values from the input pixel count and output pixel count of the filter are shown below. Set the MANT and FRAC bits in order to obtain the desired output pixel count from the number of pixels input to the CEU.

First, calculate preliminary MANT and FRAC values. The parameters needed for calculation are defined as follows:

$$\alpha = \text{MANT} \times 4096 + \text{FRAC} \quad \dots \text{Formula 1}$$

$$\text{SCL (scaling factor)} = \frac{4096}{\alpha} \quad \dots \text{Formula 2}$$

Assuming an operator $\lfloor x \rfloor$ which discards fractions of an integer x , the MANT and FRAC values can be temporarily set as follows, according to formula 1 and formula 2.

$$\text{MANT} = \left\lfloor \frac{1}{\text{SCL}} \right\rfloor, \text{FRAC} = \left\lfloor 512 \times \left(\frac{1}{\text{SCL}} - \text{MANT} \right) \right\rfloor \times 8$$

Here, the scaled-down filter output size (SIZE_D) can be calculated using the input image size S_{in} (8-bit interface: half of the CAPWR setting, 16-bit interface: CAPWR setting) in the following formula.

$$\text{SIZE}_D = \left\lfloor 1 + \left(\left\lfloor \frac{1}{2} + \frac{S_{in}-1}{\text{MANT}_{pre}} \right\rfloor - 1 \right) \times \frac{\text{MANT}_{pre} \times 4096}{\alpha} \right\rfloor \quad \dots \text{Formula 3}$$

$$\begin{cases} \text{MANT}_{pre} = 1 \rightarrow (0 \leq \text{MANT} < 2) \\ \text{MANT}_{pre} = 2 \rightarrow (2 \leq \text{MANT} < 4) \\ \text{MANT}_{pre} = 4 \rightarrow (4 \leq \text{MANT} < 8) \\ \text{MANT}_{pre} = 8 \rightarrow (8 \leq \text{MANT}) \end{cases}$$

The number of output pixels can be obtained by substituting the temporarily calculated MANT, FRAC, and input image size into these formulas. If the calculated number of output pixels is smaller than the number of output pixels used to obtain the preliminary MANT and FRAC values, recalculate with a smaller FRAC (α) value, and set the MANT and FRAC values in this register so that a pixel value greater than the desired number of output pixels can be obtained.

Example: Scale down 640 pixels to 480 pixels

$\text{SCL} = 480/640 = 3/4$, and the preliminary settings of $\text{MANT} = 1$, $\text{MANT}_{pre} = 1$, and $\text{FRAC} = 0x550$ are made. Substituting these in the following formula results in an output pixel count of 479.

$$\text{SIZE}_D = \left\lfloor 1 + \left(\left\lfloor \frac{1}{2} + \frac{S_{in}-1}{\text{MANT}_{pre}} \right\rfloor - 1 \right) \times \frac{\text{MANT}_{pre} \times 4096}{\alpha} \right\rfloor \quad \dots \text{Formula 3}$$

Since this output pixel count is smaller than the desired output pixel count of 480, the formula is recalculated with a FRAC value of $0x548$, a value eight less than the previous time. The obtained result of output pixel count = 480 is equal to the desired output pixel count of 480, so this register is set as $\text{MANT} = 1$ and $\text{FRAC} = 0x548$.

Table 61.8 Setting examples for each scale-down filter factor

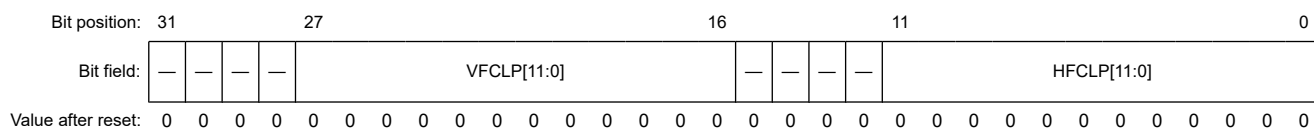
Scale-down factor	FRAC		MANT	Input pixel count	Output pixel count	Clipping size (CFSZR)
	Decimal	Hexadecimal				
7/8	576	0x240	1	640	560	560
3/4	1352	0x548	1	640	480	480
5/8	2448	0x990	1	640	400	400
1/2	0	0x0	2	640	320	320
3/8	2728	0xAA8	2	640	240	240
1/3	0.0	0x0	3	640	213	212
1/4	0.0	0x0	4	640	160	160
1/5	0.0	0x0	5	640	128	128
1/6	0.0	0x0	6	640	107	104
1/7	0.0	0x0	7	640	91	88
1/8	0.0	0x0	8	640	80	80
1/16	4088	0xFF8	15	640	40	40

Note: This scale-down filter uses a VGA-size line memory for scale-down. Therefore, when an image larger than the VGA size is input for scale-down, settings must be made so that the output image size is equal to or larger than the SubQCIF size and equal to or smaller than the VGA size. When an image is not scaled down (same size output), this restriction does not apply.

61.2.11 CFSZR, CFSZR_x : Capture Filter Size Clip Register (x = B, M)

Base address: CEU = 0x4034_8000
CEU_NS = 0x5034_8000

Offset address: 0x0034 (CFSZR)
0x1034 (CFSZR_B)
0x2034 (CFSZR_M)



Bit	Symbol	Function	R/W
11:0	HFCLP[11:0]	Specify the horizontal clipping value of the filter output size (8-pixel units). The lower 3 bits should be written 0.	R/W
15:12	—	These bits are read as 0. The write value should be 0.	R/W
27:16	VFCLP[11:0]	Specify the vertical clipping value of the filter output size (4-pixel units). The value of writing are ignored on the lower 2 bits, so write 0 to the lower 2 bits.	R/W
31:28	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

CFSZR sets the clipping size for fine adjustment of the image size output from the filter, and must be set in combination with CFLCR. When clipping the output size of the filter, set the clipping size as a number of pixels, and the setting unit should be four pixels. CFSZR must be set even when scale-down is not performed (same size output).

This register is not used, during data enable fetch mode.

In data synchronous fetch mode, set CFSZR according to the setting of CAPWR.

The scale-down filter in the CEU may output an odd number of pixels or lines depending on the settings. To adjust the output size of the filter, the CEU clips the destination image by using the number of pixels specified in CFSZR, as shown in [Figure 61.33](#). The clipping size must be specified vertically in 4-pixel units and horizontally in 8-pixel units.

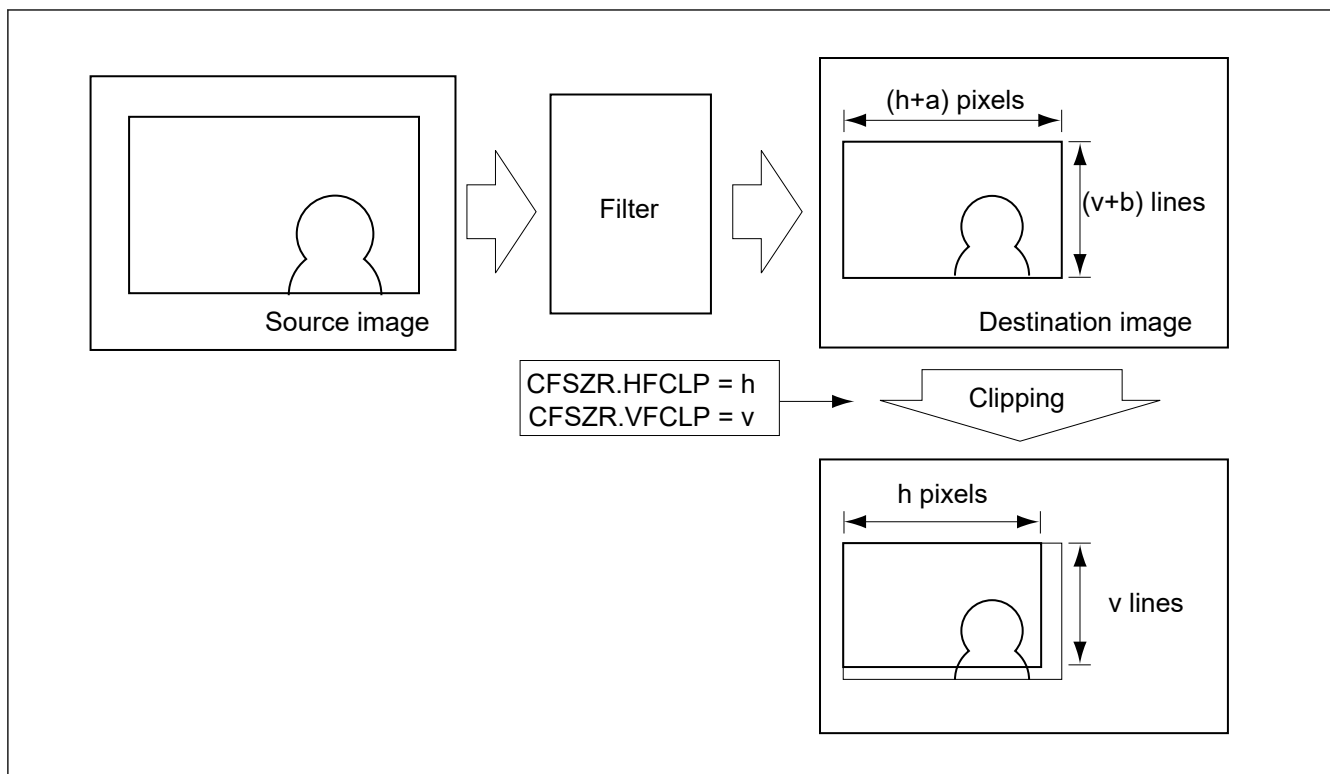


Figure 61.33 Clipping of image output from filter

The pixels to be clipped are counted from the top-left corner of a display. The pixels located to the right of the specified number of pixels or below the specified number of lines are discarded by the clipping function. If the number of pixels specified in CFSZR is larger than that output from the filter, correct operation cannot be guaranteed. To avoid this, the clipping size specified in CFSZR must be equal to or smaller than the number of pixels output from the filter.

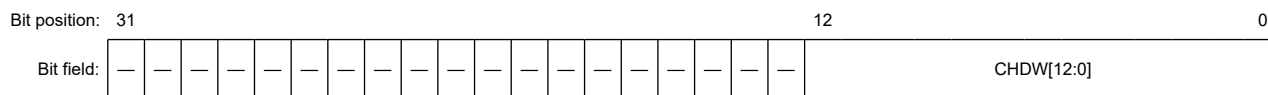
Note: In data synchronous fetch mode, the following settings are required. Data cannot be fetched correctly unless the following settings are made.

- 8-bit interface:
 $VFCLP = CAPWR.VWDTH$
 $HFCLP = CAPWR.HWDTH/2$
- 16-bit interface:
 $VFCLP = CAPWR.VWDTH$
 $HFCLP = CAPWR.HWDTH$

61.2.12 CDWDR, CDWDR_x : Capture Destination Width Register (x = B, M)

Base address: CEU = 0x4034_8000
 CEU_NS = 0x5034_8000

Offset address: 0x0038 (CDWDR)
 0x1038 (CDWDR_B)
 0x2038 (CDWDR_M)



Value after reset: 0

Bit	Symbol	Function	R/W
12:0	CHDW[12:0]	Specify the horizontal image size in the memory area where the captured image is to be stored (8-byte units). The image data captured by the CEU is stored in the memory. In data synchronous fetch mode, set as follows: 8-bit interface: CHDW = CAPWR.HWDTH 16-bit interface: CHDW = CAPWR.HWDTH × 2 The lower 3 bits should be written 0.	R/W
31:13	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

CDWDR sets the horizontal image size in the memory area where the captured image is to be output in 8-byte units (8-pixel units).

In data synchronous fetch mode, set CDWDR according to the setting of CAPWR.

This register is not used, during data enable fetch mode.

61.2.13 CDAYR, CDAYR_x : Capture Data Address Y Register (x = B, M)

Base address: CEU = 0x4034_8000
CEU_NS = 0x5034_8000

Offset address: 0x003C (CDAYR)
0x103C (CDAYR_B)
0x203C (CDAYR_M)

Bit position: 31

0

Bit field:

CAYR[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	CAYR[31:0]	Capture Data Address Y - Frame image capture: These bits set the address for storing the luminance component data of the captured data (8-pixel units). - One-field image capture: These bits set the address for storing the luminance component data of the captured data (8-pixel units). - Both-field image capture: These bits set the address for storing the luminance component data of the captured top-field data (8-pixel units). - Data fetch: These bits set the address for storing data (8-byte units). - Data enable fetch bundle write: These bits set the address for storing data (32-byte units). The lower 3 bits should be written 0.	R/W

Note: S-TYPE-3, P-TYPE-3

CDAYR specifies the address where the luminance component of the captured data is to be stored in frame image capture or one-field image capture, the address where the luminance component of the captured top field is to be stored in both-field image capture, and the address where the fetched data is to be stored in data fetch. The CEU separates the captured image data into the luminance component data (Y) and the chrominance component data (C), and stores them in the memory via the bus. In frame image capture or one-field image capture, set the start address of the memory area where the luminance component of the captured data is to be stored by CDAYR. In both-field image capture, set the start address of the memory area where the luminance component of the captured top-field image is to be stored by CDAYR. In data fetch, set the start address of the memory area where data is to be stored by CDAYR.

Because the address must be specified in 32 bits, the address set by CDAYR must be in longword units.

Set the address of the starting point of the memory area where the fetched data is to be stored in this register, as shown in [Figure 61.34](#).

- Frame image capture: Set the address of the starting point of the memory area where the luminance component of the captured image is to be stored.
- One-field image capture: Set the address of the starting point of the memory area where the luminance component of the captured image is to be stored.

- Both-field image capture: Set the address of the starting point of the memory area where the luminance component of the captured top-field image is to be stored.
- Data fetch: Set the address of the starting point of the memory area where the fetched data is to be stored. In data fetch mode, the data is simply stuffed in order from the start address so the end address becomes as follows:
End address = CDAYR + number of fetched bytes
- Data enable fetch bundle write: Set the address in 32-byte units.

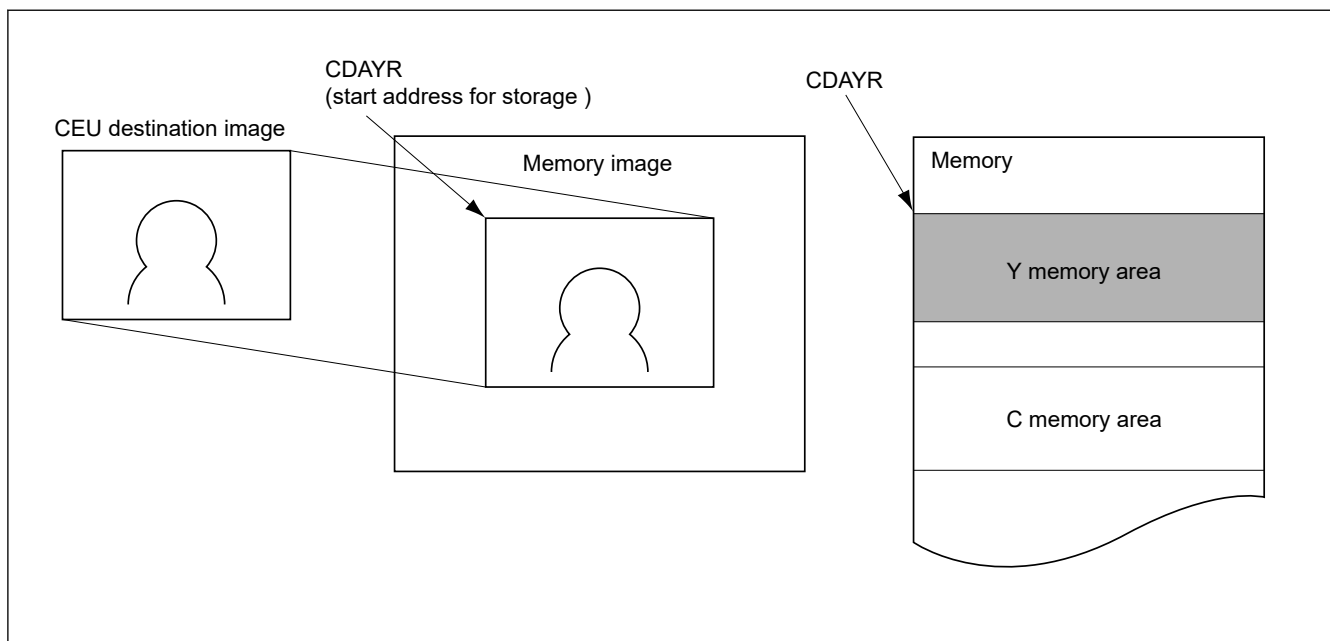


Figure 61.34 Relationship between captured image and luminance component memory area

61.2.14 CDACR, CDACR_x : Capture Data Address C Register (x = B, M)

Base address: CEU = 0x4034_8000
CEU_NS = 0x5034_8000

Offset address: 0x0040 (CDACR)
0x1040 (CDACR_B)
0x2040 (CDACR_M)

Bit position: 31

0

Bit field:

CACR[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	CACR[31:0]	Capture Data Address C • Frame image capture: These bits set the address for storing the chrominance component data of the captured data (8-pixel units). • One-field image capture: These bits set the address for storing the chrominance component data of the captured data (8-pixel units). • Both-field image capture: These bits set the address for storing the chrominance component data of the captured top-field data (8-pixel units). The lower 3 bits should be written 0.	R/W

Note: S-TYPE-3, P-TYPE-3

CDACR specifies the address where the chrominance component of the captured data is to be stored in frame image capture or one-field image capture, and the address where the chrominance component of the captured top field is to be stored in both-field image capture. The CEU separates the captured image data into the luminance component data (Y) and the chrominance component data (C), and stores them in the memory via the bus. In frame image capture or onefield image capture, set the start address of the memory area where the chrominance component of the captured data is to be stored by

CDACR. In both-field image capture, set the start address of the memory area where the chrominance component of the captured top-field image is to be stored by CDACR. CDACR is not used in data fetch.

Because the address must be specified in 32 bits, the address set by CDACR must be in longword units.

Set the address of the starting point of the memory area where the chrominance component of the captured image is to be stored in this register, as shown in Figure 61.35. The chrominance component has an output data format like that in Figure 61.36, and is saved in the memory in this format.

- Frame image capture: Set the address of the starting point of the memory area where the chrominance component of the captured image is to be stored.
- One-field image capture: Set the address of the starting point of the memory area where the chrominance component of the captured image is to be stored.
- Both-field image capture: Set the address of the starting point of the memory area where the chrominance component of the captured top-field image is to be stored.

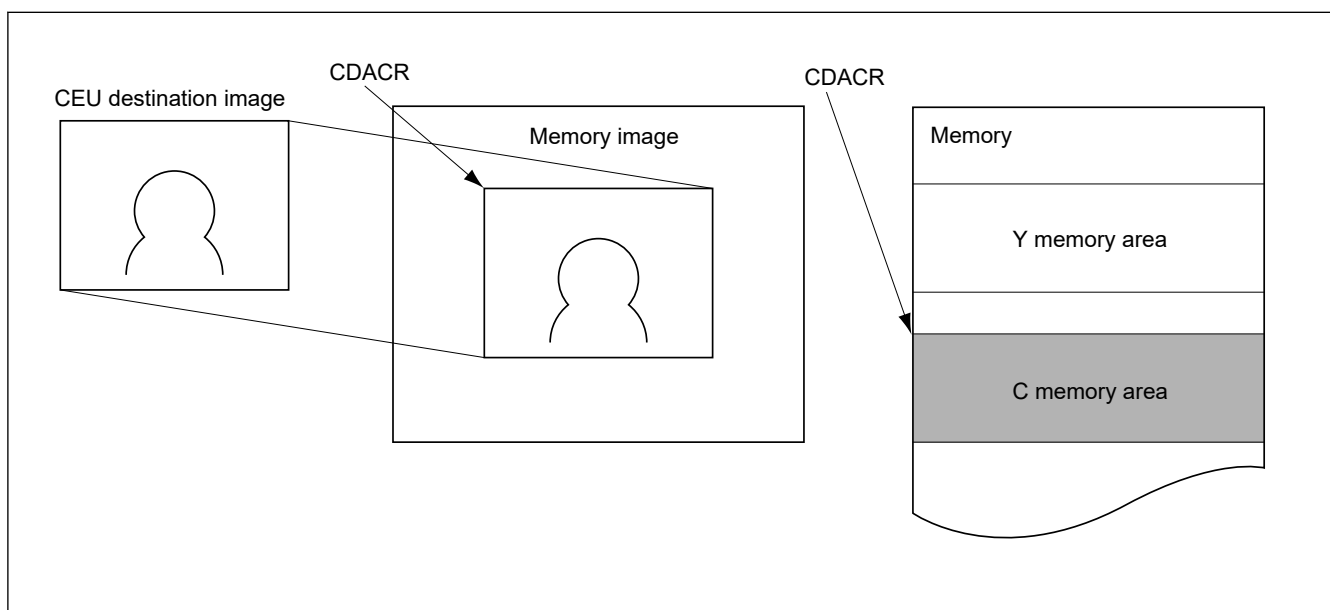


Figure 61.35 Relationship between captured image and chrominance component memory area

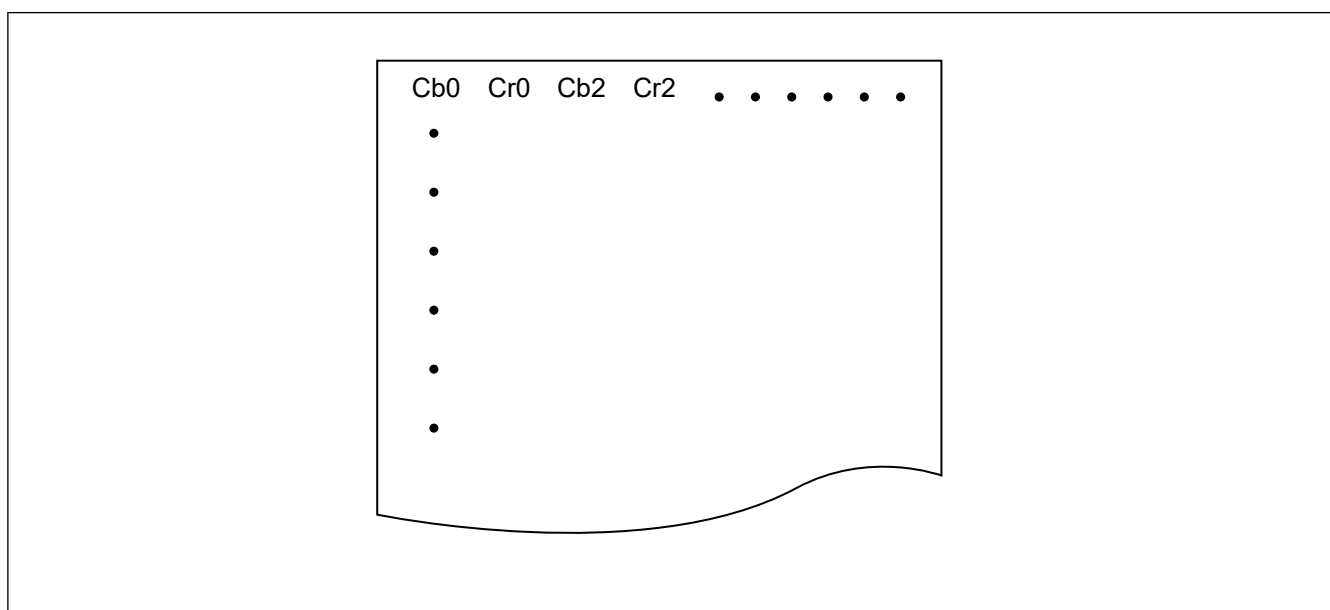


Figure 61.36 Image of storing chrominance components in memory

61.2.15 CDBYR, CDBYR_x : Capture Data Bottom-Field Address Y Register (x = B, M)

Base address: CEU = 0x4034_8000
CEU_NS = 0x5034_8000

Offset address: 0x0044 (CDBYR)
0x1044 (CDBYR_B)
0x2044 (CDBYR_M)

Bit position: 31

0

Bit field:

CBYR[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	CBYR[31:0]	Set the address for storing the luminance component data of the captured bottom-field data (8-pixel units). The lower 3 bits should be written 0.	R/W

Note: S-TYPE-3, P-TYPE-3

CDBYR specifies the address where the luminance component of the captured bottom-field data is to be stored in both-field image capture. The CEU separates the captured image data into the luminance component data (Y) and the chrominance component data (C), and stores them in the memory via the bus. Set the start address of the memory area where the luminance component of the captured bottom-field image is to be stored by CDBYR. CDBYR is not used in frame image capture, one-field image capture, or data fetch.

Because the address must be specified in 32 bits, the address set by CDBYR must be in longword units.

Set the address of the starting point of the memory area where the luminance component of the captured bottom-field image is to be stored in this register, as shown in Figure 61.37.

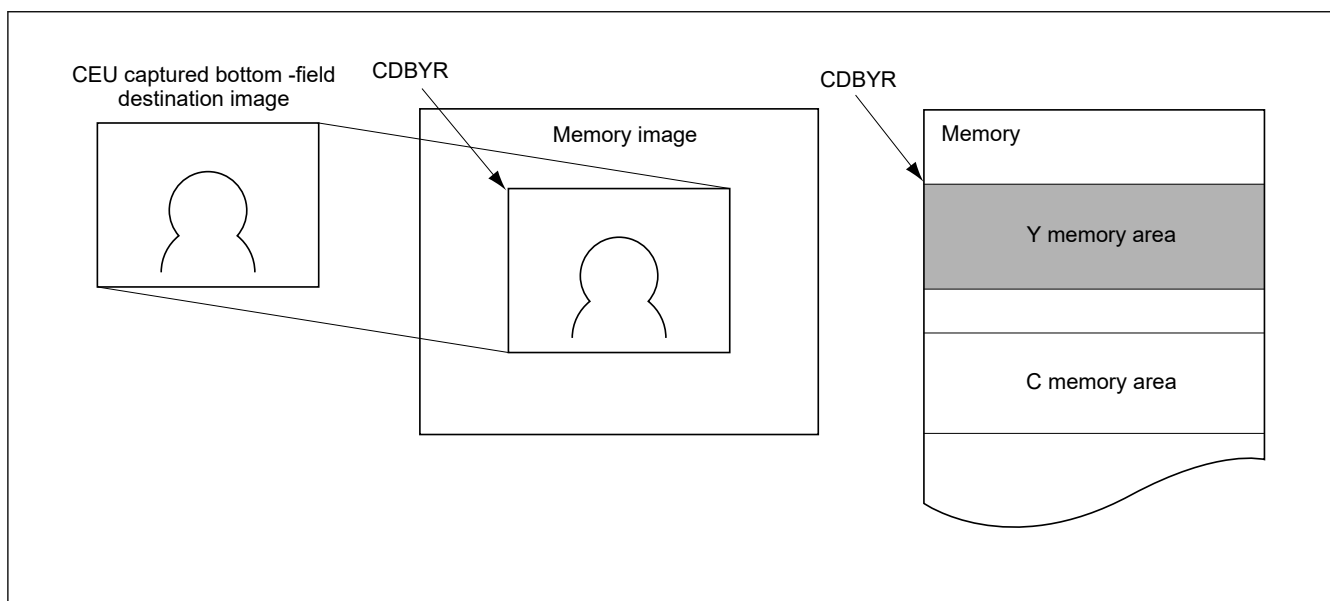


Figure 61.37 Relationship between captured bottom-field image and luminance component memory area

61.2.16 CDBCR, CDBCR_x : Capture Data Bottom-Field Address C Register (x = B, M)

Base address: CEU = 0x4034_8000
CEU_NS = 0x5034_8000

Offset address: 0x0048 (CDBCR)
0x1048 (CDBCR_B)
0x2048 (CDBCR_M)

Bit position: 31

0

Bit field:

CBCR[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	CBCR[31:0]	Set the address for storing the chrominance component data of the captured bottom-field data (8-pixel units). The lower 3 bits should be written 0.	R/W

Note: S-TYPE-3, P-TYPE-3

CDBCR specifies the address where the chrominance component of the captured bottom-field data is to be stored in both-field image capture. The CEU separates the captured image data into the luminance component data (Y) and the chrominance component data (C), and stores them in the memory via the bus. Set the start address of the memory area where the chrominance component of the captured bottom-field image is to be stored by CDBCR. CDBCR is not used in frame image capture, one-field image capture, or data fetch.

Because the address must be specified in 32 bits, the address set by CDBCR must be in longword units.

Set the address of the starting point of the memory area where the chrominance component of the captured bottom-field image is to be stored in this register, as shown in Figure 61.38. The chrominance component has an output data format like that in Figure 61.39, and is saved in the memory in this format.

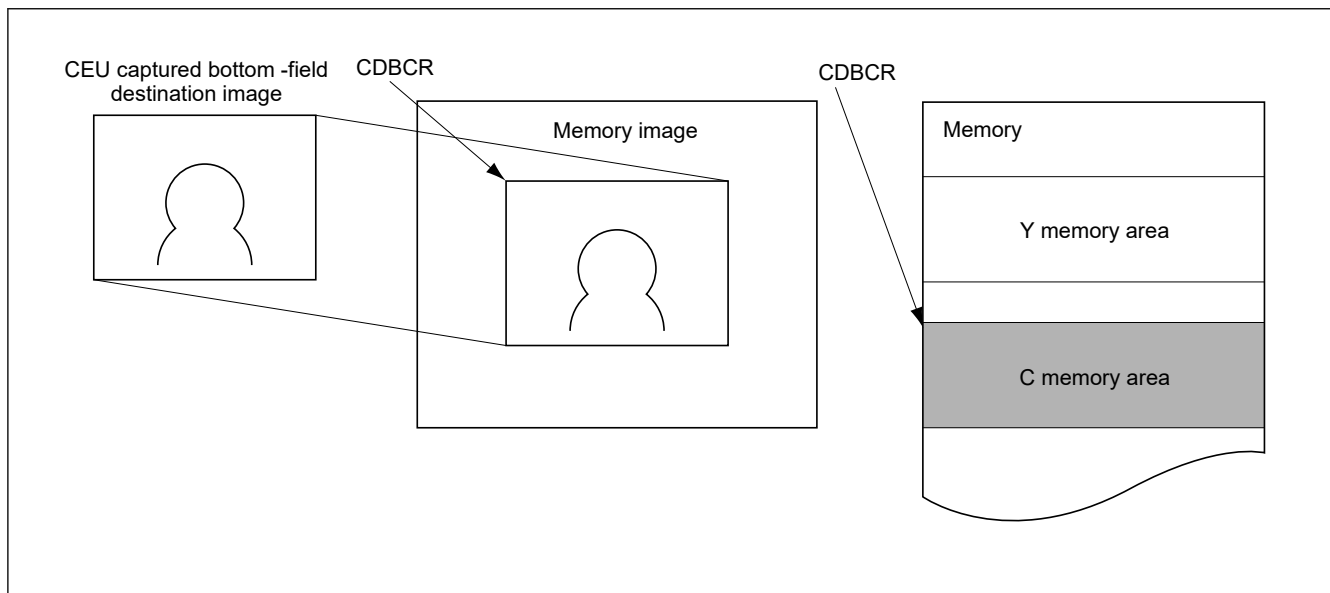


Figure 61.38 Relationship between captured bottom-field image and chrominance component memory area

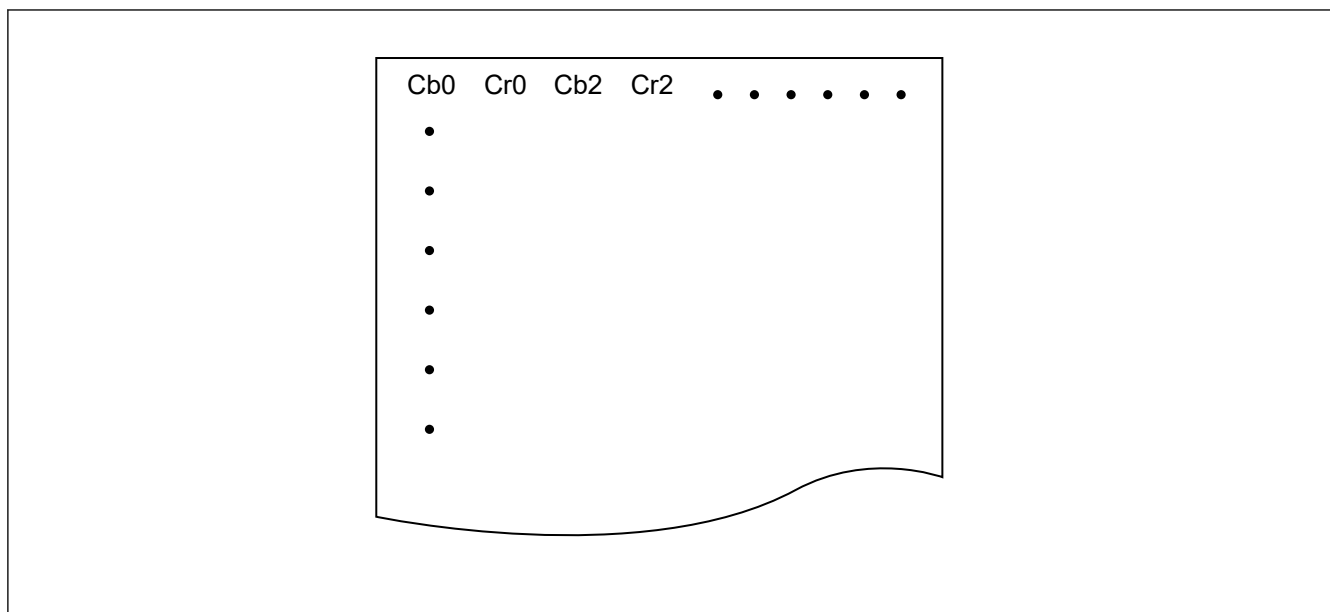


Figure 61.39 Image of storing chrominance components in memory

61.2.17 CBDSR, CBDSR_x : Capture Bundle Destination Size Register (x = B, M)

Base address: CEU = 0x4034_8000
CEU_NS = 0x5034_8000

Offset address: 0x004C (CBDSR)
0x104C (CBDSR_B)
0x204C (CBDSR_M)

Bit position: 31 22 0

Bit field: — — — — — — — — — — CBVS[22:0]

Value after reset: 0

Bit	Symbol	Function	R/W
22:0	CBVS[22:0]	Select the number of lines or number of bytes for output to the memory in a bundle write. Image capture and data synchronous fetch: Number of lines for output to the memory in a bundle write. Unit: 8 lines, min.: 8 lines, max.: 1,920 lines (0x780) Data enable fetch: Number of bytes for output to the memory in a bundle write. Unit: 32 bytes, min.: 512 bytes, max.: 6291456 bytes (0x600000) The value of writing are ignored on the lower 3 bits, so write 0 to the lower 3 bits.	R/W
31:23	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

CBDSR sets the size of output to memory in a bundle write. The number of output lines should be specified for image capture or data synchronous fetch. The number of bytes should be specified for data enable fetch.

(a) Image capture and data synchronous fetch

Set the number of lines of captured data to be written to the memory by a bundle write as a multiple of eight. This register is valid only when CDOCR.CBE = 1. When CDOCR.CBE = 1 and this register cleared to 0, this module operates with the number of lines of captured data to be written to the memory as eight. The maximum number of lines that can be set is 1,920 (0x780). Only bits CBVS[11:3] are valid.

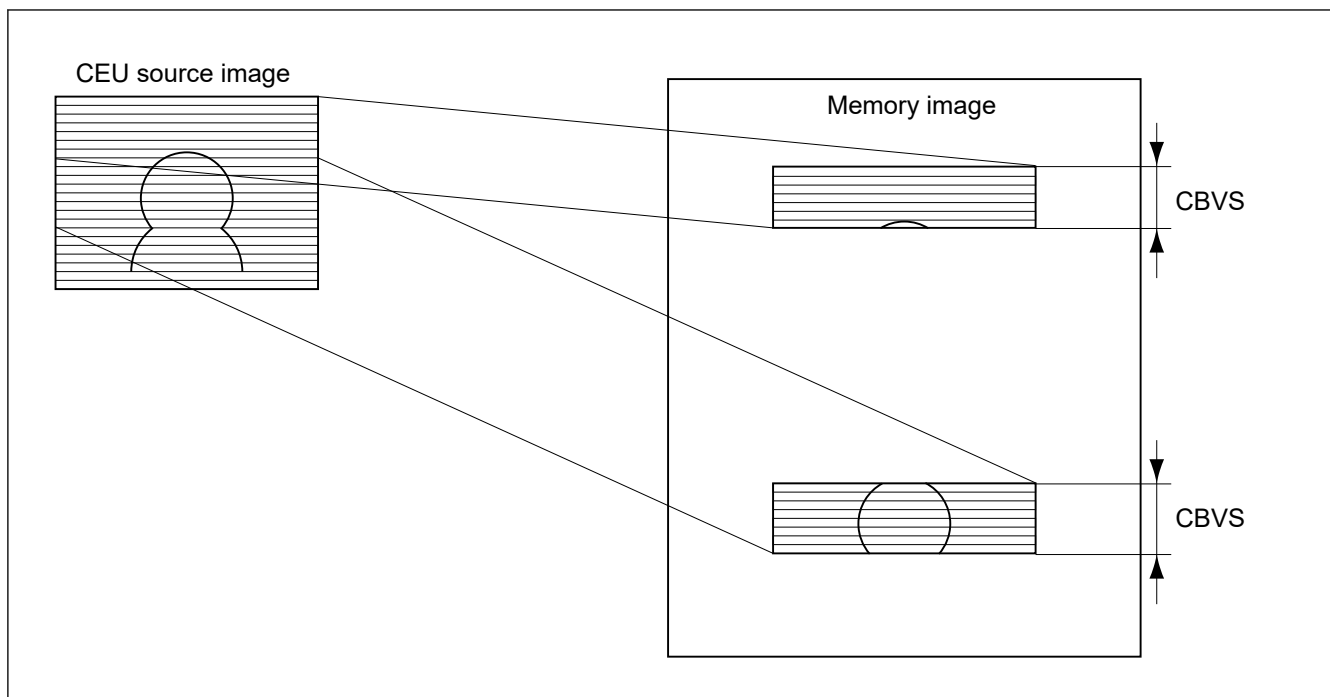


Figure 61.40 Image of storing captured image in memory by bundle write

(b) Data enable fetch

Set the number of bytes of captured data to be written to the memory by a bundle write as a multiple of 32. This register is valid only when CDOCR.CBE = 1. The minimum settable size is 512 bytes. When a number smaller than 512 bytes is specified, operation is not guaranteed.

61.2.18 CFWCR : Firewall Operation Control Register

Base address: CEU = 0x4034_8000
CEU NS = 0x5034_8000

Offset address: 0x005C

Bit position: 31

Bit field:

FWV[26:0]

5

0

Value after reset: 0 1 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
0	FWE	Firewall Operation With the setting of FWE = 1, when a write address exceeds the value set with FWV, the address is retained and an interrupt source FWF is set. After this, the address is not incremented and data is overwritten on the upper limit address. 0: Firewall is not activated. 1: Firewall is activated.	R/W
4:1	—	These bits are read as 0. The write value should be 0.	R/W
31:5	FWV[26:0]	Specify the upper limit of a write address. Specify the upper 27 bits of the 32-bit address. The upper limit of an address is $FWV[26:0] \ll 5 + 0x1F$.	R/W

Note: S-TYPE-3, P-TYPE-3

CFWCR specifies the upper limit of a write address in data enable fetch. When the VD input from an external module does not go low and end notification is not given, this register can prevent writing to memory from being out of control.

This register is enabled only in data enable fetch.

61.2.19 CLFCR, CLFCR_x : Capture Low-Pass Filter Control Register (x = B, M)

Base address: CEU = 0x4034_8000
CEU_NS = 0x5034_8000

Offset address: 0x0060 (CLFCR)
0x1060 (CLFCR_B)
0x2060 (CLFCR_M)

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	LPF
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	LPF	Enables or disables operation of the low-pass filter. The low-pass filter removes high-frequency components from the destination image in the horizontal direction. Clear this bit to 0 in data fetch mode. 0: Low-pass filter not used 1: Low-pass filter used (only in the horizontal direction)	R/W
31:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

CLFCR specifies whether or not to operate the low-pass filter. In data fetch mode, clear the LPF bit to 0.

The characteristic of the low-pass filter installed in the CEU causes the phase location of the image processed by the low-pass filter to be shifted right by one pixel compared to the raw image.

61.2.20 CDOCR, CDOCR_x : Capture Data Output Control Register (x = B, M)

Base address: CEU = 0x4034_8000
CEU_NS = 0x5034_8000

Offset address: 0x0064 (CDOCR)
0x1064 (CDOCR_B)
0x2064 (CDOCR_M)

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	CBE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	CDS	—	COLS	COWS	COBS
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	COBS	Controls swapping in 8-bit units for data output from the CEU. 0: Data is not swapped in 8-bit units 1: Data is swapped in 8-bit units	R/W
1	COWS	Controls swapping in 16-bit units for data output from the CEU. 0: Data is not swapped in 16-bit units 1: Data is swapped in 16-bit units	R/W
2	COLS	Controls swapping in 32-bit units for data output from the CEU. 0: Data is not swapped in 32-bit units 1: Data is swapped in 32-bit units	R/W

Bit	Symbol	Function	R/W
3	—	This bit is read as 0. The write value should be 0.	R/W
4	CDS	Sets the image format when outputting the image data captured in the YCbCr422 format to the memory. When 0 is written to this bit, only the luminance component (Y) is output and no chrominance components (Cb and Cr) are output for the odd-numbered lines. With an interlace source image, similarly only the luminance component (Y) is output and no chrominance components (Cb and Cr) are output for the odd-numbered lines of the field. In data fetch mode, set this bit to 1. 0: Converts the YCbCr422 format to the YCbCr420 format before outputting data to the memory 1: Outputs data in the YCbCr422 format to the memory without conversion	R/W
15:5	—	These bits are read as 0. The write value should be 0.	R/W
16	CBE	Controls the number of lines of captured data to be written to the memory. - Image capture This bit controls the number of lines of captured data to be written to the memory. When bundle write is set by this register, captured data is written in line units specified by CBDSR to the addresses specified by CDAYR and CDACR, and CDAYR2 and CDACR2 (CDBYR and CDBCR, and CDBYR2 and CDBCR2 for the bottom field in both-field capture) alternately (Figure 61.41). When captured data has been written for the number of lines set by CBDSR, a write end interrupt corresponding to each address setting register occurs. However, after write for one-frame (one-field) capture ends, a bundle write end interrupt does not occur even when bundle write has finished. - Data synchronous fetch This bit controls the number of lines of captured data to be written to the memory. When bundle write is set by this register, captured data is written in line units specified by CBDSR to the addresses specified by CDAYR and CDAYR2 alternately. When captured data has been written for the number of lines set by CBDSR, a write end interrupt corresponding to each address setting register occurs. However, after write for one-frame capture ends, a bundle write end interrupt does not occur even when bundle write has finished. - Data enable fetch This bit controls the number of bytes of captured data to be written to the memory. When bundle write is set by this register, captured data is written in byte units specified by CBDSR to the addresses specified by CDAYR and CDAYR2 alternately. When captured data has been written for the number of bytes set by CBDSR, a write end interrupt corresponding to each address setting register occurs. Also, only in data enable fetch, a bundle write end interrupt occurs when bundle write has finished after write for one-frame capture ends. Table 61.9 shows the correspondence between address setting registers and write end interrupt sources. Figure 61.42 shows the timing of write end interrupts in image capture and data synchronous fetch. Figure 61.43 shows the timing of write end interrupts in data enable fetch. 0: Normal write 1: Bundle write	R/W
31:17	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

CDOCR sets the format for outputting captured data to the memory. In data fetch mode, set the CDS bit to 1.

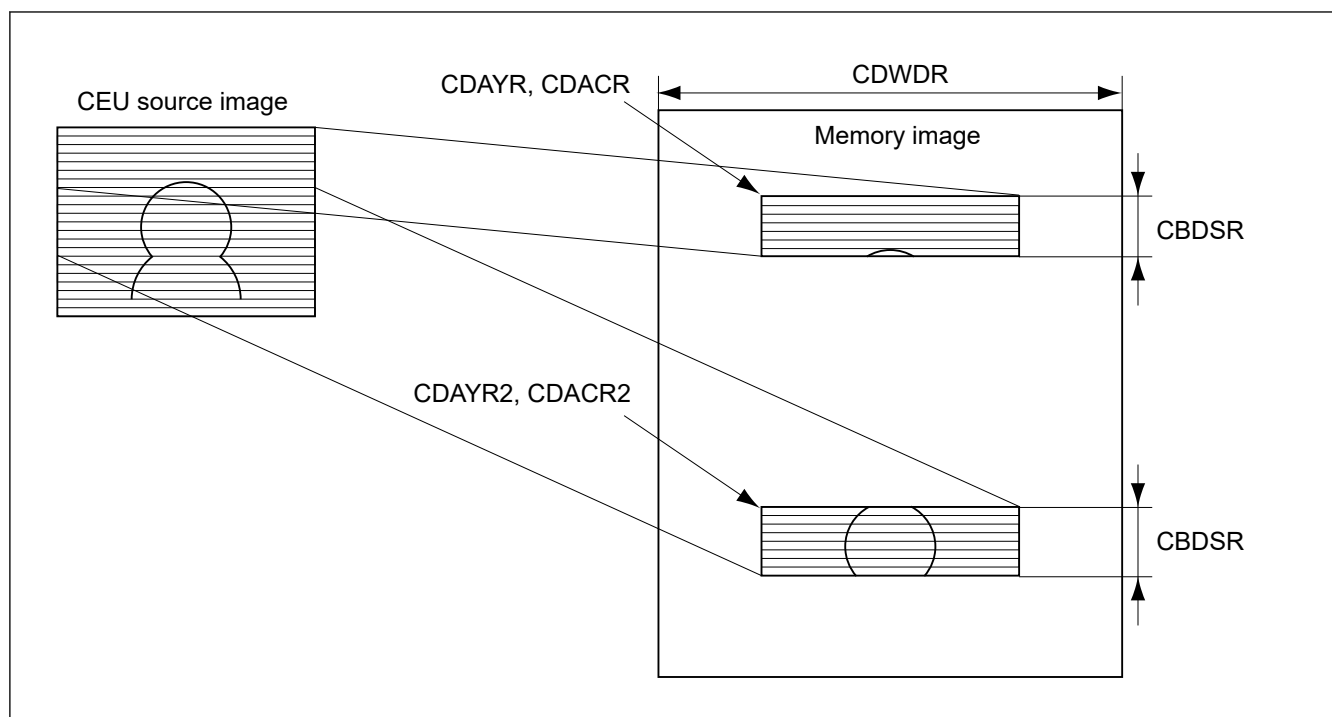


Figure 61.41 Image of bundle write to memory

Table 61.9 Correspondence between address setting registers and write end interrupt sources

Address setting registers	Bundle write end interrupt source
CDAYR, CDACR	CPBE1 bit in CETCR
CDAYR2, CDACR2	CPBE2 bit in CETCR
CDBYR, CDBCR	CPBE3 bit in CETCR
CDBYR2, CDBCR2	CPBE4 bit in CETCR

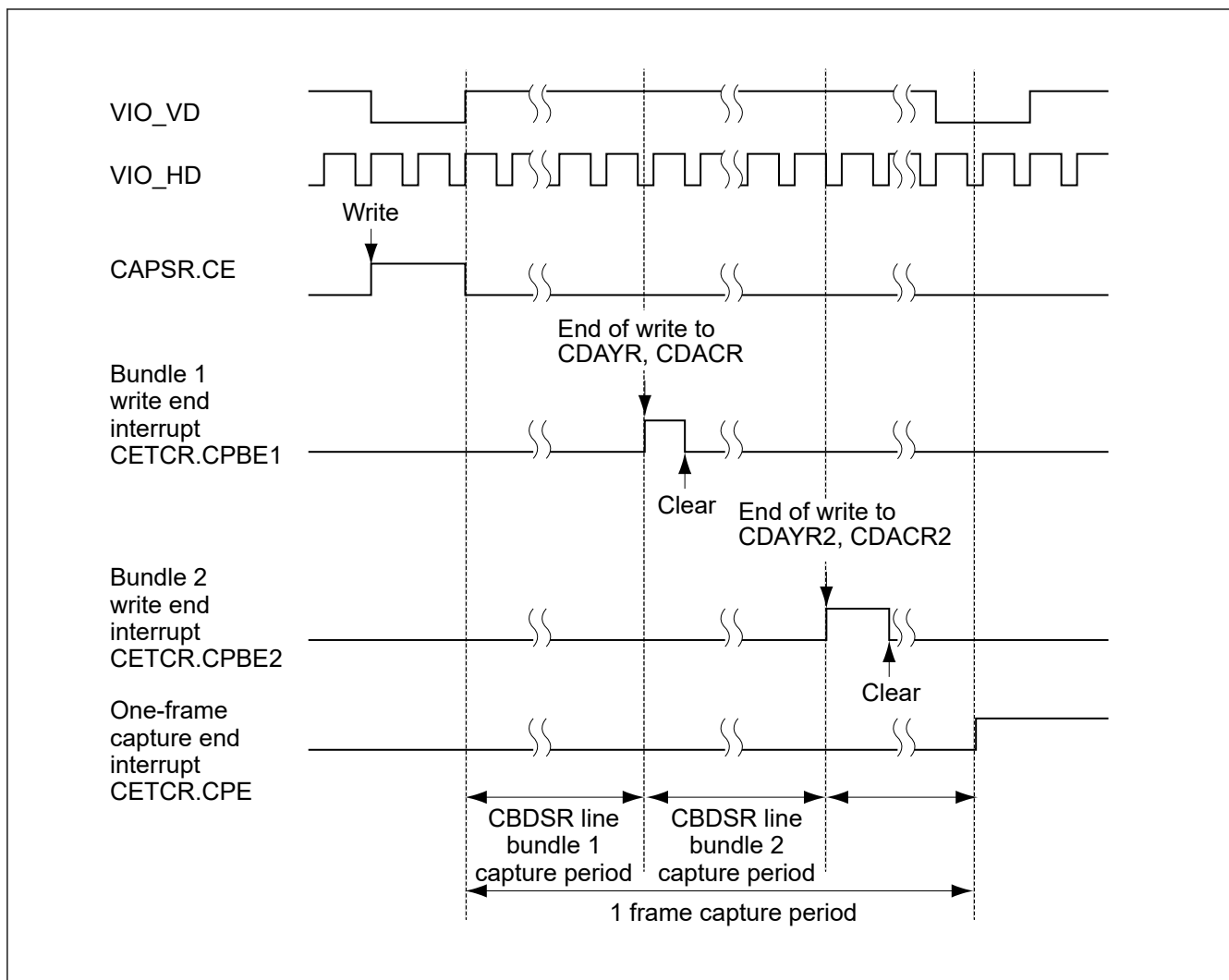


Figure 61.42 Timing of write end interrupts (Image capture, data synchronous fetch)

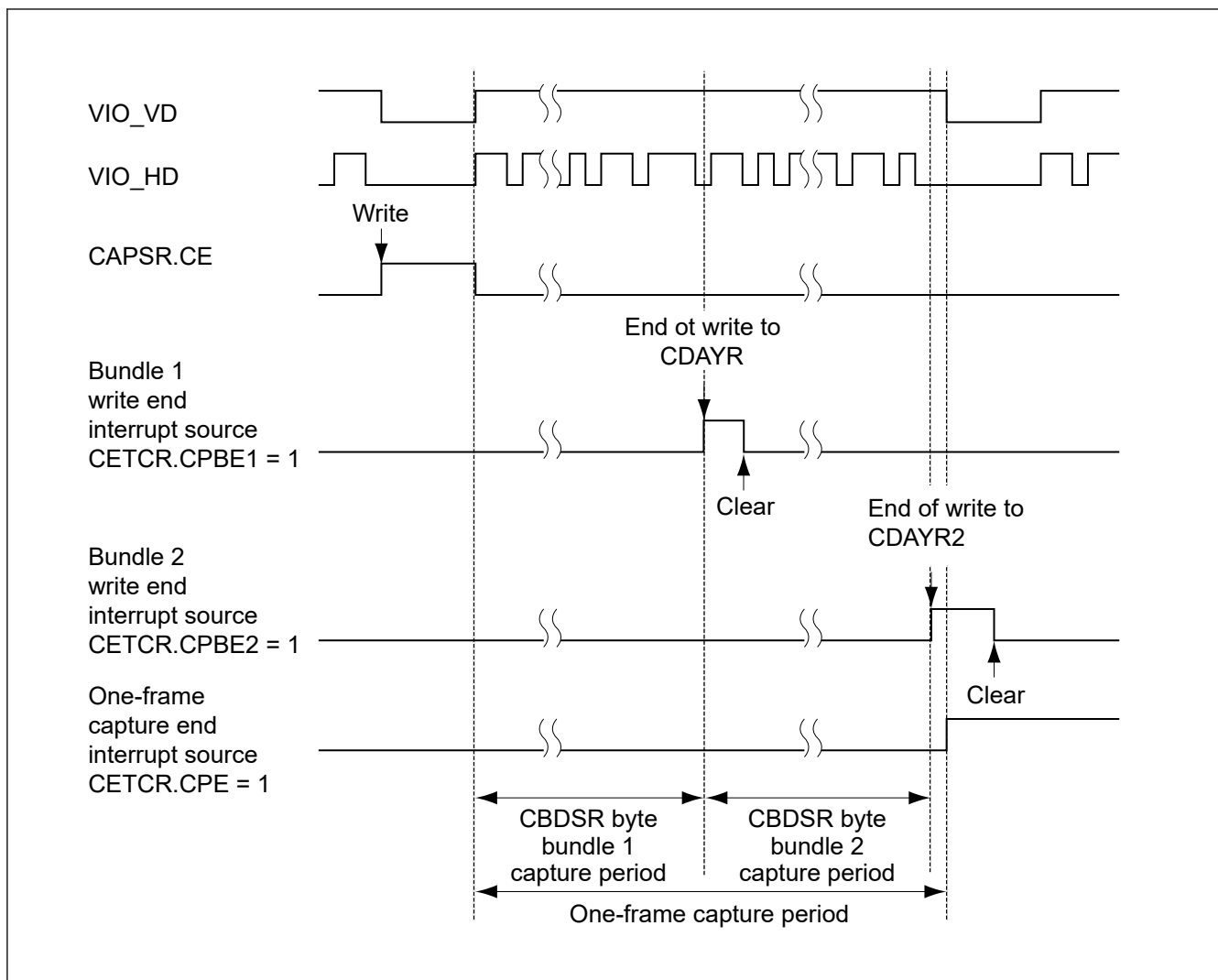


Figure 61.43 Timing of write end interrupts (data enable fetch)

For data output from the CEU, the COLS, COWS, and COBS bits control swapping in 32-bit, 16-bit, and 8-bit units, respectively. Set these bits when data is misaligned because of a difference in endian. The data swapping bits are shown below. These bits can be set similarly in data fetch mode.

Data can be swapped in 8-bit, 16-bit, 32-bit units, or in 32 bits, 16 bits and 8 bits, as shown in [Figure 61.44](#). To enable data swapping, set the corresponding control bit to 1.

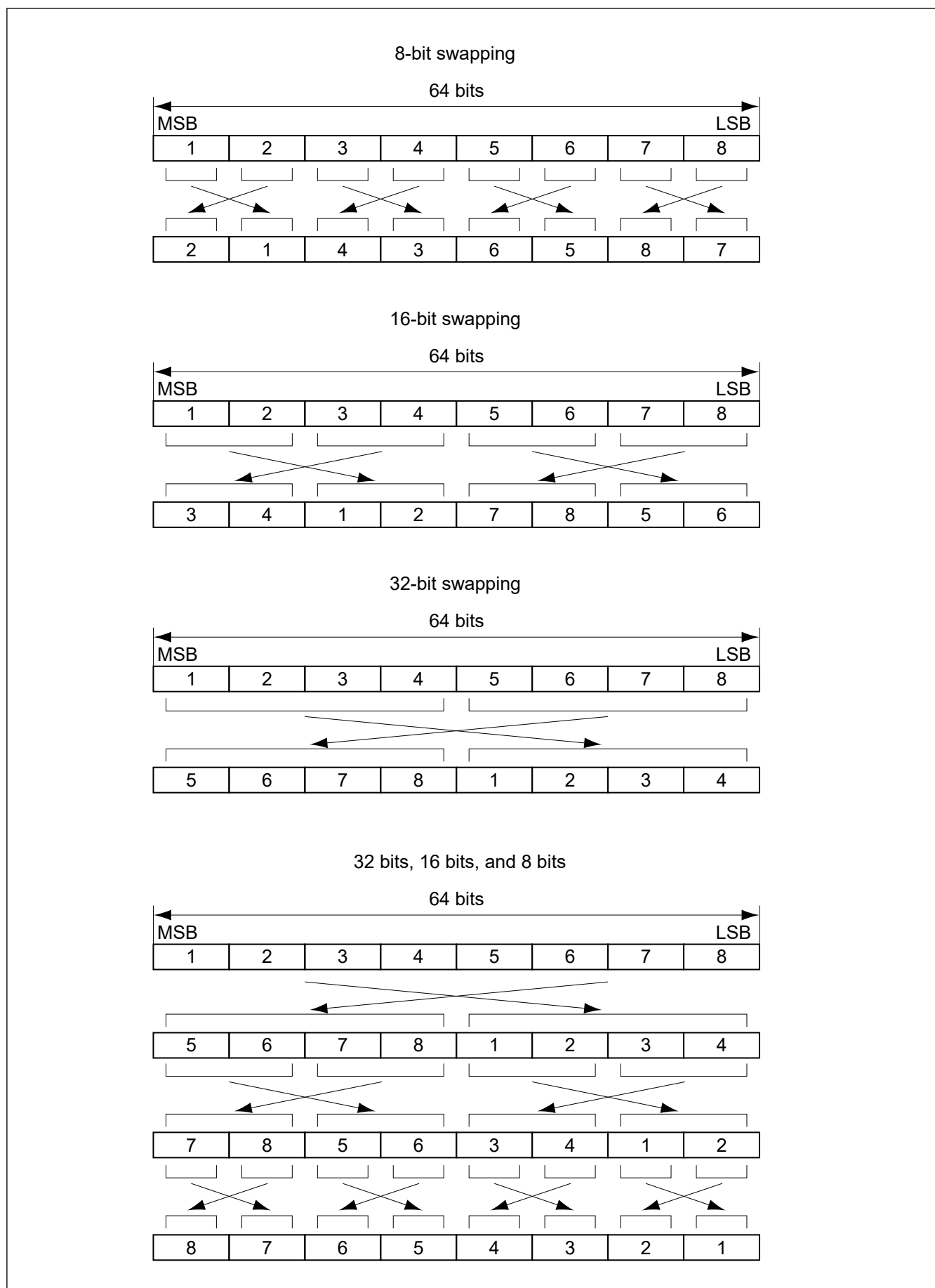


Figure 61.44 Data swapping by data aligner

61.2.21 CEIER : Capture Event Interrupt Enable Register

Base address: CEU = 0x4034_8000
CEU_NS = 0x5034_8000

Offset address: 0x0070

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	NVDIE	NHDIE	FWFIE	—	—	VBPIE	—	IGVSI E	IGHSI E	CDTO FIE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	CPBE 4IE	CPBE 3IE	CPBE 2IE	CPBE 1IE	—	—	VDIE	HDIE	—	—	—	IGRWI E	—	—	CFEIE	CPEIE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	CPEIE	One-Frame Capture End Interrupt Enable 0: Disables a one-frame capture end interrupt 1: Enables a one-frame capture end interrupt	R/W
1	CFEIE	CFE Interrupt Enable 0: Disables a CFE interrupt 1: Enables a CFE interrupt	R/W
3:2	—	These bits are read as 0. The write value should be 0.	R/W
4	IGRWIE	Register-Access-During-Capture Interrupt Enable 0: Disables a register-access-during-capture interrupt 1: Enables a register-access-during-capture interrupt	R/W
7:5	—	These bits are read as 0. The write value should be 0.	R/W
8	HDIE	HD Interrupt Enable 0: Disables an HD interrupt 1: Enables an HD interrupt	R/W
9	VDIE	VD Interrupt Enable 0: Disables a VD interrupt 1: Enables a VD interrupt	R/W
11:10	—	These bits are read as 0. The write value should be 0.	R/W
12	CPBE1IE	CPBE1 Interrupt Enable 0: Disables a CPBE1 interrupt 1: Enables a CPBE1 interrupt	R/W
13	CPBE2IE	CPBE2 Interrupt Enable 0: Disables a CPBE2 interrupt 1: Enables a CPBE2 interrupt	R/W
14	CPBE3IE	CPBE3 Interrupt Enable 0: Disables a CPBE3 interrupt 1: Enables a CPBE3 interrupt	R/W
15	CPBE4IE	CPBE4 Interrupt Enable 0: Disables a CPBE4 interrupt 1: Enables a CPBE4 interrupt	R/W
16	CDTOFIE	CDTOF Interrupt Enable 0: Disables a CDTOF interrupt 1: Enables a CDTOF interrupt	R/W
17	IGHSE	IGHS Interrupt Enable 0: Disables an IGHS interrupt 1: Enables an IGHS interrupt	R/W
18	IGVSE	IGVS Interrupt Enable 0: Disables an IGVS interrupt 1: Enables an IGVS interrupt	R/W

Bit	Symbol	Function	R/W
19	—	This bit is read as 0. The write value should be 0.	R/W
20	VBPIE	VBP Interrupt Enable 0: Disables a VBP interrupt 1: Enables a VBP interrupt	R/W
22:21	—	These bits are read as 0. The write value should be 0.	R/W
23	FWFIE	FWF Interrupt Enable 0: Disables a FWF interrupt 1: Enables a FWF interrupt	R/W
24	NHDIE	Non-HD Interrupt Enable Disable this interrupt (NHDIE = 0) for data enable fetch. 0: Disables a non-HD interrupt 1: Enables a non-HD interrupt	R/W
25	NVDIE	Non-VD Interrupt Enable Disable this interrupt (NVDIE = 0) for data enable fetch. 0: Disables a non-VD interrupt 1: Enables a non-VD interrupt	R/W
31:26	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

CEIER enables or disables interrupts of the event flag register that generates CEU interrupts.

61.2.22 CETCR : Capture Event Flag Clear Register

Base address: CEU = 0x4034_8000
CEU_NS = 0x5034_8000

Offset address: 0x0074

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	NVD	NHD	FWF	—	—	VBP	—	IGVS	IGHS	CDTO F
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	CPBE 4	CPBE 3	CPBE 2	CPBE 1	—	—	VD	HD	—	—	—	IGRW	—	—	CFE	CPE
Value after reset:	0	0	0	0	0	0	x	x	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	CPE	An interrupt indicating that capturing of one frame from an external module has finished. This interrupt is output when the last captured data has been transferred and the end notification received, regardless of the next VD input. This interrupt indicates that capturing of one frame has finished. This bit is set to 1 when the image of the size set in CAPWR is captured and the last data transfer to the bus finished (see Figure 61.46).	R/W
1	CFE	An interrupt indicating that capturing of one field from an external module has finished. This interrupt is output when the last captured data has been transferred and the end notification received, regardless of the next VD input (see Figure 61.45). This interrupt occurs only in both-field capture mode.	R/W
3:2	—	These bits are read as 0. The write value should be 0.	R/W
4	IGRW	An interrupt indicating that during capturing, access was attempted to a register to which writing during operation is prohibited. Among the CEU registers, writing during capturing is prohibited for some registers. Table 61.10 shows which registers can/cannot be written to during capturing. This bit is set to 1 when a register to which writing during capturing is prohibited has been written to.	R/W
7:5	—	These bits are read as 0. The write value should be 0.	R/W

Bit	Symbol	Function	R/W
8	HD	An interrupt indicating that HD (horizontal sync signal) was input from an external module. This bit is set to 1 when an HD input from an external module is detected. Immediately after the HDPOL bit in CAMCR is modified, a pseudo HD is input and this bit is set to 1. The HD interrupt after the HDPOL bit is modified should be ignored.	R/W
9	VD	An interrupt indicating that VD (vertical sync signal) was input from an external module. In data enable fetch mode, this bit is set to 1 when a VD input from an external module is detected. In image capture mode and data synchronous fetch mode, this module generates a VD interrupt in response to the first detection of the active level of HD following detection of the active level of VD from an external module. Note that, when VD and HD are asserted and detected at the same time, this module generates a VD interrupt at that time. Immediately after the VDPOL bit in CAMCR is modified, a pseudo VD is input and this bit is set to 1. The VD interrupt after the VDPOL bit is modified should be ignored.	R/W
11:10	—	These bits are read as 0. The write value should be 0.	R/W
12	CPBE1	An interrupt indicating that writing to CDAYR and CDACR in a bundle write has finished. This interrupt is output when the last captured data has been transferred and the end notification received, regardless of the next HD input. This bit is set to 1 when data for the number of lines (number of bytes in data enable fetch) set in CBDSR has been captured and the last data transfer to the bus has completed. However, in image capture or data synchronous fetch, this interrupt does not occur when the last captured data in a bundle write is the last captured data of a frame (field).	R/W
13	CPBE2	An interrupt indicating that writing to CDAYR2 and CDACR2 in a bundle write has finished. This interrupt is output when the last captured data has been transferred and the end notification received, regardless of the next HD input. This bit is set to 1 when data for the number of lines (number of bytes in data enable fetch) set in CBDSR has been captured and the last data transfer to the bus has completed. However, in image capture or data synchronous fetch, this interrupt does not occur when the last captured data in a bundle write is the last captured data of a frame (field).	R/W
14	CPBE3	An interrupt indicating that writing to CDBYR and CDBCR in a bundle write has finished. This interrupt is output when the last captured data has been transferred and the end notification received, regardless of the next HD input. This bit is set to 1 when data for the number of lines set in CBDSR has been captured and the last data transfer to the bus has completed. However, this interrupt does not occur when the last captured data in a bundle write is the last captured data of a frame (field).	R/W
15	CPBE4	An interrupt indicating that writing to CDBYR2 and CDBCR2 in a bundle write has finished. This interrupt is output when the last captured data has been transferred and the end notification received, regardless of the next HD input. This bit is set to 1 when data for the number of lines set in CBDSR has been captured and the last data transfer to the bus has completed. However, this interrupt does not occur when the last captured data in a bundle write is the last captured data of a frame (field).	R/W
16	CDTOF	An interrupt indicating that data overflowed in the CRAM of the write buffer. Since data is input at realtime from an external module in capture operations, the frame image is overwritten unless the captured data is transferred from the CEU internal buffer to the memory at a certain or higher transfer rate. This bit is set to 1 when writing the data in the CRAM of the CEU internal write buffer to the bus is not performed within time and data has overflowed.	R/W
17	IGHS	An interrupt generated when the number of HD cycles set in CMCYR differ from the number of HD cycles input from an external module. This bit is set to 1 when there is an illegal HD input from an external module. This bit is set to 1 when the number of clock cycles for the HD input to the CEU differs from the value set in the HCYL bits in CMCYR. Note however that when the HCYL bits are cleared to 0, this interrupt does not occur.	R/W
18	IGVS	An interrupt generated when the number of VD cycles set in CMCYR differ from the number of VD cycles input from an external module. This bit is set to 1 when there is an illegal VD input from an external module. This bit is set to 1 when the number of HD cycles for the VD input to the CEU differs from the value set in the VCYL bits in CMCYR. Note however that when the VCYL bits are cleared to 0, this interrupt does not occur.	R/W
19	—	This bit is read as 0. The write value should be 0.	R/W

Bit	Symbol	Function	R/W
20	VBP	An interrupt indicating that VD has been input while the CEU holds data (insufficient vertical-sync front porch). The conditions for a VBP interrupt to occur are as follows: - Condition 1 VD is input when there is captured data within the CEU - Condition 2 The last transfer data cannot be internally detected due to a write buffer overflow or an illegal HD so that the end timing is unclear until the next VD (By generating a VBP interrupt at the VD input timing, capture fail can be announced.) When a VBP interrupt occurs, a capture end interrupt (CPE bit in CETCR) does not occur and the image of that frame is not captured correctly. Though a capture end interrupt (CPE bit) will occur on rare occasions, it should be ignored in this case. Capturing cannot be performed until the next VD (even if the CE bit (capture reservation signal) in CAPSR is 1, capturing does not start). In the case of condition 2, instead of waiting for a VBP interrupt to occur, execute a software reset (CPKIL bit in CAPSR) to stop capturing and then restart capturing. In this case, since capture operation is terminated without waiting for the next VD, a VBP interrupt does not occur and capturing can be performed from the next VD.	R/W
22:21	—	These bits are read as 0. The write value should be 0.	R/W
23	FWF	The interrupt is generated when data is written to the address that exceeds the value specified with CFWCR.FMV. This bit is set to 1 when data is written to the address that exceeds the value specified with CFWCR.FMV while CFWCR.FWE = 1.	R/W
24	NHD	An interrupt indicating that no HD was input. The timing for a non-HD interrupt to occur differs depending on the bit width of the digital image input pins. 8-bit digital image input pins: Occurs when the 11-bit internal counter that is incremented every eight cycles becomes full. Accordingly, this bit is set to 1 when no HD has been input for at least 16,376 cycles since the last HD was input. 16-bit digital image input pins: Occurs when the 12-bit internal counter that is incremented every four cycles becomes full. Accordingly, this bit is set to 1 when no HD has been input for at least 16,380 cycles since the last HD was input. When connecting a camera whose HD is fixed low when VD is low, this bit may be set to 1. Ignore this interrupt during data enable fetch.	R/W
25	NVD	An interrupt indicating that no VD was input. A non-VD interrupt occurs when the 14-bit internal counter becomes full. Accordingly, this bit is set to 1 when no VD has been input for at least 16,383 lines since the last VD was input.	R/W
31:26	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

CETCR notifies the CPU of the source of an interrupt that is generated in the CEU. The flags of CETCR can be used as interrupt signals. When the corresponding interrupt is enabled, an interrupt (CEU_CEUI) is generated. To clear the interrupt, clear the bit corresponding to the interrupt source to 0. After several cycles have passed after modifying the bit, the interrupt is cleared.

To clear the bit corresponding to the interrupt source to be cleared to 0 and retain that state, write 1 to that bit. For example, to clear only the CPE bit to 0, write 0xFFFFF0FE to CETCR.

In CETCR, only bits to which 0 is written are cleared. Bits to which 1 is written retain their current values. To clear an interrupt source, write 0 only to the bit corresponding to the interrupt source to be cleared, and 1 to the other bits.

Note: Since the CETCR value becomes undefined in the following cases, clear all bits in CETCR to 0.

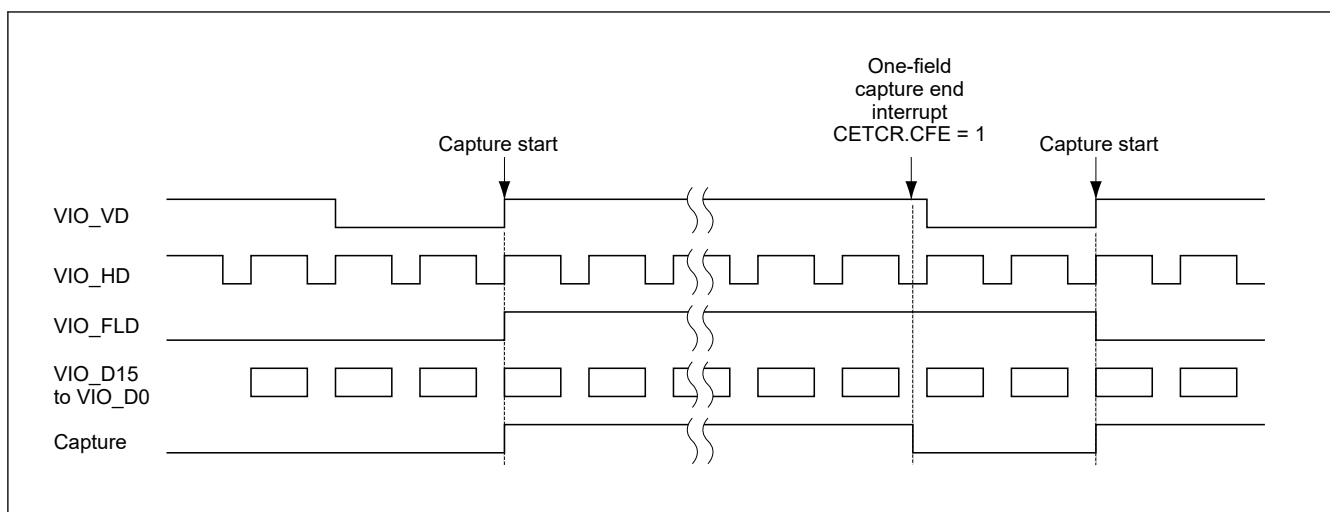
- VD and HD bits immediately after system reset or Software Standby mode is entered.
- VD and HD bits after the polarities of the capture interface sync signals are changed.

Table 61.10 Registers that can/cannot be modified during capturing (1 of 2)

Register name	Modification during capturing
CAPSR	Possible
CAPCR	Prohibited
CAMCR	Prohibited

Table 61.10 Registers that can/cannot be modified during capturing (2 of 2)

Register name	Modification during capturing
CMCYR	Prohibited
CAMOR	Possible
CAPWR	Possible
CAIFR	Prohibited
CRCNTR	Possible
CRCMPR	Prohibited
CFLCR	Possible
CFSZR	Possible
CDWDR	Possible
CDAYR	Possible
CDACR	Possible
CDBYR	Possible
CDBCR	Possible
CBDSR	Possible
CFWCR	Possible
CLFCR	Possible
CDOCR	Possible
CEIER	Possible
CETCR	Possible
CSTSR	Prohibited
CDSSR	Prohibited
CDAYR2	Possible
CDACR2	Possible
CDBYR2	Possible
CDBCR2	Possible

**Figure 61.45** CFE generation timing

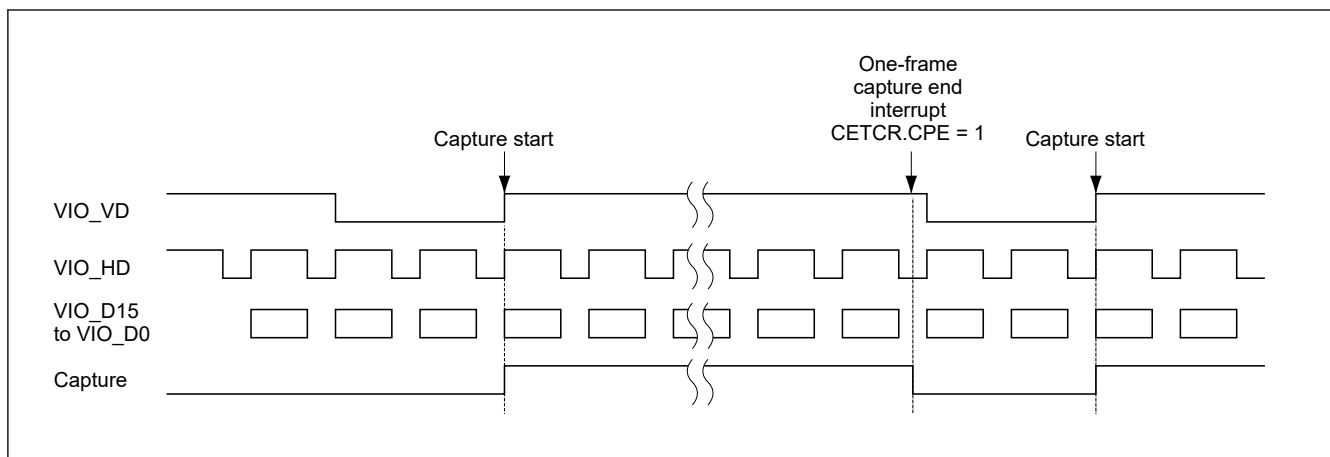


Figure 61.46 CPE generation timing

61.2.23 CSTSR : Capture Status Register

Base address: CEU = 0x4034_8000
CEU_NS = 0x5034_8000

Offset address: 0x007C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	CRST	—	—	—	—	—	—	—	CPFLD
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	CPTON
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	CPTON	Indicates that the CEU is operating. This bit retains 1 during the period that starts from the internal VD at capture start and ends when a one-frame capture end interrupt occurs. Figure 61.47 shows the CEU operating period.	R
15:1	—	These bits are read as 0.	R
16	CPFLD	Indicates which field is being captured. 0: Bottom field is being captured 1: Top field is being captured	R
23:17	—	These bits are read as 0.	R
24	CRST	Indicates which register plane is currently used. 0: Plane A of the register is being used 1: Plane B of the register is being used	R
31:25	—	These bits are read as 0.	R

Note: S-TYPE-3, P-TYPE-3

CSTSR indicates the internal status of the CEU. CSTSR differs from CETCR in that no interrupt is generated for the events indicated in CSTSR.

The CEU operating/halt state can be confirmed using CSTSR. To confirm the halt state of the CEU, make sure that the status bit (bit 0) indicating that the CEU is operating is cleared to 0 for sure.

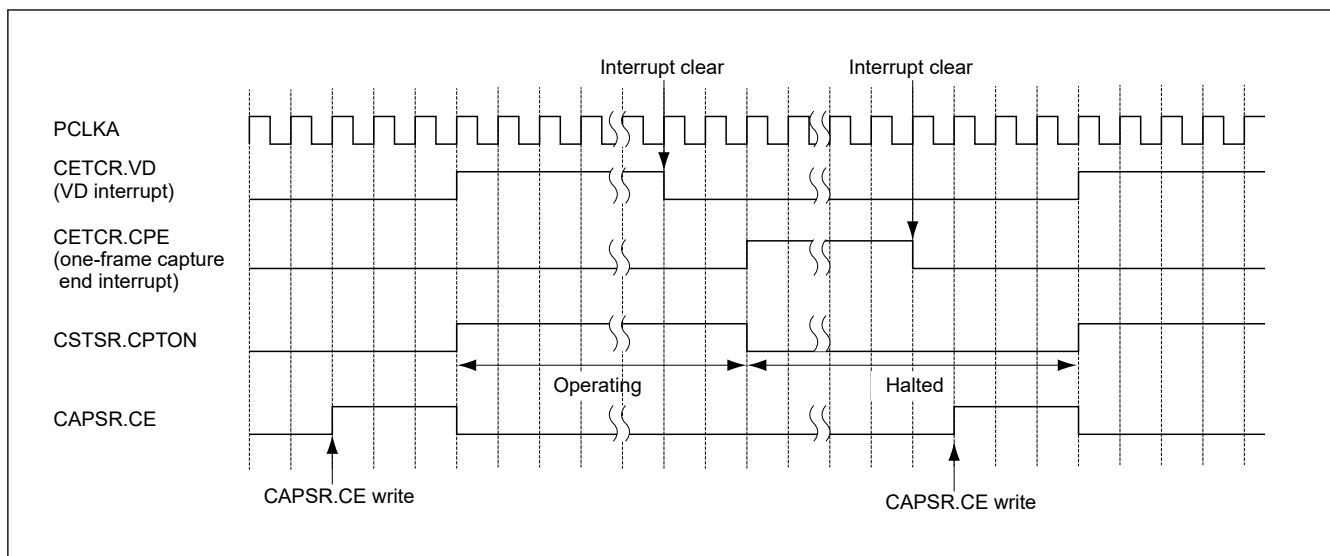


Figure 61.47 Operating status during capturing

61.2.24 CDSSR : Capture Data Size Register

Base address: CEU = 0x4034_8000
CEU_NS = 0x5034_8000

Offset address: 0x0084

Bit position: 31

0

Bit field:

CDSS[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	CDSS[31:0]	Indicate the size of data written to the memory in data enable fetch. In a bundle write, size of data written to the selected address at the end of one-frame capture is indicated. In a bundle write, as soon as the number of bytes specified by CBDSR is transferred to the bus, address to which data is written is switched. Therefore, if one-frame capture is completed at the same time as a bundle write is completed, this register indicates 0x00000000. Figure 61.48 and Figure 61.49 show the overall timing of the CDSSR operation in a bundle write.	R

Note: S-TYPE-3, P-TYPE-3

CDSSR indicates the size of data written to the memory in data enable fetch. As this register indicates a correct value at the end of capture, confirm this register when capture is completed.

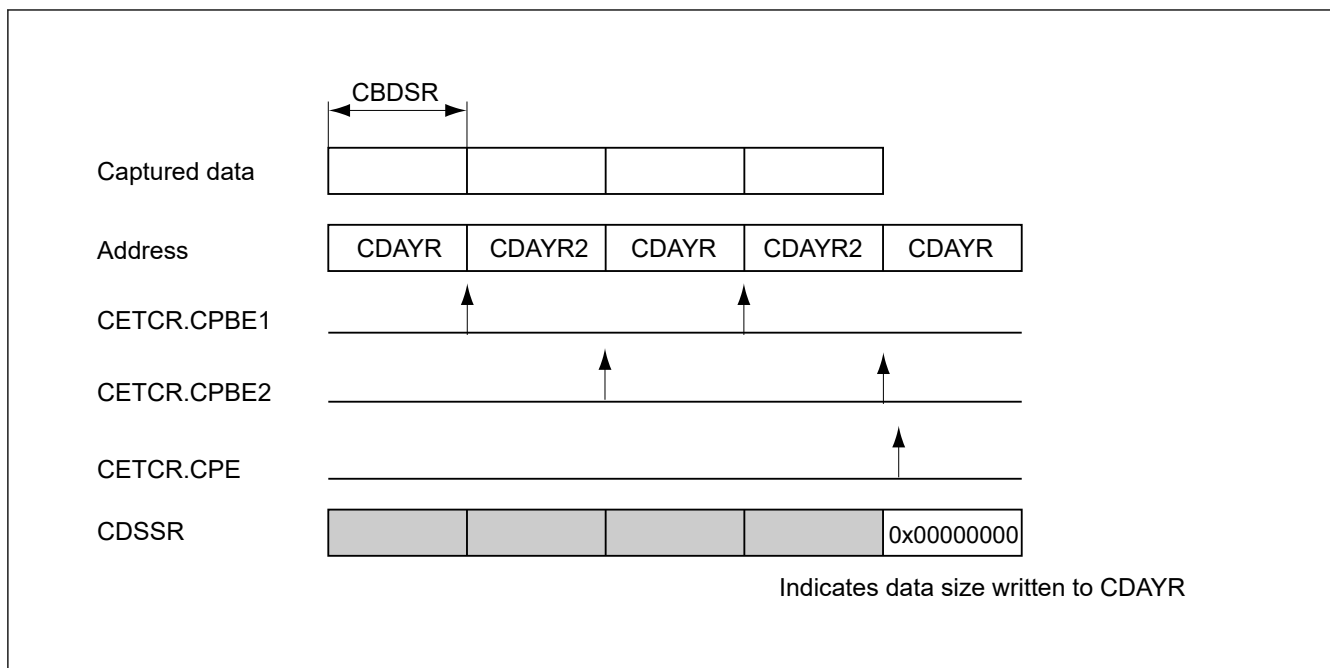


Figure 61.48 Overall timing of CDSSR operation in bundle write (when bundle write end and capture end coincide)

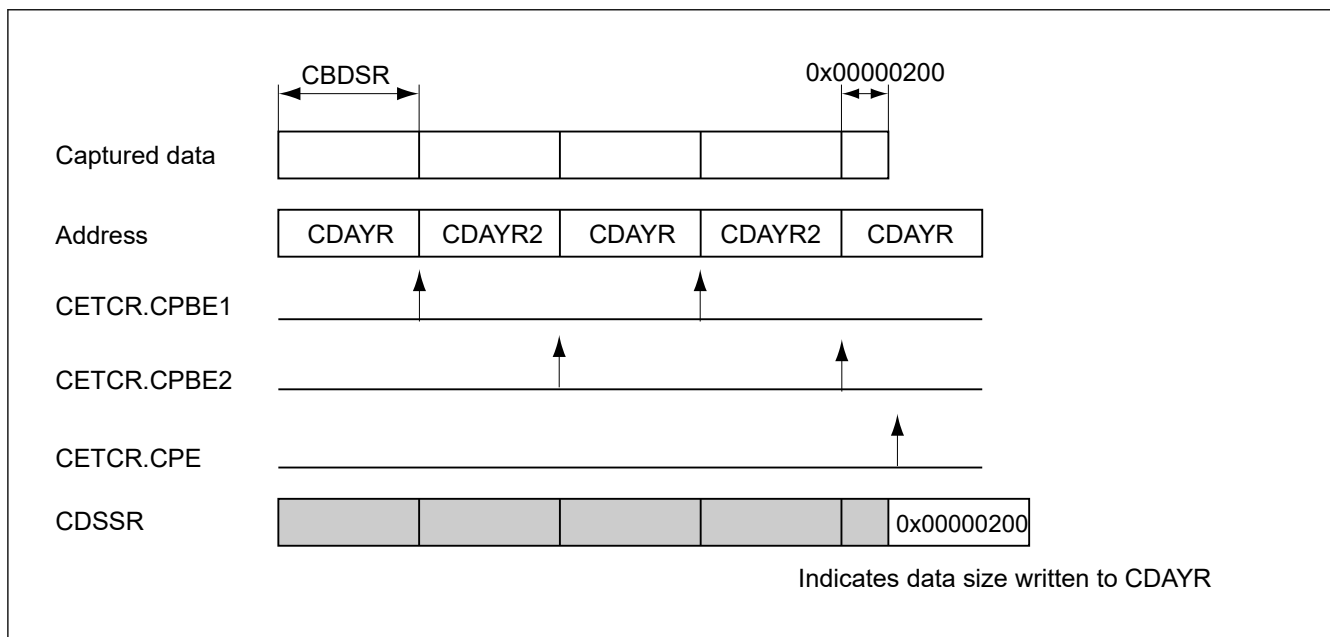


Figure 61.49 Overall timing of CDSSR operation in bundle write (when bundle write end and capture end do not coincide)

61.2.25 CDAYR2, CDAYR2_x : Capture Data Address Y Register 2 (x = B, M)

Base address: CEU = 0x4034_8000
CEU_NS = 0x5034_8000

Offset address: 0x0090 (CDAYR2)
0x1090 (CDAYR2_B)
0x2090 (CDAYR2_M)

Bit position: 31

0

Bit field:

CAYR2[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	CAYR2[31:0]	Capture Data Address Y Frame image capture: These bits set the address for storing the luminance component data of the captured data (8-pixel units). - One-field image capture: These bits set the address for storing the luminance component data of the captured data (8-pixel units). - Both-field image capture: These bits set the address for storing the luminance component data of the captured top-field data (8-pixel units). - Data synchronous fetch: These bits set the address for storing data (8-byte units). - Data enable fetch: These bits set the address for storing data (32-byte units). The lower 3 bits should be written 0.	R/W

Note: S-TYPE-3, P-TYPE-3

CDAYR2 specifies the address for the luminance component used in a bundle write and the address for data storage in a bundle write in data fetch. CDAYR2 is used only in a bundle write.

CDAYR2 specifies the address where the luminance component of the captured data is to be stored in frame image capture or one-field image capture, the address where the luminance component of the captured top field is to be stored in both-field image capture, and the address where data is to be stored in data fetch. The CEU separates the captured image data into the luminance component data (Y) and the chrominance component data (C), and stores them in the memory via the bus. In frame image capture or one-field image capture, set the start address of the memory area where the luminance component of the captured data is to be stored by CDAYR2. In both-field image capture, set the start address of the memory area where the luminance component of the captured top-field image is to be stored by CDAYR2. In data fetch, set the start address of the memory area to be used for data storage.

The address specified by this register must be in 32-bit units.

Set the address of the starting point of the memory area where the luminance component of the captured image is to be stored in this register, as shown in [Figure 61.50](#).

- Frame image capture: Set the address of the starting point of the memory area where the luminance component of the captured image is to be stored.
- One-field image capture: Set the address of the starting point of the memory area where the luminance component of the captured image is to be stored.
- Both-field image capture: Set the address of the starting point of the memory area where the luminance component of the captured top-field image is to be stored.
- Data synchronous fetch: Set the address of the starting point of the memory area where the captured data is to be stored.
- Data enable fetch: Set the address of the starting point of the memory area where the captured data is to be stored in 32-byte units.

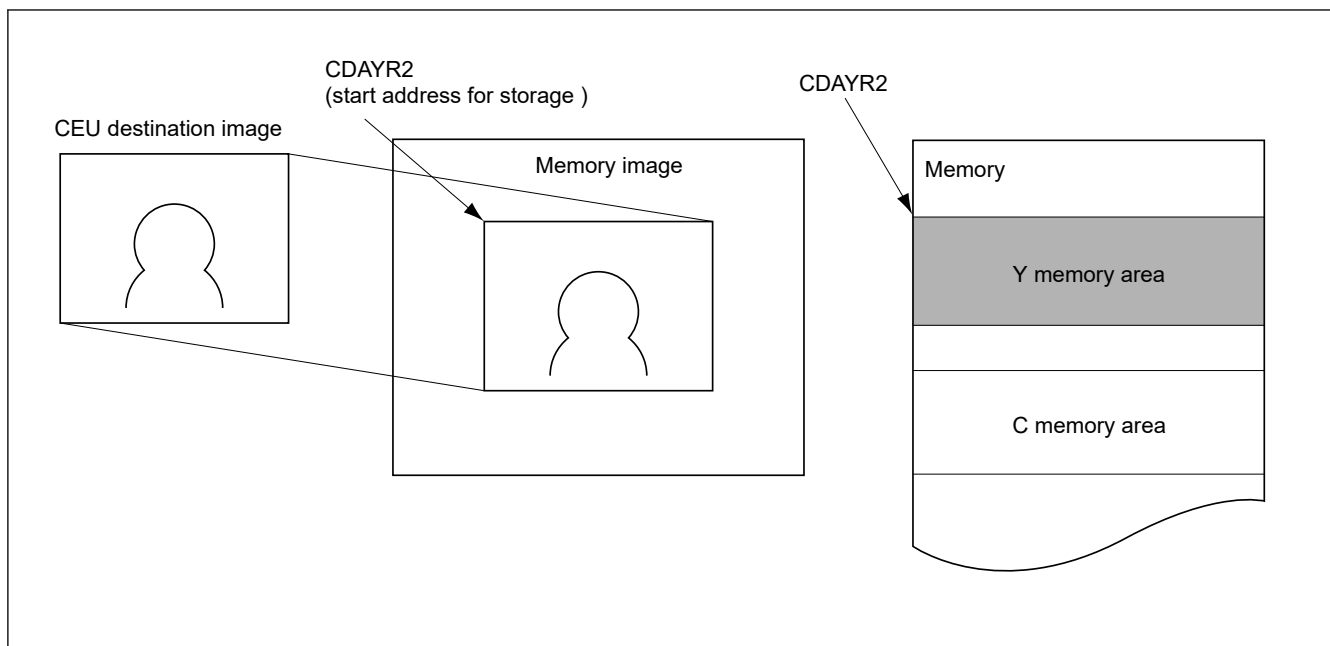


Figure 61.50 Relationship between captured image and luminance component memory area

61.2.26 CDACR2, CDACR2_x : Capture Data Address C Register 2 (x = B, M)

Base address: CEU = 0x4034_8000
CEU_NS = 0x5034_8000

Offset address: 0x0094 (CDACR2)
0x1094 (CDACR2_B)
0x2094 (CDACR2_M)

Bit position: 31

0

Bit field:

CACR2[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	CACR2[31:0]	Capture Data Address C - Frame image capture: These bits set the address for storing the chrominance component data of the captured data (8-pixel units). - One-field image capture: These bits set the address for storing the chrominance component data of the captured data (8-pixel units). - Both-field image capture: These bits set the address for storing the chrominance component data of the captured top-field data (8-pixel units). The lower 3 bits should be written 0.	R/W

Note: S-TYPE-3, P-TYPE-3

CDACR2 specifies the address for the chrominance component used in a bundle write. CDACR2 is used only in a bundle write.

CDACR2 specifies the address where the chrominance component of the captured data is to be stored in frame image capture or one-field image capture, and the address where the chrominance component of the captured top field is to be stored in both-field image capture. The CEU separates the captured image data into the luminance component data (Y) and the chrominance component data (C), and stores them in the memory via the bus. In frame image capture or one-field image capture, set the start address of the memory area where the chrominance component of the captured data is to be stored by CDACR2. In bothfield image capture, set the start address of the memory area where the chrominance component of the captured top-field image is to be stored by CDACR2. CDACR2 is not used in data fetch.

The address specified by this register must be in 32-bit units.

In this register, set the address of the starting point of the memory area where the captured data is to be stored in a bundle write, as shown in [Figure 61.51](#).

- Frame image capture: Set the address of the starting point of the memory area where the chrominance component of the captured image is to be stored.
- One-field image capture: Set the address of the starting point of the memory area where the chrominance component of the captured image is to be stored.
- Both-field image capture: Set the address of the starting point of the memory area where the chrominance component of the captured top-field image is to be stored.

The chrominance component has an output data format as shown in [Figure 61.52](#), and is saved in the memory in this format.

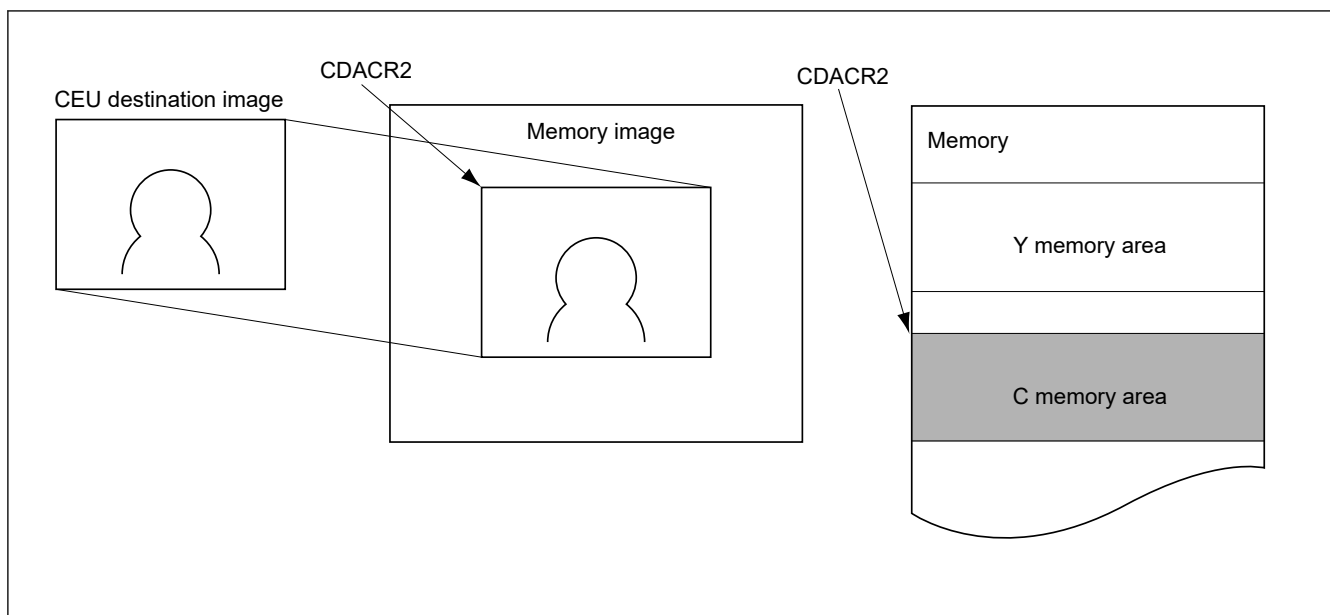


Figure 61.51 Relationship between captured image and chrominance component memory area

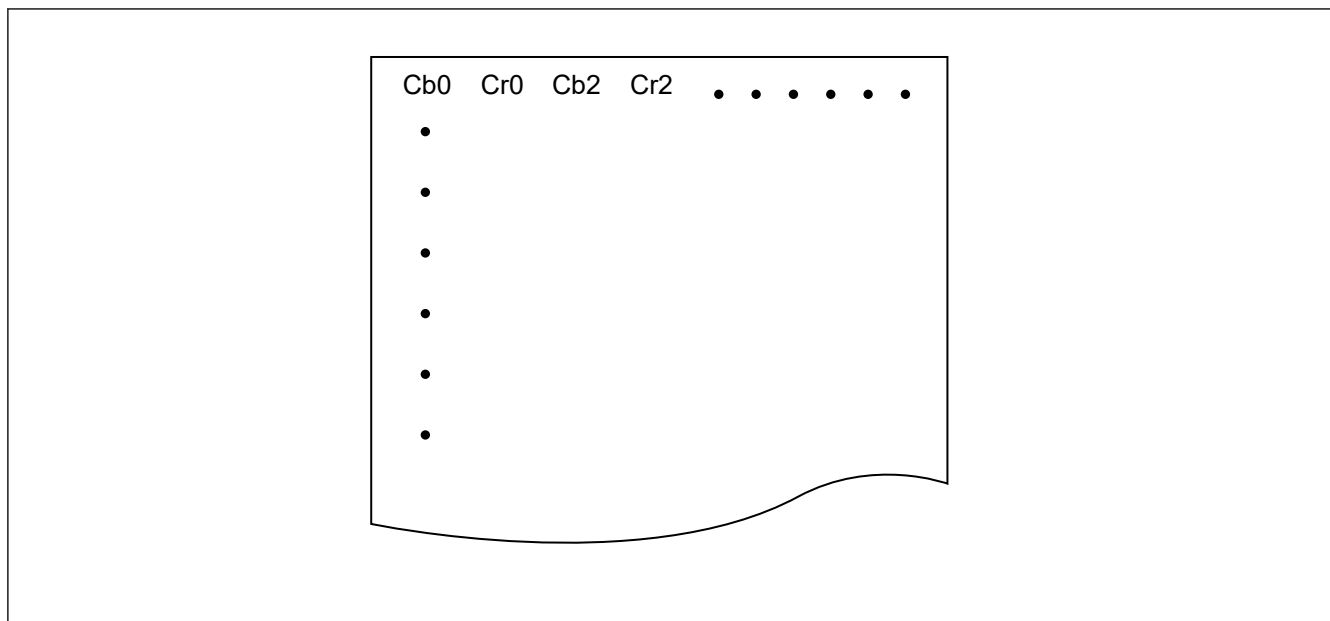


Figure 61.52 Image of storing chrominance components in memory

61.2.27 CDBYR2, CDBYR2_x : Capture Data Bottom-Field Address Y Register 2 (x = B, M)

Base address: CEU = 0x4034_8000
CEU_NS = 0x5034_8000

Offset address: 0x0098 (CDBYR2)
0x1098 (CDBYR2_B)
0x2098 (CDBYR2_M)

Bit position: 31

0

Bit field:

CBYR2[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	CBYR2[31:0]	Set the address for storing the luminance component data of the captured bottom-field data (8-pixel units). The lower 3 bits should be written 0.	R/W

Note: S-TYPE-3, P-TYPE-3

CDBYR2 specifies the address for the luminance component of the bottom field used in a bundle write. CDBYR2 is used only in a bundle write.

CDBYR2 specifies the address where the luminance component of the captured bottom-field data is to be stored in both-field image capture. The CEU separates the captured image data into the luminance component data (Y) and the chrominance component data (C), and stores them in the memory via the bus. Set the start address of the memory area where the luminance component of the bottom-field image captured in both-field image capture is to be stored by CDBYR2. CDBYR2 is not used in frame image capture, one-field image capture, or data fetch.

The address specified by this register must be in 32-bit units.

In this register, set the address of the starting point of the memory area where the luminance component of the captured bottomfield image is to be stored in a bundle write, as shown in [Figure 61.53](#).

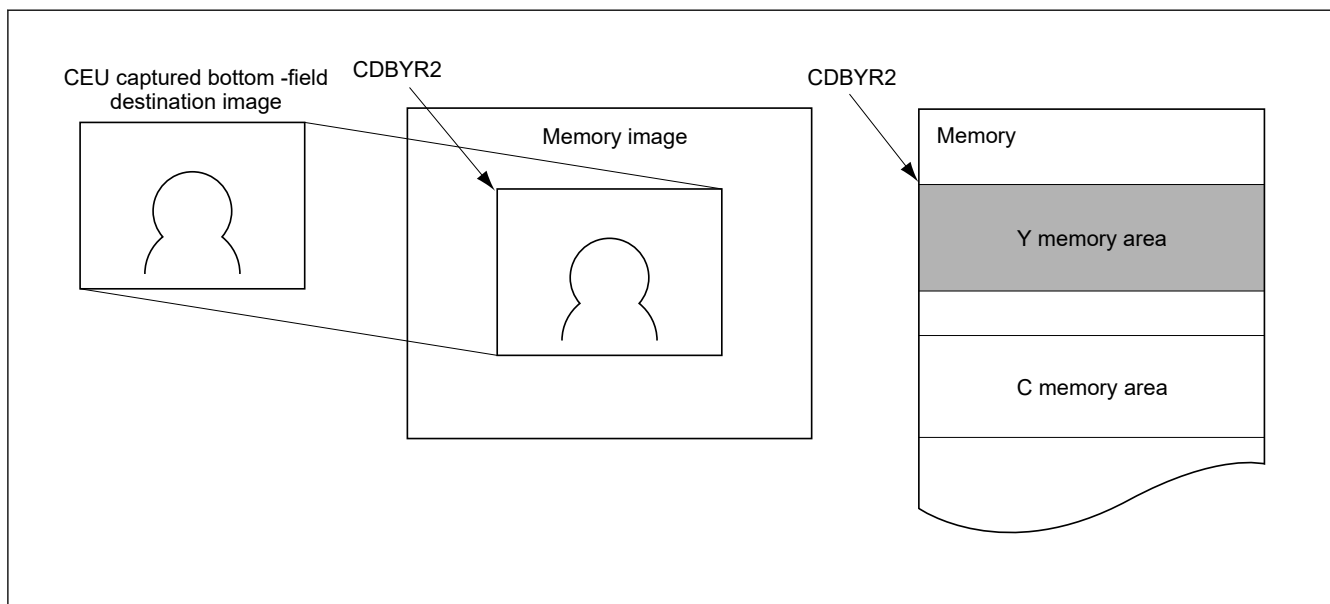
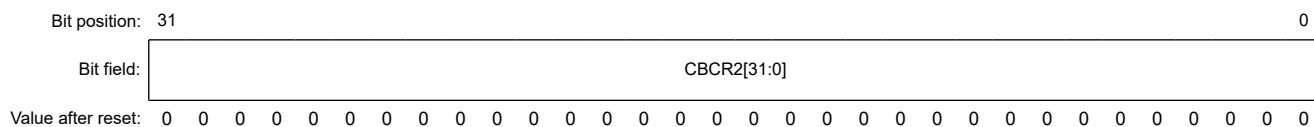


Figure 61.53 Relationship between captured bottom-field image and luminance component memory area

61.2.28 CDBCR2, CDBCR2_x : Capture Data Bottom-Field Address C Register 2 (x = B, M)

Base address: CEU = 0x4034_8000
CEU_NS = 0x5034_8000

Offset address: 0x009C (CDBCR2)
0x109C (CDBCR2_B)
0x209C (CDBCR2_M)



Bit	Symbol	Function	R/W
31:0	CBCR2[31:0]	Set the address for storing the chrominance component data of the captured bottom-field data (8-pixel units). The lower 3 bits should be written 0.	R/W

Note: S-TYPE-3, P-TYPE-3

CDBCR2 specifies the address for the chrominance component of the bottom field used in a bundle write. CDBCR2 is used only in a bundle write.

CDBCR2 specifies the address where the chrominance component of the captured bottom-field data is to be stored in both-field image capture. The CEU separates the captured image data into the luminance component data (Y) and the chrominance component data (C), and stores them in the memory via the bus. Set the start address of the memory area where the chrominance component of the bottom-field image captured in both-field image capture is to be stored by CDBCR2. CDBCR2 is not used in frame image capture, one-field image capture, or data fetch.

The address specified by this register must be in 32-bit units.

In this register, set the address of the starting point of the memory area where the chrominance component of the captured bottomfield image is to be stored in a bundle write, as shown in [Figure 61.54](#). The chrominance component has an output data format as shown in [Figure 61.55](#), and is saved in the memory in this format.

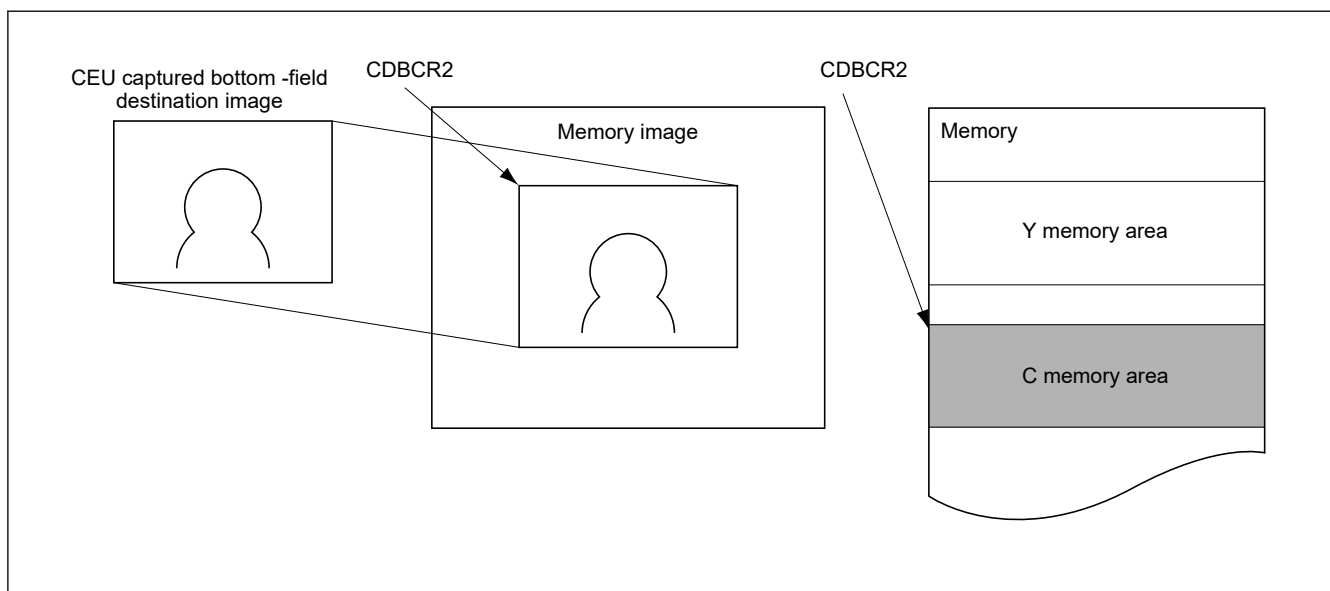


Figure 61.54 Relationship between captured bottom-field image and chrominance component memory area

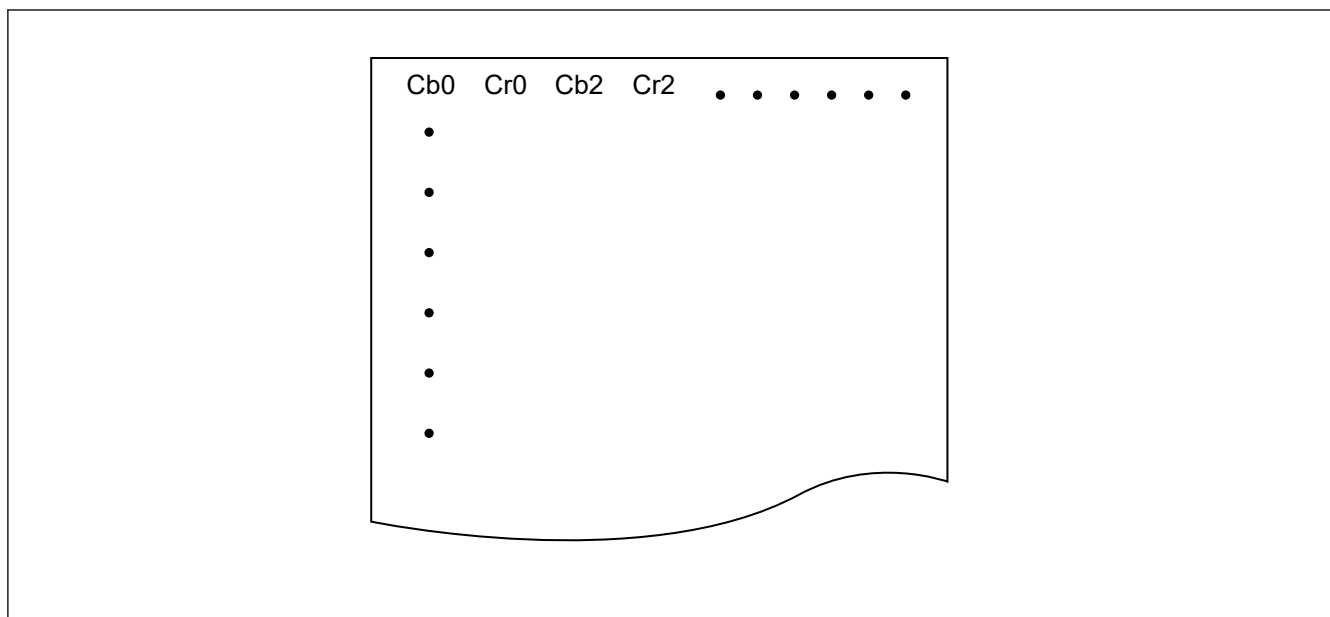


Figure 61.55 Image of storing chrominance components in memory

61.2.29 CBWER : CEU Bufferable Write Enable Register

Base address: CEU = 0x4034_8000
CEU NS = 0x5034 8000

Offset address: 0x00A0

Bit position: 31

Bit field:

[illegible][illegible]

Bit	Symbol	Function	R/W
0	BWE	Bufferable Write Enable 0: Disables Bufferable Write 1: Enables Bufferable Write	R/W
31:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

BWE bit (Bufferable Write Enable)

BWE bit indicates bufferable write is either enabled or disabled.

Note that if this bit is 1, even if the write access as CEU is completed, the actual slave write may not have ended.

Note: When this bit is 1, it requests an early response by writing write data to the temporary buffer if possible to the slave. Therefore, even if write access as CEU is completed, the actual slave write may not be finished. For details on slave groups that support bufferable writes, see [section 15.7.2. Operations When a Bus Error Occurs](#) and respective module sections.

61.3 Usage Notes

61.3.1 Settings for the Module-Stop Function

CEU operation can be disabled or enabled using Module Stop Control Register C (MSTPCRC). The CEU is initially stopped after reset. Releasing the module-stop state enables access to the registers. For details, see [section 11, Low Power Mode](#).

61.3.2 Conditions for Connection to an External Module

(1) Clock Frequency

The external input clock (VIO_CLK) should have a frequency at most the same as the CEU operating clock (PCLKA) frequency, with jitter on both sides included.

$$PCLKA \geq VIO_CLK$$

(2) Blanking Period

The period from the last valid pixel in each line to the next horizontal sync signal HD must be at least 20 cycles.

(3) Fixed Period of Field Identification Signal

The field identification signal FLD should be fixed for at least 1-HD period since a VD input

61.3.3 Restrictions on Input/Output Functions

Table 61.11 lists the restrictions regarding the CEU input/output functions.

Table 61.11 Restrictions on CEU Input/Output Functions

Item	Restrictions
External module interface	The operating clock of the external module (VIO_CLK) should always have a frequency at most the same as that of the CEU operating (PCLKA), with jitter on both sides included.
	Selecting the interface, or modifying the frequency of the external module operating clock or HD/VD polarity must be done when capture operations are halted for sure.
	The capture horizontal size in image capture must be specified as follows: 8-bit interface: 16-cycle units 16-bit interface: 8-cycle units
	The capture horizontal size in data fetch must be specified as follows: 8-bit interface: 8-cycle units 16-bit interface: 4-cycle units
	The capture vertical size must be specified in 4-line units.
	The maximum number of cycles in the horizontal sync signal period should be 16,375 cycles of external input clock for 8-bit digital image input pins, or 16,379 cycles of external input clock for 16-bit digital image input pins.
	The maximum number of lines (HD count) in the vertical sync signal should be 16,382 lines.
	The minimum number of captured pixels should be sub-QCIF (128 × 96).
	The maximum number of captured pixels should be 5 megapixels (2,560 × 1,920).
	The capture size in data enable fetch must be controlled by VD and HD in 4-byte units. The range of the capture size in data enable fetch that can be input is as follows. Maximum: 6 MB (2,048 × 1,536 × 2) Minimum: 16 bytes
Memory output	The output address must be specified in 32-bit units.
	The horizontal size of the destination image (memory) must be specified in 8-pixel units.
	The number of horizontal output pixels (= horizontal clipping size) must be specified in 8-pixel units.
	The number of vertical output lines (HD) (= vertical clipping size) must be specified in 4-line (HD) units.
	In data enable fetch bundle write, the output address must be specified in 32-byte units.
Internal processing	The filter clipping size must be specified as a value equal to or lower than the actual output size of the filter.